Scalar order and separable convergence for primal-first Chambolle–Pock

Abstract

Scalar order strengthens primal-first Chambolle–Pock at extrapolation 1/8. For countably scalar-separable objectives in matched diagonal coordinates, KKT existence gives strong paired convergence through $r_* = (88{+}16\sqrt{91})/81$ and a summable coordinate envelope with squared constant 10^{20}, including zero and arbitrarily small couplings. We make scalar continuation effective with $\eta_{\mathrm{eff}} = 10^{-118}$. A genuine finite-shadow state Lyapunov function then gives whole-orbit KKT convergence and square-summable actual increments for scalar parameters $r_n \in [m, r_* + \eta_{\mathrm{eff}}]$, $m > 0$, of finite total variation, also with summable full-state errors or the stated two-proximal-output errors. The objectives remain fixed; no smallness of variation or total error is required. The proof combines exact four-output order, robust anchored axis geometry, positive finite-shadow energies and an overlapping parameter partition. The original fixed-step matched-Hilbert extension and its envelope $\max\{10^{20}, 3B_{\mathrm{cont}}^2\}$ are retained with complete proofs. Separately, every scalar orbit whose period divides six is constant when $297/100 \le r < 3$, without a KKT assumption; four- and five-period exclusions force every nonconstant periodic orbit there to have least period at least seven. We retain all-real-extrapolation trapping and the complete period-three classification for nonnegative extrapolation. The explicit extension is an effective existence result. No convergence rate, uniform robustness as $m \downarrow 0$, variable-parameter Hilbert theorem, or convergence throughout $(r_*, 3)$ is asserted. Complete analytic proofs and solver-free finite verification data are supplied.

Keywords: Chambolle–Pock iteration; scalar order; countably separable objectives; strong endpoint convergence; uniform coordinate bounds; period-three onset; finite-shadow Lyapunov functions; summable parameter variation

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1 The iteration and principal conclusions

For real Hilbert spaces X, Y, proper lower-semicontinuous convex functions f, h, a bounded linear map $L : X \to Y$, positive steps τ, σ, and a fixed real extrapolation θ, we study exactly

$$\begin{aligned} x_n &= \operatorname{prox}_{\tau f}(x_{n-1} - \tau L^* y_{n-1}), \\ y_n &= \operatorname{prox}_{\sigma h}\big(y_{n-1} + \sigma L(x_n + \theta(x_n - x_{n-1}))\big), \qquad n \ge 1. \end{aligned} \tag{1}$$

The proximal map minimizes $f(x) + \|x - w\|^2/(2\tau)$. The function h is the actual dual-side function and may be a conjugate in saddle-point notation. A KKT pair satisfies

$$-L^* y_* \in \partial f(x_*), \qquad L x_* \in \partial h(y_*). \tag{2}$$

Initialization is arbitrary; in particular, finite initial objective values are never required. The update order and extrapolated point in (1) are fixed throughout.

The main question is how one-dimensional order and matched coordinate separability alter critical dynamics. The scalar barriers below work for all real θ, while the original quantitative strong-convergence theorem fixes

$$\theta = \frac{1}{8}, \qquad \tau\sigma\|L\|^2 \le r_*, \qquad r_* = \frac{88 + 16\sqrt{91}}{81} < 3. \tag{3}$$

The countable result concerns a specific objective class, which we now define precisely. Write $\mathcal{H} = \ell^2(\mathbb{N};\mathbb{R})$. For a sequence of nonnegative extended-real numbers, the notation $\sum_i q_i$ means $\sup_m \sum_{i=1}^m q_i$. Thus no conditional series or difference of two infinite quantities is implicit below.

Let

$$\begin{aligned} f(x) &= \alpha + \langle a, x\rangle + \sum_{i=1}^{\infty} \phi_i(x_i), \\ h(y) &= \gamma + \langle b, y\rangle + \sum_{i=1}^{\infty} \psi_i(y_i), \end{aligned} \tag{4}$$

where $\alpha, \gamma \in \mathbb{R}$, $a, b \in \mathcal{H}$, and every $\phi_i, \psi_i : \mathbb{R} \to [0, +\infty]$ is proper, lower-semicontinuous, and convex. Assume each sum in (4) is finite at some point of $\mathcal{H}$. The functions f, h are then proper lower-semicontinuous convex functions on $\mathcal{H}$. Let $L : \mathcal{H} \to \mathcal{H}$ be the bounded real diagonal operator

$$(Lx)_i = \ell_i x_i, \qquad \sup_i |\ell_i| < \infty.$$

In particular $L^* = L$. Assume a KKT pair in $\mathcal{H} \times \mathcal{H}$ exists as in (2).

Theorem 1 (Strong convergence)**.** *Assume* (4)*, a bounded diagonal L, a KKT pair, and* (3)*. Then every actual orbit* (1) *converges strongly in $\mathcal{H} \times \mathcal{H}$ to a KKT pair. The conclusion includes endpoint equality, zero diagonal entries, and arbitrary initial points in the Hilbert product space.*

Theorem 2 (Summable coordinate envelope)**.** *Under the hypotheses of Theorem 1, for every initialization $(x_0, y_0) \in \mathcal{H} \times \mathcal{H}$, the actual orbit* (1) *satisfies, for each i and $n \ge 0$,*

$$\frac{|x_{n,i} - x_{*,i}|^2}{\tau} + \frac{|y_{n,i} - y_{*,i}|^2}{\sigma} \le C\left(\frac{|x_{0,i} - x_{*,i}|^2}{\tau} + \frac{|y_{0,i} - y_{*,i}|^2}{\sigma}\right), \qquad C = 10^{20}. \tag{5}$$

In particular,

$$\sum_{i=1}^{\infty} \sup_{n\ge 0}\left(\frac{|x_{n,i} - x_{*,i}|^2}{\tau} + \frac{|y_{n,i} - y_{*,i}|^2}{\sigma}\right) \le C\left(\frac{\|x_0 - x_*\|^2}{\tau} + \frac{\|y_0 - y_*\|^2}{\sigma}\right) < \infty. \tag{6}$$

Thus the entire paired orbit is bounded. No finite initial objective value is required. The constant is independent of the objectives, the diagonal entries, the step sizes, the KKT pair, and the initialization. If $L = 0$, both conclusions hold with $C = 1$.

The operator norm need not be attained by any diagonal entry, and nonzero entries may tend to zero. Both objectives must separate in the same matched scalar coordinates that diagonalize L. Separability into coupled blocks of dimension greater than one is not a hypothesis of this theorem.

Equivalently, the functions may be supplied in the anchored form

$$\begin{aligned} f(x) &= f(x_*) - \langle L^* y_*, x - x_* \rangle + \sum_i F_i(x_i - x_{*,i}), \\ h(y) &= h(y_*) + \langle L x_*, y - y_* \rangle + \sum_i H_i(y_i - y_{*,i}), \end{aligned} \tag{7}$$

where the displayed anchor values are finite and $F_i, H_i : \mathbb{R} \to [0, +\infty]$ are proper lower-semicontinuous convex functions with $F_i(0) = H_i(0) = 0$. This form itself defines proper lower-semicontinuous convex functions and supplies (2). Section 4 proves that (4) has this representation at every KKT pair.

The two theorems remain valid on finite matched orthonormal scalar products. The general all-real-θ bounded-increment implication is stated separately in Theorem 5; its parameter scope is not subsumed by (3). The complementary period-three classification, Theorem 7, has exact onset 3 for every $\theta \geq 0$, with no KKT assumption in the exclusion direction. The onset 3 is a period-three threshold, not a convergence threshold. The scalar theorem below admits the effective refinement $\eta_{\text{eff}} = 10^{-118}$; the full interval $(r_*, 3)$ remains unresolved for two arbitrary scalar objectives. The separate six-period restriction in Theorem 8 gives no convergence implication.

1.1 Scalar continuation and its effective refinement

The scalar endpoint admits a strict right-hand continuation. The following theorem concerns the same actual update, now with $L = I$, $\sigma = 1$ and $\tau = r$. Its proof uses the KKT anchor as well as scalar order; it does not follow from excluding finitely many periods.

Theorem 3 (Uniform scalar continuation)**.** *There exist constants*

$$0 < \eta < 3 - r_*, \qquad B_{\text{cont}} < \infty, \qquad H_{\text{cont}} < \infty$$

with the following property. Let $f, h : \mathbb{R} \to (-\infty, +\infty]$ be two fixed proper lower-semicontinuous convex functions for which

$$\mathcal{Z} = \{(x, y) : -y \in \partial f(x),\ x \in \partial h(y)\}$$

is nonempty. Fix any $r \in [r_*, r_* + \eta]$ and any initial pair $z_0 = (x_0, y_0) \in \mathbb{R}^2$. Then the actual iteration

$$x_n = \text{prox}_{rf}(x_{n-1} - r y_{n-1}), \qquad y_n = \text{prox}_h\left(y_{n-1} + \frac{9x_n - x_{n-1}}{8}\right) \tag{8}$$

converges as a pair to a member of $\mathcal{Z}$. For every $z_* \in \mathcal{Z}$ and every integer $m \geq 0$ it satisfies

$$\sup_{n \geq m} \|z_n - z_*\| \leq B_{\mathrm{cont}} \|z_m - z_*\|, \tag{9}$$

$$\sum_{n=m+1}^{\infty} \|z_n - z_{n-1}\|^2 \leq H_{\mathrm{cont}} \|z_m - z_*\|^2. \tag{10}$$

The same three constants work for all functions, anchors, initializations, fixed parameters r in the displayed interval, and restart indices m. No finite initial objective value is required.

The original compactness proof of this theorem is existential. Theorem 9 independently certifies the concrete choice $\eta_{\mathrm{eff}} = 10^{-118}$. In particular the theorem does not establish convergence at $r = 2.99$ or throughout $(r_*, 3)$. The parameter in this theorem is fixed along each orbit; its uniformity alone does not authorize varying r with n. Such variation is treated separately in Theorem 4. Endpoint convergence was already included in Theorem 1; the additional endpoint conclusion here is square summability starting with the first increment. Combined with the scalar case of the earlier theorem and the fixed scalar normalization, this gives general scalar convergence for $0 < r \leq r_* + \eta$. The constants in (9)–(10) are asserted only on $[r_*, r_* + \eta]$.

A separate matched-separable consequence, Corollary 7.3, uses the same η on the original objective class (4). Its new squared envelope constant is $\max\{10^{20}, 3B_{\mathrm{cont}}^2\}$; the original numerical constants remain available on their original range.

1.2 Effective continuation and scalar robustness

Theorem 9 below gives the concrete choice $\eta_{\mathrm{eff}} = 10^{-118}$ in the scalar continuation theorem. More importantly, finite frozen shadows provide a strict state energy that permits parameters to vary over any compact positive scalar band.

Theorem 4 (Scalar robustness on a compact positive band)**.** *Fix $0 < m \leq r_* + 10^{-118}$ and two fixed proper lower-semicontinuous convex functions $f, h : \mathbb{R} \to (-\infty, +\infty]$ with nonempty KKT set*

$$\mathcal{Z} = \{(x, y) : -y \in \partial f(x),\ x \in \partial h(y)\}.$$

For $r > 0$, define $T_r(x, y) = (u, v)$ by

$$u = \mathrm{prox}_{rf}(x - ry), \qquad v = \mathrm{prox}_h\left(y + \frac{9u - x}{8}\right). \tag{11}$$

Let $z_0 \in \mathbb{R}^2$ be arbitrary and suppose

$$\begin{aligned} z_n &= T_{r_n}(z_{n-1}) + b_n, \qquad r_n \in [m, r_* + 10^{-118}], \\ \sum_{n \geq 1} |r_{n+1} - r_n| &< \infty, \qquad \sum_{n \geq 1} \|b_n\| < \infty. \end{aligned} \tag{12}$$

Then the actual whole trajectory converges to a pair in $\mathcal{Z}$ and

$$\sum_{n\geq 1} \|z_n - z_{n-1}\|^2 < \infty. \tag{13}$$

Neither total variation nor total error must be small. The conclusion includes $b_n = 0$, requires no finite initial objective value, and permits repeated crossings of r_*. Constants may depend on m, total variation and total error; none is asserted uniform as $m \downarrow 0$.

Corollary 1.1 (Two summable proximal-output errors)**.** *Under the same function and parameter assumptions, the conclusion of Theorem 4 also holds for the literal implementation*

$$\begin{aligned} x_n &= \operatorname{prox}_{r_n f}(x_{n-1} - r_n y_{n-1}) + e_n^x, \\ y_n &= \operatorname{prox}_h\left(y_{n-1} + \frac{9x_n - x_{n-1}}{8}\right) + e_n^y, \qquad \sum_{n\geq 1}(|e_n^x| + |e_n^y|) < \infty. \end{aligned} \tag{14}$$

In particular the second update uses the possibly perturbed current x_n.

The new theorem concerns the normalized scalar map with dual step and coupling equal to one, fixed objectives, and extrapolation 1/8. It does not permit independent arbitrary variation of two physical steps, changing objectives, or general inexact-solver criteria. Finite variation does not mean eventual constancy or monotonicity, and no assumption $\sum_n |r_n - r_\infty| < \infty$ is needed. The original fixed-step matched-Hilbert results, including arbitrarily small and zero couplings and the 10^{20} envelope, remain separate assertions with their original constants. No countably matched variable-parameter theorem follows from the compact scalar result.

1.3 Context and proof architecture

Chambolle and Pock [1] introduced the underlying method. For the same primal-first order, Banert, Upadhyaya, and Giselsson [2], Theorem 2, give weak convergence for $\theta > 1/2$ and $\tau\sigma\|L\|^2 < 4/(1 + 2\theta)$. The version-pinned Theorem 1.2 of Upadhyaya [3] gives weak convergence for $0 < \theta \leq 1/2$ under

$$\tau\sigma\|L\|^2 < \frac{4\theta(2-\theta)}{1 - 2\theta + 9\theta^2 - 4\theta^3}.$$

At $\theta = 1/8$ this is $120/113 < r_*$. These particular parameter ranges do not supply the endpoint conclusion here. The higher-dimensional threshold in report [4] and the scalar period-three onset concern different admissible classes; r_* is not claimed to be the optimal scalar or separable convergence threshold. This is a targeted comparison, not exhaustive novelty clearance.

We first establish scalar trapping and periodic obstructions. One shared exact coefficient table gives the no-anchor Hilbert increment theorem, the original coupling-uniform scalar envelope, and the endpoint kernel. Genuine coordinate proximals, actual forward

limiting orbits and summable tails prove the original countable theorem, including its full Hilbert KKT conclusion. We then continue the scalar mechanism: four-output order reduces endpoint kernel histories to axis ramps; explicit local descent controls their degenerate constant-axis directions; a compactness argument supplies uniform compensation; and exact primal-block continuation permits a boundary-complete telescoping estimate. This proves Theorem 3 independently of the restricted separable transfer in Corollary 7.3. Period-six exclusion is a separate restriction on possible obstructions, not an ingredient replacing any step of the convergence argument. We next make the compensation quantitative, construct a strict finite-shadow state energy, prove a compact subcritical counterpart, and join the two on a genuine overlap. The resulting positive energy controls finite parameter variation and summable state errors. All necessary finite data, short-period bridges and analytic passages are included here.

2 Explicit scalar trapping from bounded increments

In this section $X = Y = \mathbb{R}$ and L is a nonzero real number. The normalized statement separates the one-dimensional argument from the endpoint parameter choice.

Theorem 5 (Scalar bounded-increment trapping)**.** *Let $F, H : \mathbb{R} \to (-\infty, +\infty]$ be proper lower-semicontinuous convex functions such that*

$$0 \in \partial F(0), \qquad 0 \in \partial H(0).$$

Let $k > 0$ and $\theta \in \mathbb{R}$. From arbitrary $x_0, y_0 \in \mathbb{R}$, run

$$\begin{aligned} x_n &= \operatorname{prox}_F(x_{n-1} - k y_{n-1}), \\ y_n &= \operatorname{prox}_H\big(y_{n-1} + k(x_n + \theta(x_n - x_{n-1}))\big), \qquad n \geq 1. \end{aligned} \tag{15}$$

Suppose finite $A, B \geq 0$ satisfy

$$|x_n - x_{n-1}| \leq A, \qquad |y_n - y_{n-1}| \leq B, \qquad n \geq 1. \tag{16}$$

Define

$$\begin{aligned} C &= \frac{B}{k} + |\theta| A, \qquad & X &= A + \max\{|x_0|, C\}, \\ z_0 &= y_0 - \theta k x_0, \qquad & T &= \frac{A}{k} + B + |\theta| k X, \\ Z &= \max\{|z_0|, T\}, & Y &= Z + |\theta| k X. \end{aligned} \tag{17}$$

Then, for every $n \geq 0$,

$$|x_n| \leq X, \qquad |y_n - \theta k x_n| \leq Z, \qquad |y_n| \leq Y. \tag{18}$$

No finiteness of $F(x_0)$ or $H(y_0)$ is required.

2.1 Subgradient signs and output indices

If $p \in \partial F(x)$, the subgradient inequality at x and the minimizing property of zero give

$$F(0) \geq F(x) - px, \qquad F(x) \geq F(0), \qquad px \geq 0.$$

Similarly, $q \in \partial H(y)$ implies $qy \geq 0$. In particular,

$$\begin{aligned} x > 0 &\Longrightarrow p \geq 0, \quad x < 0 \Longrightarrow p \leq 0, \\ p > 0 &\Longrightarrow x \geq 0, \quad p < 0 \Longrightarrow x \leq 0. \end{aligned} \tag{19}$$

The last line deliberately has weak conclusions. A positive subgradient may occur at zero, for example at a domain boundary. This distinction allows flat pieces and multivalued subdifferentials.

Write $a_n = x_n - x_{n-1}$ and $b_n = y_n - y_{n-1}$. The actual proximal updates give

$$p_n = -a_n - ky_{n-1} \in \partial F(x_n), \qquad q_n = -b_n + kx_n + \theta k a_n \in \partial H(y_n), \quad n \geq 1. \tag{20}$$

Both points at index $n \geq 1$ are proximal outputs and belong to the relevant subdifferential domains. No subgradient at index zero is used.

2.2 The primal barrier

The increment hypothesis and the definition of C yield

$$k(x_n - C) \leq q_n \leq k(x_n + C), \qquad n \geq 1. \tag{21}$$

Suppose $x_n > C$ at an index $n \geq 1$. Then $q_n > 0$, so $y_n \geq 0$. If $x_{n+1} > 0$, then $p_{n+1} \geq 0$ and

$$x_{n+1} - x_n = -p_{n+1} - ky_n \leq 0.$$

If $x_{n+1} \leq 0$, it is already smaller than x_n. Thus $x_n > C$ implies $x_{n+1} \leq x_n$. On the other hand, if $x_n \leq C$, then $x_{n+1} \leq C + A$. Consequently,

$$x_{n+1} \leq \max\{x_n, C + A\}, \qquad n \geq 1. \tag{22}$$

For the lower bound, $x_n < -C$ gives $q_n < 0$ and $y_n \leq 0$. If $x_{n+1} < 0$, then $p_{n+1} \leq 0$ and $x_{n+1} - x_n \geq 0$. If $x_{n+1} \geq 0$, the same lower bound is immediate. When $x_n \geq -C$, the increment bound gives $x_{n+1} \geq -C - A$. Hence

$$x_{n+1} \geq \min\{x_n, -C - A\}, \qquad n \geq 1. \tag{23}$$

The constant X satisfies

$$|x_0| \leq X, \qquad |x_1| \leq |x_0| + A \leq X, \qquad C + A \leq X.$$

Induction in (22)–(23) proves

$$|x_n| \leq X \qquad (n \geq 0). \tag{24}$$

This handles the first output separately, because (21) is not available at index zero. It also covers overshoots across either threshold and crossings of zero.

2.3 The shifted dual barrier

Define $z_n = y_n - \theta k x_n$. The dual inclusion gives the exact identity

$$z_n - z_{n-1} = b_n - \theta k a_n = k x_n - q_n, \qquad n \geq 1. \tag{25}$$

Set $D = A/k + B$. If $y_n > D$ at an index $n \geq 1$, then

$$y_{n-1} \geq y_n - B > A/k, \qquad p_n = -a_n - k y_{n-1} \leq A - k y_{n-1} < 0.$$

Thus $x_n \leq 0$. Also $y_n > 0$ implies $q_n \geq 0$, and (25) yields $z_n \leq z_{n-1}$. Reversing the signs gives the companion implication. Explicitly, if $y_n < -D$, then $y_{n-1} < -A/k$, $p_n > 0$, $x_n \geq 0$, and $q_n \leq 0$, so $z_n \geq z_{n-1}$. We have proved

$$y_n > D \Longrightarrow z_n \leq z_{n-1}, \qquad y_n < -D \Longrightarrow z_n \geq z_{n-1}. \tag{26}$$

Since $T = D + |\theta| k X$ and (24) holds,

$$z_n > T \Longrightarrow y_n = z_n + \theta k x_n > D, \qquad z_n < -T \Longrightarrow y_n < -D.$$

Let $Z = \max\{|z_0|, T\}$. If n were the first index with $z_n > Z$, the first implication in (26) would give $z_n \leq z_{n-1} \leq Z$, a contradiction. The first index with $z_n < -Z$ is excluded in the same way. Therefore $|z_n| \leq Z$ for every $n \geq 0$, and

$$|y_n| = |z_n + \theta k x_n| \leq Z + |\theta| k X = Y.$$

Together with (24), this proves Theorem 5. Only magnitude estimates use $|\theta|$; the exact shift and identity retain the signed θ. Every real extrapolation parameter is therefore covered. ∎

2.4 Normalization at a scalar KKT pair

Theorem 6 (Bounded positions in the original scalar variables)**.** *Assume* (2)*. If the successive primal and dual increments of* (1) *are bounded, then both position sequences are bounded, for any $\tau, \sigma > 0$ and any fixed $\theta \in \mathbb{R}$. More precisely, put*

$$s = \operatorname{sign}(L), \qquad u_n = \frac{x_n - x_*}{\sqrt{\tau}}, \qquad v_n = \frac{s(y_n - y_*)}{\sqrt{\sigma}}, \qquad k = \sqrt{\tau\sigma}\,|L|. \tag{27}$$

Choose normalized increment bounds A, B and calculate X, Z, Y from (17) *using u_0, v_0 instead of x_0, y_0. Then, for every $n \geq 0$,*

$$|x_n - x_*| \leq \sqrt{\tau} X, \qquad |v_n - \theta k u_n| \leq Z, \qquad |y_n - y_*| \leq \sqrt{\sigma} Y.$$

Proof. Define

$$\begin{aligned} F(u) &= f(x_* + \sqrt{\tau} u) - f(x_*) + \sqrt{\tau} L y_* u, \\ H(v) &= h(y_* + s\sqrt{\sigma} v) - h(y_*) - s\sqrt{\sigma} L x_* v. \end{aligned} \tag{28}$$

The KKT conditions imply finite $f(x_*), h(y_*)$, and their subgradient inequalities give $F, H \geq 0$ and $F(0) = H(0) = 0$. Thus $0 \in \partial F(0)$ and $0 \in \partial H(0)$. Translation, nonzero affine coordinate changes, and addition of an affine function preserve properness, lower semicontinuity, and convexity. In particular, the sign reversal when $L < 0$ causes no difficulty.

The original proximal inclusions are

$$\frac{x_{n-1} - x_n}{\tau} - Ly_{n-1} \in \partial f(x_n),$$
$$\frac{y_{n-1} - y_n}{\sigma} + L(x_n + \theta(x_n - x_{n-1})) \in \partial h(y_n).$$

Multiply the first by $\sqrt{\tau}$ and add $\sqrt{\tau}Ly_*$. Multiply the second by $s\sqrt{\sigma}$ and subtract $s\sqrt{\sigma}Lx_*$. The affine subgradient chain rule gives

$$-(u_n - u_{n-1}) - kv_{n-1} \in \partial F(u_n),$$
$$-(v_n - v_{n-1}) + k(u_n + \theta(u_n - u_{n-1})) \in \partial H(v_n).$$

These are exactly the proximal inclusions for (15). For example, if A_0, B_0 bound the original increments, one may take $A = A_0/\sqrt{\tau}$ and $B = B_0/\sqrt{\sigma}$. Theorem 5 and (27) prove all three estimates. ∎

The scalar theorem uses no smoothness, strong convexity, bounded domain, finite piecewise description, or uniqueness of the KKT pair. It does not say that increments are bounded at every parameter choice. Section 5.3 supplies that remaining hypothesis at the exact endpoint.

3 Scalar order and periodic obstructions

To classify scalar three-cycles, use $L = 1$, $\sigma = 1$, $\tau = r > 0$ and write the normalized map as

$$x_{n+1} = \operatorname{prox}_{rf}(x_n - ry_n), \qquad y_{n+1} = \operatorname{prox}_h\big(y_n + (1+\theta)x_{n+1} - \theta x_n\big). \tag{29}$$

Here f, h are any fixed proper lower-semicontinuous convex scalar functions. We use

$$z = \operatorname{prox}_{\lambda f}(w) \quad \Longleftrightarrow \quad (w - z)/\lambda \in \partial f(z). \tag{30}$$

A KKT pair for this normalization obeys $-y_* \in \partial f(x_*)$ and $x_* \in \partial h(y_*)$. A three-periodic orbit satisfies $(x_{n+3}, y_{n+3}) = (x_n, y_n)$; it is genuine when its least period is three. Since three is prime, every nonconstant three-periodic orbit is genuine.

Theorem 7 (Exact scalar period three onset)**.** *For every fixed $\theta \geq 0$, the normalized iteration* (29) *has the following properties.*

1. *If $0 < r < 3$, every three-periodic actual orbit is constant, for every pair of fixed proper lower-semicontinuous convex functions f, h. No KKT existence assumption is needed.*

2. *For every $r \geq 3$, the functions*

$$f_{r,\theta}(x) = \max\left\{\left(\frac{1}{r} - \frac{1}{3} - \theta\right)x, \left(\frac{1}{3} - \frac{1}{r}\right)x\right\}, \qquad h(y) = \frac{y}{3} \tag{31}$$

possess the KKT pair

$$(x_*, y_*) = \left(\frac{1}{3}, \frac{1}{r} - \frac{1}{3}\right) \tag{32}$$

and realize the genuine three-cycle

$$(0,0) \longmapsto \left(0, -\frac{1}{3}\right) \longmapsto \left(1, \frac{1}{3} + \theta\right) \longmapsto (0,0). \tag{33}$$

Consequently the set of r admitting such fixed scalar objectives, a KKT pair, and a genuine period-three actual orbit is exactly $[3, +\infty)$, independently of $\theta \geq 0$. The boundary case $\theta = 0$, $r = 3$ is included; there $f_{3,0} \equiv 0$ and the three states are still distinct.

3.1 Actual residuals and scalar order

All indices in the proof are read modulo three. Suppose $(x_i, y_i)_{i=0}^2$ is a three-periodic actual orbit. Define

$$p_i = \frac{x_{i-1} - x_i}{r} - y_{i-1}, \qquad q_i = y_{i-1} - y_i + (1+\theta)x_i - \theta x_{i-1}. \tag{34}$$

By (30),

$$p_i \in \partial f(x_i), \qquad q_i \in \partial h(y_i). \tag{35}$$

For any convex function g on $\mathbb{R}$, the subgradient inequalities at two points imply

$$a \in \partial g(s),\ b \in \partial g(t) \quad \Longrightarrow \quad (s-t)(a-b) \geq 0.$$

Applying this fact to the same two fixed functions in (35) gives

$$(x_i - x_j)(p_i - p_j) \geq 0, \qquad (y_i - y_j)(q_i - q_j) \geq 0 \qquad (i, j \in \{0,1,2\}). \tag{36}$$

Only this necessary condition is used to exclude cycles. When two coordinates coincide, the corresponding product is zero and imposes no ordering on their selected subgradients. This observation is important in both tie cases below.

3.1.1 Constant primal triples

If $x_0 = x_1 = x_2 = c$, then $q_i = y_{i-1} - y_i + c$. The three dual inequalities in (36) sum to

$$0 \le \sum_{i<j} (y_i - y_j)(q_i - q_j) = -\frac{3}{2} \sum_{i=0}^{2} (y_i - y_{i-1})^2. \tag{37}$$

For completeness, setting $\bar{y} = (y_0 + y_1 + y_2)/3$ yields

$$\sum_{i<j} (y_i - y_j)(q_i - q_j) = 3 \sum_{i} (y_i - \bar{y}) q_i = 3 \left(\sum_i y_i y_{i-1} - \sum_i y_i^2 \right),$$

which is the last expression in (37). Therefore all three dual values also coincide. Every nonconstant three-periodic state orbit has a nonconstant primal triple.

3.1.2 Normalization of the necessary inequalities

The conditions (36), with residuals (34), are invariant under cyclic rotation, translation of either coordinate triple, and a common nonzero rescaling of the two coordinate triples. To see the last two assertions explicitly, let

$$x_i' = \frac{x_i - a}{d}, \qquad y_i' = \frac{y_i - b}{d}, \qquad d \neq 0.$$

The residuals formed from these new triples are

$$p_i' = \frac{p_i + b}{d}, \qquad q_i' = \frac{q_i - a}{d}.$$

Thus every product in (36) is divided by $d^2 > 0$. These transformations normalize necessary inequalities; it is unnecessary to assert that the transformed triples remain orbits of the original functions.

If the three primal values are distinct, cyclic rotation and a simultaneous sign reversal, when needed, arrange their increasing order along the cycle. Translate and scale to obtain

$$(x_0, x_1, x_2) = (0, t, 1), \quad 0 < t < 1, \qquad (ry_0, ry_1, ry_2) = (0, u, v). \tag{38}$$

Indeed cyclic rotation supplies three orders and reversal of the value order supplies the other three. If exactly two primal values coincide, make the repeated value the smaller one by a simultaneous sign reversal if necessary, and rotate so that the repeated entries occupy indices zero and one. The resulting normalization is

$$(x_0, x_1, x_2) = (0, 0, 1), \qquad (ry_0, ry_1, ry_2) = (0, u, v). \tag{39}$$

These cases exhaust nonconstant primal triples. The second case is $t = 0$ in the common residual formulas

$$\begin{aligned} A_0 &= 1 - v, \quad & A_1 &= -t, & A_2 &= t - 1 - u, \\ B_0 &= v - r\theta, \quad & B_1 &= -u + r(1+\theta)t, & B_2 &= u - v + r(1 + \theta - \theta t), \end{aligned} \tag{40}$$

where $A_i = rp_i$ and $B_i = rq_i$ after normalization. Because $r > 0$, the required order of the B_i is the order of the coordinates ry_i whenever those coordinates are distinct.

3.2 Three distinct primal values

Assume (38). Primal monotonicity forces $A_0 \le A_1 \le A_2$, so (40) gives

$$v \ge 1 + t, \qquad u \le 2t - 1. \tag{41}$$

In particular $v > 0$ and $v - u \ge 2 - t > 0$. Hence the only possible dual tie is $u = 0$; the following split also includes that case.

3.2.1 Nonnegative middle dual value

Suppose $u \ge 0$. Then (41) implies $t \ge 1/2$. Since $v > u$, dual monotonicity requires $B_2 \ge B_1$ even if $u = 0$. Consequently

$$r\big[(1+\theta) - (1+2\theta)t\big] \ge v - 2u \ge 3(1-t) > 0. \tag{42}$$

The second inequality follows from

$$v - 2u - 3(1-t) = (v - 1 - t) + 2(2t - 1 - u) \ge 0.$$

Set $D = (1+\theta) - (1+2\theta)t$. Then $D > 0$, and division in (42) gives

$$r \ge \frac{3(1-t)}{D} = 3 + \frac{3\theta(2t-1)}{D} \ge 3. \tag{43}$$

The last inequality uses $\theta \ge 0$ and $t \ge 1/2$. If $u = 0$, no relation between B_0 and B_1 has been imposed. Only their separate relation to the strictly larger dual coordinate could be used, and the displayed argument needs just $B_2 \ge B_1$.

3.2.2 Negative middle dual value

Suppose $u < 0$. Now $u < 0 < v$, so dual monotonicity gives $B_1 \le B_0 \le B_2$. Formula (40) yields

$$u + v \ge r\big[\theta + (1+\theta)t\big], \qquad 2v - u \le r\big[(1+2\theta) - \theta t\big]. \tag{44}$$

Taking twice the first lower bound minus the second upper bound gives

$$3u \ge r\big[(2+3\theta)t - 1\big]. \tag{45}$$

Because $u < 0$ and $r > 0$, this forces $t < 1/(2+3\theta) \le 1/2$. Combining (45) with $u \le 2t-1$ gives

$$3(2t-1) \ge r\big[(2+3\theta)t - 1\big].$$

Here $(2+3\theta)t - 1 < 0$ and $2t - 1 < 0$. Reversing the inequality when dividing by the first negative quantity gives

$$r \ge \frac{3(1-2t)}{1-(2+3\theta)t} = 3 + \frac{9\theta t}{1-(2+3\theta)t} \ge 3. \tag{46}$$

For $\theta > 0$ the last inequality is strict, while for $\theta = 0$ it becomes equality of the lower-bound expressions. Both subcases imply $r \ge 3$ for a nonconstant orbit with three distinct primal values.

3.3 Two equal primal values

Assume (39), hence $t = 0$ in (40). Primal monotonicity compares $x_2 = 1$ with each occurrence of zero, giving $A_2 \geq A_0$ and $A_2 \geq A_1$. It does not order A_0 and A_1. The two valid inequalities are

$$u \leq -1, \qquad v - u \geq 2. \tag{47}$$

In particular $u < 0$. If $v \leq 0$, the strict comparison $ry_1 = u < ry_0 = 0$ would require $B_1 \leq B_0$. But

$$B_1 = -u > 0, \qquad B_0 = v - r\theta \leq 0,$$

which is impossible. The weak inequality for B_0 uses only $\theta \geq 0$. This argument includes the potential dual tie $v = 0$. Therefore $v > 0$, and all dual values are strictly ordered as $u < 0 < v$.

The inequalities $B_1 \leq B_0 \leq B_2$, with $t = 0$ in (40), become

$$u + v \geq r\theta, \qquad 2v - u \leq r(1 + 2\theta).$$

Eliminating v also cancels θ, giving $3u \geq -r$. Combining with (47),

$$r \geq -3u \geq 3. \tag{48}$$

Together with the constant-primal argument and Section 3.2, this proves the exclusion part of Theorem 7. No KKT pair, finite-valued objective, differentiability, or strict convexity was assumed. In particular the proof applies to extended-real objectives and arbitrary three-periodic initial states.

3.4 An explicit fixed-function realization

Fix $\theta \geq 0$ and $r \geq 3$, and abbreviate

$$\alpha = \frac{1}{r} - \frac{1}{3} - \theta, \qquad \beta = \frac{1}{3} - \frac{1}{r}. \tag{49}$$

Then

$$\beta \geq 0, \qquad \alpha = -\beta - \theta \leq 0, \qquad \beta - \alpha = 2\beta + \theta \geq 0. \tag{50}$$

The functions in (31) are finite, continuous, proper, and convex. Their subgradients are

$$\partial f_{r,\theta}(x) = \begin{cases} \{\alpha\}, & x < 0, \\ [\alpha, \beta], & x = 0, \\ \{\beta\}, & x > 0, \end{cases} \qquad \partial h(y) = \{1/3\}. \tag{51}$$

The actual proximal maps are therefore

$$\operatorname{prox}_{rf_{r,\theta}}(w) = \begin{cases} w - r\alpha, & w < r\alpha, \\ 0, & r\alpha \leq w \leq r\beta, \\ w - r\beta, & w > r\beta, \end{cases} \qquad \operatorname{prox}_h(w) = w - \frac{1}{3}. \tag{52}$$

The formulas agree at both breakpoints, including the degenerate case $\alpha = \beta = 0$ when $\theta = 0$ and $r = 3$. In that case the primal objective is identically zero and its proximal map is the identity.

We can check (33) directly. From $(0, 0)$ the primal input is zero, which lies in $[r\alpha, r\beta]$, so the next state is $(0, -1/3)$. At that state the primal input is $r/3$ and

$$\frac{r}{3} - r\beta = 1 > 0.$$

Thus the next primal point is one, and the next dual point is

$$-\frac{1}{3} + (1 + \theta) - \frac{1}{3} = \frac{1}{3} + \theta.$$

At $(1, 1/3 + \theta)$ the primal input equals

$$1 - r\left(\frac{1}{3} + \theta\right) = r\alpha,$$

so the next primal point is zero. The next dual point is

$$\left(\frac{1}{3} + \theta\right) - \theta - \frac{1}{3} = 0.$$

This proves all three updates with the same fixed proximal maps.

Equivalently, substitution into (34) gives the exact cyclic residuals

$$(p_0, p_1, p_2) = (\alpha, 0, \beta), \qquad (q_0, q_1, q_2) = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right). \tag{53}$$

They lie in the corresponding subgradients from (51). Strong convexity of each proximal minimization ensures uniqueness, so there is no choice of minimizer hidden in the construction. The three state pairs are distinct.

Finally $x_* = 1/3 > 0$ and $y_* = -\beta$ imply

$$-y_* = \beta \in \partial f_{r,\theta}(x_*), \qquad x_* = \frac{1}{3} \in \partial h(y_*).$$

Hence (32) is a KKT pair. At $r = 3$, the function is $f_{3,\theta}(x) = \max\{-\theta x, 0\}$. At $\theta = 1/8$ this specializes to

$$f_{3,1/8}(x) = \max\{0, -x/8\}, \qquad h(y) = y/3, \qquad (x_*, y_*) = (1/3, 0).$$

The minimum admissible product is attained. This completes the proof of Theorem 7.

3.5 Exclusion of every period dividing six

We now fix $\theta = 1/8$ in (29), and denote this literal paired-state map by T_r. The following restriction is independent of the convergence argument and does not require an anchor.

Theorem 8 (No nonconstant six-periodic histories)**.** *Let $f, h : \mathbb{R} \to \mathbb{R} \cup \{+\infty\}$ be fixed proper lower-semicontinuous convex functions. If*

$$\frac{297}{100} \le r < 3,$$

then every state satisfying $T_r^6 z = z$ is a fixed point of T_r. Equivalently, every six-periodic actual paired orbit is constant. There is no KKT-existence assumption, no distinct-coordinate assumption, and no assumption that the least period is six.

3.5.1 Cyclic residuals and closed scalar order cones

All indices in this subsection are modulo six. Every state of a periodic orbit, including index zero, is a proximal output. Thus the same fixed functions supply the inclusions

$$\begin{aligned} p_i &= \frac{x_{i-1} - x_i}{r} - y_{i-1} \in \partial f(x_i), \\ q_i &= y_{i-1} - y_i + \frac{9x_i - x_{i-1}}{8} \in \partial h(y_i). \end{aligned} \tag{54}$$

Set, only for this cyclic argument, $w_i = ry_i$, $P_i = rp_i$, and $Q_i = 8rq_i$. These are the exact residual normalizations

$$P_i = x_{i-1} - x_i - w_{i-1}, \qquad Q_i = 8(w_{i-1} - w_i) + r(9x_i - x_{i-1}). \tag{55}$$

Since $r > 0$, scalar monotonicity implies

$$(x_i - x_j)(P_i - P_j) \ge 0, \qquad (w_i - w_j)(Q_i - Q_j) \ge 0 \quad (0 \le i, j < 6). \tag{56}$$

We use only these necessary inequalities. In particular, we never replace the two fixed proximal maps by independently selectable linear gains.

Lemma 1 (Complete weak-order coverage)**.** *Up to cyclic relabeling and common sign reversal, every history satisfying* (56) *belongs to one of* 46 080 *specified closed linear cones. These are indexed by* 64 *primal orders and all* 720 *dual orders, and include every tied, flat, and constant-coordinate case.*

Proof. For any finite collection (a_i, b_i) of monotone scalar graph pairs, sort the a_i increasingly and, within each block of equal a_i, sort the b_i increasingly. Between distinct blocks, monotonicity forces every ordinate in the earlier block to be no larger than every ordinate in the later block. Hence some permutation orders both lists weakly increasingly. Conversely, such a common weak order makes all pairwise products $(a_i - a_j)(b_i - b_j)$ nonnegative. This argument uses no division by a coordinate difference and imposes no false ordering at a vertical graph piece: the order within its tied block is selected from the actual ordinates.

Choose a common order π of x, P and a common order ρ of w, Q. A cyclic relabeling makes $\pi_0 = 0$, leaving $5! = 120$ possible primal orders. The dual order can still be any

permutation of $\{0,\dots,5\}$. Independent translations $x_i \mapsto x_i - c$, $w_i \mapsto w_i - d$ replace P_i by $P_i + d$ and Q_i by $Q_i - 8rc$, so they preserve every order difference. A simultaneous sign reversal negates all four lists. Reversing their graph orders and rotating the old last primal index to zero therefore sends the normalized primal order to

$$R(\pi)_k = (\pi_{5-k} - \pi_5) \pmod 6. \tag{57}$$

This reverses value order, not the direction of time in the recurrence. The corresponding dual order is $\rho'_k = (\rho_{5-k} - \pi_5) \pmod 6$, which remains among the 720 allowed dual orders.

The map R is an involution. A fixed order has $\pi_5 = 3$, since $\pi_5 = -\pi_5 \pmod 6$ and $\pi_5 \neq 0$. Its remaining entries pair the two sets $\{1,4\}$ and $\{2,5\}$ across positions $(1,4)$ and $(2,3)$. There are two assignments of these sets and two orientations for each, hence exactly $2 \cdot 2^2 = 8$ fixed orders. Taking the lexicographically smaller member of each R-orbit leaves $(120+8)/2 = 64$ representatives. Together with all 720 dual orders they give 46 080 cones. Before normalization there are $(6!)^2 = 518\,400$ order pairs. The $8 \cdot 720 = 5760$ labels with fixed primal order have six raw preimages each; the other $56 \cdot 720 = 40\,320$ labels have twelve each. Thus

$$5760 \cdot 6 + 40\,320 \cdot 12 = 518\,400.$$

This also explains the exhaustive coverage test in the independent checker. All inequalities are weak, so their boundaries are retained throughout. ∎

For a retained pair (π, ρ), work in all twelve original coordinates

$$\mathbf{z} = (x_0, \dots, x_5, w_0, \dots, w_5)^{\mathsf{T}}.$$

Let $M_{\pi,\rho}(r)$ be the 20×12 matrix whose rows, in order, represent the five consecutive differences of x along π, those of P along π, those of w along ρ, and those of Q along ρ. For example the first group is $x_{\pi_{k+1}} - x_{\pi_k}$, $0 \le k < 5$. Equations (55) specify every coefficient, and show that $M(r) = M_0 + rM_1$. Write

$$d_\pi = e^x_{\pi_5} - e^x_{\pi_0}, \qquad d_\pi^{\mathsf{T}}\mathbf{z} = x_{\pi_5} - x_{\pi_0},$$

where e^x_i is the coordinate vector selecting x_i. The cone is $M(r)\mathbf{z} \ge 0$, and $d_\pi^{\mathsf{T}}\mathbf{z} \ge 0$ is its primal range. No translation or division by that range is needed in this twelve-coordinate representation.

3.5.2 An exact interval certificate and its equality case

The finite rational certificate supplied with the source gives, for each of the 46 080 cones, two nonnegative vectors $u, v \in \mathbb{Q}^{20}$ with the polynomial identity

$$M(r)^{\mathsf{T}}\big[(300 - 100r)u + (100r - 297)v\big] = -(300 - 100r)d_\pi. \tag{58}$$

Appendix C specifies every row, label, rational weight and coefficient test needed to replay this finite assertion; no solver output is a premise. The important parameter dependence

is visible in (58): both scalar coefficients are nonnegative on $[297/100, 3]$, and $300 - 100r$ is strictly positive on $[297/100, 3)$. Thus this is an interval certificate, rather than checks at sampled products.

Proof of Theorem 8. The actual residuals satisfy (56). By Lemma 1, apply the permitted rotation and common sign change to place them in a retained cone. Its nonnegative multiplier in (58) gives

$$0 \le \big[(300 - 100r)u + (100r - 297)v\big]^{\mathsf{T}} M(r)\mathbf{z} = -(300 - 100r)(x_{\pi_5} - x_{\pi_0}) \le 0.$$

Since $r < 3$, the primal range is zero. All ordered primal coordinates therefore coincide. Undoing the relabeling and sign change preserves this conclusion.

It remains to rule out a nonconstant dual sequence over a constant primal sequence. If $x_i = c$, then $Q_i = 8(w_{i-1} - w_i) + 8rc$. With $\bar{w} = \frac{1}{6}\sum_i w_i$, sum the dual inequalities in (56) over unordered pairs:

$$\begin{aligned} 0 &\le \sum_{i<j} (w_i - w_j)(Q_i - Q_j) = 6 \sum_i (w_i - \bar{w}) Q_i \\ &= 48 \sum_i w_i (w_{i-1} - w_i) = -24 \sum_i (w_i - w_{i-1})^2. \end{aligned} \tag{59}$$

Hence every adjacent dual difference vanishes. Since $r > 0$, $y_i = w_i/r$ is constant as well. In particular, any nonconstant paired cyclic history would necessarily have had a nonconstant primal sequence. This equality argument covers a constant species, repeated graph points, flat residuals and all other degeneracies without any normalization by a nonzero range. Finally, a constant actual orbit satisfies $-y \in \partial f(x)$ and $x \in \partial h(y)$ by (54); KKT existence is a consequence of that orbit, not an assumption of the theorem. ∎

Remark (Gap containment and the open endpoint). The exact positive margins

$$91 \cdot 1600^2 - 15257^2 = 183951, \qquad 155^2 - 91 \cdot 16^2 = 729$$

prove $297/100 < r_* < 3$, so the theorem includes the full interval $(r_*, 3)$. At $r = 3$, the already verified functions (31) with $\theta = 1/8$, their actual proximal maps (52), and the orbit (33) give

$$(0, 0) \longmapsto (0, -1/3) \longmapsto (1, 11/24) \longmapsto (0, 0).$$

Repeated twice, this nonconstant orbit satisfies $T_3^6 z = z$. Consequently the open right endpoint in Theorem 8 is sharp. The witness has least period three; it makes no assertion about the onset of least period six. The certificate loses its positive range coefficient precisely at this endpoint.

3.6 The remaining short-period exclusions

The six-period theorem already treats periods two and three. To draw a conclusion about the least period of an arbitrary periodic orbit, periods four and five must also

be excluded. We give the necessary arguments here, so that the resulting corollary has no external mathematical dependency. These arguments use cyclic subgradient inequalities, which retain more information than pairwise monotonicity in a general temporal decomposition.

For an m-periodic orbit of (29), put $X_i = x_i/\sqrt{r}$, $Y_i = y_i$, and $k = \sqrt{r}$. With $\widehat{f}(X) = f(\sqrt{r}X)$ the actual scaled residuals are

$$\begin{aligned} \alpha_i &= X_{i-1} - X_i - kY_{i-1} \in \partial\widehat{f}(X_i), \\ \beta_i &= Y_{i-1} - Y_i + k\big((1+\theta)X_i - \theta X_{i-1}\big) \in \partial h(Y_i). \end{aligned} \tag{60}$$

Every cyclic state is an output, so all these function values are finite. For any displacement j, the subgradient inequalities with targets X_{i+j} and Y_{i+j} telescope on summation to

$$\frac{1}{m}\sum_i \alpha_i(X_{i+j} - X_i) \le 0, \qquad \frac{1}{m}\sum_i \beta_i(Y_{i+j} - Y_i) \le 0. \tag{61}$$

Subtracting the means of X, Y does not change these sums: it changes each residual by a constant, and each sum of target differences is zero. On the centered temporal space use $\langle U, V\rangle_m = m^{-1}\sum_i U_i V_i$ and its norm. Let F be the forward shift and $S = F^{-1}$ the backward shift. The residual sequences in (61) are therefore $(S-I)X - kSY$ and $(S-I)Y + k((1+\theta)I - \theta S)X$.

Lemma 2 (The four-period restriction)**.** *For $\theta = 1/8$, an actual scalar orbit of least period four requires $r > 16/5$. No KKT assumption is required.*

Proof. More generally let $\theta > 0$. For $m = 4$, the centered temporal space is the orthogonal sum of the alternating subspace, where $S = -I$, and the quarter-frequency subspace, where $S^2 = -I$ and $S^* = -S$. Write x, y for the latter components of X, Y, and u, v for their alternating components. In this proof set

$$\begin{aligned} U &= \|x\|_4^2, \quad V = \|y\|_4^2, \quad A = \langle y, kx\rangle_4, \quad B = \langle Sy, kx\rangle_4, \\ U_2 &= \|u\|_4^2, \quad V_2 = \|v\|_4^2, \quad A_2 = \langle v, ku\rangle_4, \quad \alpha = 1 + 2\theta. \end{aligned}$$

On the quarter-frequency subspace $F^2 - I = -2I$, while it vanishes on the alternating subspace. Substituting $j = 2$ in (61) and dividing by two consequently gives

$$U + B \le 0, \qquad V - (1+\theta)A - \theta B \le 0. \tag{62}$$

Thus $B \le -U$ and $(1+\theta)A \ge V + \theta U$. If $U = 0$, then $A = B = 0$ and $V = 0$; if $V = 0$, then $B = 0$ and $U = 0$. A least-four paired orbit has a nonzero quarter-frequency component, since otherwise both coordinates have period dividing two. It follows that $U, V > 0$ and $A > 0$.

For $j = 1$, $F = -S$ on the quarter-frequency subspace. Its own-variable terms vanish by skew-adjointness of S; the two cross terms are $A + B$ and $-A - \alpha B$. On the alternating subspace the residuals are $-2u + kv$ and $-2v + \alpha ku$, and the target differences are

$-2u, -2v$. Thus (61) gives exactly

$$A + B + 4U_2 - 2A_2 \le 0, \qquad -A - \alpha B + 4V_2 - 2\alpha A_2 \le 0. \tag{63}$$

Multiply the first inequality by α and add the second:

$$(\alpha - 1)A + 4\alpha U_2 + 4V_2 - 4\alpha A_2 \le 0.$$

If $r \le 4/\alpha$, Cauchy–Schwarz gives $\alpha A_2 \le \alpha\sqrt{rU_2V_2} \le 2\sqrt{\alpha U_2 V_2}$. Consequently the left side is at least

$$2\theta A + 4\big(\sqrt{\alpha U_2} - \sqrt{V_2}\big)^2 > 0,$$

a contradiction. This estimate never divides by U_2 or V_2, and includes their zero cases and equality $r = 4/\alpha$. For $\theta = 1/8$ it proves $r > 4/(1 + 2\theta) = 16/5$. ∎

Lemma 3 (The five-period exclusion)**.** *For $\theta = 1/8$ and $0 < r \le 31/10$, every five-periodic actual scalar paired orbit is constant, without a KKT-existence assumption.*

Proof. Take $m = 5$ in (61). If the primal sequence is constant, the dual residual has the form $\beta_i = Y_{i-1} - Y_i + c$. Using displacement $j = 4$, for which $F^4 = S$, gives

$$0 \ge \frac{1}{5}\sum_i \beta_i(Y_{i-1} - Y_i) = \frac{1}{5}\sum_i (Y_{i-1} - Y_i)^2.$$

The dual sequence is then constant. We may therefore suppose that the centered primal sequence is nonzero.

The centered five-term real temporal space is the orthogonal sum of two modes indexed by $k = 1, 2$. To avoid confusing this mode index with the scalar coupling, write $\kappa = \sqrt{r}$ for the latter in the rest of this proof. Put $\phi_k = 2\pi k/5$. On mode k,

$$F = \cos\phi_k I + \sin\phi_k J_k, \qquad J_k^* = -J_k, \qquad J_k^2 = -I.$$

For the orthogonal components X_k, Y_k define

$$A_k = r\|X_k\|_5^2, \quad B_k = \|Y_k\|_5^2, \quad c_k = \langle \kappa X_k, Y_k\rangle_5, \quad d_k = \langle \kappa J_k X_k, Y_k\rangle_5. \tag{64}$$

The vectors κX_k and $\kappa J_k X_k$ are orthogonal and have equal norm. Bessel's inequality, with its zero-vector case included, therefore gives

$$A_k, B_k \ge 0, \qquad c_k^2 + d_k^2 \le A_k B_k. \tag{65}$$

Since $X \ne 0$, $A_1 + A_2 > 0$. All forthcoming inequalities are homogeneous in these moments, so divide the moments by this positive sum to arrange $A_1 + A_2 = 1$.

For $j = 1, 2, 3, 4$, define the following explicit trigonometric coefficients:

$$\begin{aligned} D_{kj} &= \cos((j+1)\phi_k) - \cos\phi_k - \cos(j\phi_k) + 1, \\ Z_{kj} &= e^{\mathrm{i}\phi_k}(e^{\mathrm{i}j\phi_k} - 1), \\ W_{kj} &= \big((1+\theta) - \theta e^{\mathrm{i}\phi_k}\big)(e^{\mathrm{i}j\phi_k} - 1). \end{aligned} \tag{66}$$

Expanding each shift product in I, J_k in (61) yields all eight necessary inequalities

$$\begin{aligned}\mathcal{P}_j &= \sum_{k=1}^{2}\left(\frac{D_{kj}}{r}A_k - \operatorname{Re} Z_{kj}\, c_k - \operatorname{Im} Z_{kj}\, d_k\right) \le 0,\\ \mathcal{H}_j &= \sum_{k=1}^{2}\left(D_{kj}B_k + \operatorname{Re} W_{kj}\, c_k - \operatorname{Im} W_{kj}\, d_k\right) \le 0.\end{aligned} \tag{67}$$

Here the own-variable coefficient is the real part of $(e^{\mathrm{i}\phi_k}-1)(e^{\mathrm{i}j\phi_k}-1)$, which is D_{kj}. The primal cross term is $-\langle Y_k, \kappa F(F^j - I)X_k\rangle_5$, giving the two minus signs involving Z_{kj}. Moving the forward target shift from the dual variable to the primal variable takes its adjoint and gives the displayed real-part plus sign and imaginary-part minus sign for W_{kj}. These expansions justify the signs directly; explicit real-matrix identities are also given in Appendix C.3.

Set $\theta = 1/8$, $r_{\mathrm{c}} = 31/10$, and choose the nonnegative weights

$$(\lambda_1,\dots,\lambda_4) = (407, 0, 133, 0), \qquad (\mu_1,\dots,\mu_4) = (332, 2, 126, 0). \tag{68}$$

Their weighted sum in (67) equals $\sum_k (a_k A_k + b_k B_k + g_k c_k + h_k d_k)$, where

$$\begin{aligned} a_k(r) &= \frac{1}{r}\sum_j \lambda_j D_{kj}, & b_k &= \sum_j \mu_j D_{kj},\\ g_k &= \sum_j(-\lambda_j \operatorname{Re} Z_{kj} + \mu_j \operatorname{Re} W_{kj}), & h_k &= \sum_j(-\lambda_j \operatorname{Im} Z_{kj} - \mu_j \operatorname{Im} W_{kj}).\end{aligned} \tag{69}$$

Writing $\varepsilon_1 = -1$, $\varepsilon_2 = 1$, direct exact evaluation at $r = r_{\mathrm{c}}$ gives

$$\begin{aligned} a_k(r_{\mathrm{c}}) &= \frac{6750 + 2720\varepsilon_k\sqrt{5}}{31} > 0,\\ b_k &= 575 + 218\varepsilon_k\sqrt{5} > 0,\\ 4a_k(r_{\mathrm{c}})b_k - g_k^2 - h_k^2 &= \frac{3980285 + 1556275\varepsilon_k\sqrt{5}}{992} > 0.\end{aligned} \tag{70}$$

For the minus signs, positivity follows by comparing the squares of the positive rational terms; the exact positive differences are recorded in Appendix C.3. The plus signs are immediate. For $0 < r \le r_{\mathrm{c}}$, only a_k changes, and $a_k(r) = (r_{\mathrm{c}}/r)a_k(r_{\mathrm{c}}) \ge a_k(r_{\mathrm{c}})$. All three strict inequalities in (70) thus hold on the whole asserted interval.

By (65), each mode contributes at least

$$a_k A_k + b_k B_k - \sqrt{g_k^2 + h_k^2}\sqrt{A_k B_k}.$$

If both A_k, B_k are positive, this is strictly positive because $\sqrt{g_k^2 + h_k^2} < 2\sqrt{a_k b_k}$ and $a_k A_k + b_k B_k \ge 2\sqrt{a_k b_k A_k B_k}$. If precisely one is positive, strict positivity follows from $a_k, b_k > 0$; if both vanish, the mode contributes zero. Since $A_1 + A_2 = 1$, at least one contribution is strictly positive. Their sum contradicts the nonpositive weighted sum of (67). This proves the lemma, including all zero-mode and equality cases. ∎

Corollary 3.1 (Minimum possible period)**.** *For $\theta = 1/8$ and $297/100 \le r < 3$, any nonconstant periodic actual scalar paired orbit has least period at least seven. In particular this holds throughout $(r_*, 3)$, without assuming a KKT pair exists.*

Proof. Theorem 8 rules out nonconstant periods two, three and six. Lemma 2 excludes least period four, since $r < 3 < 16/5$, and Lemma 3 excludes period five, since $r < 3 < 31/10$. A least-one orbit is constant. ∎

This is a finite-period restriction. It does not establish boundedness, asymptotic regularity or convergence, and it does not rule out periods at least seven or bounded aperiodic behavior. The convergence statements elsewhere in this manuscript require their own hypotheses and proofs.

4 Anchoring and exact coordinate proximals

Each finite nonnegative sum in (4) is lower-semicontinuous and convex on $\mathcal{H}$. Its increasing supremum has the same properties. Addition of a finite continuous affine functional preserves them, and the assumed finite point proves properness.

At a KKT pair, both objective values and all component values at the anchor are finite. Varying only coordinate i in the subgradient inequalities gives

$$\alpha_i^* = -\ell_i y_{*,i} - a_i \in \partial\phi_i(x_{*,i}), \qquad \gamma_i^* = \ell_i x_{*,i} - b_i \in \partial\psi_i(y_{*,i}).$$

The two slope sequences belong to $\mathcal{H}$. Define

$$\begin{aligned} F_i(t) &= \phi_i(x_{*,i} + t) - \phi_i(x_{*,i}) - \alpha_i^* t, \\ H_i(t) &= \psi_i(y_{*,i} + t) - \psi_i(y_{*,i}) - \gamma_i^* t. \end{aligned} \tag{71}$$

These are nonnegative proper lower-semicontinuous convex functions vanishing at zero. The base component values have finite sums, and all pairings of the slope sequences with $x - x_*$ or $y - y_*$ converge absolutely by Cauchy–Schwarz. Taking limits in the finite-sum identities proves (7), including at points with infinite objective value. This uses no infinity-minus-infinity expression.

Lemma 4 (An infinite-product proximal identity)**.** *Let $Q_i : \mathbb{R} \to [0, +\infty]$ be proper lower-semicontinuous convex functions with $Q_i(0) = 0$, and put $Q(z) = \sum_i Q_i(z_i)$ on $\mathcal{H}$. For every $w \in \mathcal{H}$ and $\lambda > 0$,*

$$\big(\operatorname{prox}_{\lambda Q}(w)\big)_i = \operatorname{prox}_{\lambda Q_i}(w_i).$$

Proof. Set $p_i = \operatorname{prox}_{\lambda Q_i}(w_i)$. Since zero minimizes Q_i, its proximal map fixes zero and is nonexpansive. Hence $|p_i| \le |w_i|$ and $p \in \mathcal{H}$. Comparison with the scalar competitor zero yields

$$Q_i(p_i) + \frac{|p_i - w_i|^2}{2\lambda} \le \frac{|w_i|^2}{2\lambda}.$$

Summation proves $Q(p) < \infty$. For any $z \in \mathcal{H}$, sum the scalar minimizing inequalities, whose objective terms are nonnegative, to obtain

$$Q(p) + \frac{\|p - w\|^2}{2\lambda} \leq Q(z) + \frac{\|z - w\|^2}{2\lambda}.$$

The squared norm is strictly convex, so the minimizer is unique. This proves the identity. ∎

Completing the square for the continuous affine terms in (7) and translating by the anchor now proves exact coordinate separation for the original f, h. Thus the original proximal maps are not being replaced by a merely formal family of scalar recursions.

For $\ell_i \neq 0$ put $s_i = \operatorname{sign}(\ell_i)$; put $s_i = 1$ when $\ell_i = 0$. Normalize by

$$u_{n,i} = \frac{x_{n,i} - x_{*,i}}{\sqrt{\tau}}, \qquad v_{n,i} = \frac{s_i(y_{n,i} - y_{*,i})}{\sqrt{\sigma}}, \qquad k_i = \sqrt{\tau\sigma}|\ell_i|. \tag{72}$$

With $A_i(u) = F_i(\sqrt{\tau}u)$ and $B_i(v) = H_i(s_i\sqrt{\sigma}v)$, completing squares in each actual update gives exactly

$$u_n = \operatorname{prox}_A(u_{n-1} - kv_{n-1}), \qquad v_n = \operatorname{prox}_B\left(v_{n-1} + \frac{k}{8}(9u_n - u_{n-1})\right). \tag{73}$$

Here $A, B \geq 0$, $A(0) = B(0) = 0$, and $0 \leq k \leq \sqrt{r_*}$. It remains to prove a scalar bound uniform on this whole coupling interval.

4.1 Finite prefixes and positive index objective values

Fix one instance of (73). Write

$$d_j = (u_j^2 + v_j^2)^{1/2}, \qquad d = d_0, \qquad \Delta u_j = u_j - u_{j-1}, \quad \Delta v_j = v_j - v_{j-1}.$$

Nonexpansiveness of both proximal maps relative to their zero fixed points implies

$$|u_j| \leq |u_{j-1}| + k|v_{j-1}|, \qquad |v_j| \leq \frac{5k}{4}|u_{j-1}| + \left(1 + \frac{9k^2}{8}\right)|v_{j-1}|.$$

Since $r_* < 3 < 4$, the squared Frobenius norm of the nonnegative coefficient matrix is at most $83/2 < 64$. Consequently,

$$d_j \leq 8^j d \qquad (j \geq 0). \tag{74}$$

At every positive index, the actual proximal inclusions give

$$a_j = -\Delta u_j - kv_{j-1} \in \partial A(u_j), \qquad b_j = -\Delta v_j + \frac{k}{8}(9u_j - u_{j-1}) \in \partial B(v_j). \tag{75}$$

The anchored subgradient inequalities yield

$$0 \leq A(u_j) \leq a_j u_j, \qquad 0 \leq B(v_j) \leq b_j v_j. \tag{76}$$

In particular these values are finite. Using $k \le 2$ and (74),

$$|a_j| \le d_j + 3d_{j-1} \le 11 \cdot 8^{j-1} d, \quad |b_j| \le \frac{13}{4} d_j + \frac{5}{4} d_{j-1} \le \frac{109}{4} 8^{j-1} d.$$

Therefore

$$A(u_j) \le \frac{11}{8} 64^j d^2 \le 4 \cdot 64^j d^2, \qquad B(v_j) \le \frac{109}{32} 64^j d^2 \le 4 \cdot 64^j d^2. \tag{77}$$

No value or subgradient at index zero is used. If $d = 0$, the orbit is identically zero and every displayed bound still holds.

5 Shared exact storage and continuation

Put

$$\delta = \frac{1}{100000}, \qquad r_0 = r_* - \delta, \qquad \beta_{\mathrm{c}} = 10^{-10}. \tag{78}$$

The public coefficient file `exact_certificate.json` stores a family over $\mathbb{Q}(\sqrt{91})$. For the original uniform coordinate envelope, only the two fixed parameters $t = 0$ and $t = \delta$ are used. A serialized pair $[a, b]$ denotes $a + b\sqrt{91}$ with rational a, b. If P_0, P_1, P_2 are the symmetric 8×8 matrices stored under `P_numerator`, define

$$r = r_* - t, \qquad P(t) = \frac{P_0 + tP_1 + t^2 P_2}{r}. \tag{79}$$

Each matrix is stored as its 36 upper-triangular entries in lexicographic order (i, j), $0 \le i \le j < 8$. The two arrays under each of `c`, `lambda`, and `mu` give its constant and linear coefficients in t; their lengths are respectively 8, 50, and 32. This completely specifies $c(t), \lambda(t), \mu(t)$.

5.1 Formal slacks and identities

Order ten slots as $w = (X_0, Y_0, \ldots, X_4, Y_4)$. Write $z_{j,0} = X_j$ and $z_{j,1} = Y_j$. The formal gradients at positive indices are

$$g_{j,0} = \frac{X_{j-1} - X_j}{r} - Y_{j-1}, \qquad g_{j,1} = Y_{j-1} - Y_j + \frac{9X_j - X_{j-1}}{8}. \tag{80}$$

For an edge $e = (i, j, s)$, its formal convexity slack is

$$S_e = \phi_{j,s} - \phi_{i,s} - \langle g_{i,s}, z_{j,s} - z_{i,s} \rangle = w^{\mathsf{T}} Q_e w + v_e^{\mathsf{T}} \phi. \tag{81}$$

The anchor index $\varnothing$ has point, value, and gradient zero. No gradient at slot zero is defined or used. Within each species $s = 0, 1$, order the descent edges by

$$(\varnothing, j)\ (0 \le j \le 4), \quad (i, \varnothing)\ (1 \le i \le 4), \quad (i, j)\ (1 \le i \le 4,\ 0 \le j \le 4,\ i \ne j),$$

with the last list lexicographic. The 32 lower-bound edges are the 50-edge descent list with every edge involving index four removed. The multiplier arrays follow these orders.

Let O, N select the first and last eight slots, respectively. The certificate matrices are reconstructed by

$$\begin{aligned} D &= O^{\mathsf{T}} P O - N^{\mathsf{T}} P N - \sum_e \lambda_e Q_e, \\ C &= O^{\mathsf{T}} P O - \sum_e \mu_e Q_e, \end{aligned} \tag{82}$$

where the second sum uses the lower-bound edges. The associated exact function-value identities are

$$O^{\mathsf{T}} c - N^{\mathsf{T}} c - \sum_e \lambda_e v_e = 0, \qquad O^{\mathsf{T}} c - \sum_e \mu_e v_e = 0. \tag{83}$$

At both fixed parameters, all multipliers are nonnegative, $D \succeq 0$, and C has final two rows and columns zero. At $t = 0$, every nonzero multiplier has ordinary positive indices $i, j \geq 1$; in particular no nonzero anchor or index-zero edge occurs. This support assertion is checked separately by `check_endpoint.py` and is essential for the no-KKT application below. Denote its leading 8×8 block by C_8. The additional facts needed below are

$$\begin{aligned} &t = 0: && c = 0, \quad P Z_8 = C_8 Z_8 = 0, \quad C_6 - \beta_{\mathrm{c}} I_6 \succ 0, \\ &t = \delta: && C_8 - \beta_{\mathrm{c}} I_8 \succ 0, \\ &t \in \{0, \delta\}: && \sum_{a,b} |P_{ab}| < 3, \quad \sum_a |c_a| < 1, \quad 1 < r_0 < r_* < 3. \end{aligned} \tag{84}$$

Here $Z_8 = [I_2; I_2; I_2; I_2]$, and C_6 is the leading 6×6 block of the endpoint C_8. In particular the endpoint lower bound is semicoercive; it is not a coercive bound on positions.

5.2 Exact verification of positivity

The supplied standard-library Python verifier reconstructs (82) and (83) from the serialized coefficients. It uses exact arithmetic in $\mathbb{Q}(\sqrt{91})$, testing signs by rational square comparisons when the two terms have opposite signs. Every positive-definiteness check is an exact LDL^{T} factorization with strictly positive diagonal pivots, followed by exact reconstruction of the matrix.

For the endpoint descent matrix D, the checker uses six explicit kernel columns. Their constant and ramp rows at pair j are I_2 and jI_2. For the remaining two columns, repeat

$$(c_j, s_j) = (1, 0), \quad (-1/2, 1/2), \quad (-1/2, -1/2)$$

with period three, and use the two rows

$$(c_j, s_j), \qquad \left(\frac{9}{16} c_j + \frac{3 s_j}{2 r_*}, \ \frac{9}{16} s_j - \frac{c_j}{2 r_*} \right), \qquad 0 \leq j \leq 4.$$

If K comprises these six columns and E_4 is the first four coordinate columns, the checker verifies $DK = 0$, invertibility of $[E_4, K]$, and positivity of all four pivots of $E_4^{\mathsf{T}} D E_4$.

Congruence proves $D \succeq 0$. At $t = \delta$, it verifies that D annihilates the two constant histories $Z_{10} = [I_2; \cdots; I_2]$ and that its leading 8×8 block is positive definite. Since $[E_8, Z_{10}]$ is invertible, this also proves $D \succeq 0$. Finally, it factors $C_6 - \beta_c I_6$ at zero and $C_8 - \beta_c I_8$ at δ. Together with the exact constant-history annihilation, these checks imply $C_8 \succeq 0$ at both parameters.

The coefficient file is pinned by SHA-256

```
421ea0abb9e4bb550d19310aa0c85d90
db54531b181c1378796904bff5b97815
```

with the two lines concatenated. It is byte-identical to the public coefficient file used for the endpoint in report [5]. The strict member is checked directly at $t = \delta$; no interval-continuation claim or numerical eigenvalue test is used. The full rational pivot pairs are regenerated by the checker rather than trusted from a saved result.

5.3 Endpoint increments without a KKT pair

Proposition 5.1 (No-anchor endpoint increments)**.** For proper lower-semicontinuous convex f, h on arbitrary real Hilbert spaces and bounded linear L, every initialized orbit of (1) satisfying

$$\theta = \frac{1}{8}, \qquad \tau\sigma\|L\|^2 = r_* \tag{85}$$

has uniformly bounded successive increments. A KKT pair is not required.

Use the shared endpoint matrices of Section 5 and put $\widehat{P} = r_* P(0)$, $\widehat{D} = r_* D(0)$ and $\widehat{C}_8 = r_* C_8(0)$. The positive factor changes neither kernels nor positivity. In particular $\ker \widehat{C}_8 = \operatorname{ran} Z_8$: its leading six-slot block is positive definite, it kills Z_8, and $[E_6, Z_8]$ is invertible. The endpoint value vector is zero and every nonzero multiplier uses ordinary positive indices. These are the only coefficient facts used in the following transfer.

5.3.1 Transfer the certificate to the actual orbit

Normalize without any solution anchor:

$$\begin{aligned} X_n &= \sqrt{r_*/\tau}\, x_n, \qquad Y_n = y_n/\sqrt{\sigma}, \\ K_0 &= \sqrt{\tau\sigma/r_*}\, L, \qquad \|K_0\| = 1. \end{aligned} \tag{86}$$

Put $\mathcal{F}(X) = f(\sqrt{\tau/r_*}\, X)$ and $\mathcal{H}(Y) = h(\sqrt{\sigma} Y)$. Direct substitution into (1) gives

$$\begin{aligned} (X_{n-1} - X_n)/r_* - K_0^* Y_{n-1} &\in \partial\mathcal{F}(X_n), \\ Y_{n-1} - Y_n + K_0(9X_n - X_{n-1})/8 &\in \partial\mathcal{H}(Y_n). \end{aligned} \tag{87}$$

For a contraction K_0, the maps into $Y \oplus X$

$$JX = (K_0 X, (I - K_0^* K_0)^{1/2} X), \qquad I_Y Y = (Y, 0)$$

are isometries on their respective spaces and satisfy

$$\langle JX, I_Y Y\rangle = \langle K_0 X, Y\rangle .$$

Thus they preserve every same-variable pairing and represent every cross pairing in the formal quadratic slacks. This transfers the scalar matrix identities to the Hilbert-valued histories. In particular, each weighted formal slack becomes exactly an actual subgradient convexity slack from (87). No subgradient on the embedding space is posited.

For $k \geq 1$, let

$$h_k = (JX_k, I_Y Y_k, \dots, JX_{k+3}, I_Y Y_{k+3}), \qquad \mathcal{E}_k = \left\langle \widehat{P} h_k, h_k \right\rangle ,$$

and let w_k be the corresponding five-pair history. The value cancellations and positive-index support in (82)–(83) give

$$\begin{aligned} \mathcal{E}_k &= \left\langle \widehat{C}_8 h_k, h_k \right\rangle + r_* \sum_e \mu_e S_{e,k} \geq 0, \\ \mathcal{E}_k - \mathcal{E}_{k+1} &= \left\langle \widehat{D} w_k, w_k \right\rangle + r_* \sum_e \lambda_e S_{e,k} \geq 0. \end{aligned} \tag{88}$$

All values appearing in a nonzero slack are at proximal outputs and are finite. There is no use of a value at the initialized point or a KKT anchor. Hence $0 \leq \mathcal{E}_k \leq \mathcal{E}_1$ for $k \geq 1$.

Let $\kappa > 0$ be the smallest eigenvalue of $\widehat{C}_8$ on the orthogonal complement of its constant-history kernel. Orthogonal diagonalization of this finite scalar matrix works for Hilbert-valued slots. If $\bar{U}, \bar{V}$ are the separate means of the four primal and dual slots, it follows that

$$\left\langle \widehat{C}_8 h, h \right\rangle \geq \kappa \sum_{i=0}^{3} (\left\| U_i - \bar{U} \right\|^2 + \left\| V_i - \bar{V} \right\|^2).$$

Since $\|a-b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$ and both embeddings are isometries, (88) implies

$$\mathcal{E}_k \geq \frac{\kappa}{2} \big(\|X_{k+3} - X_{k+2}\|^2 + \|Y_{k+3} - Y_{k+2}\|^2 \big). \tag{89}$$

This bounds all tail increments. The finite initial prefix consists of finite Hilbert-space points. Undoing the fixed positive scalings proves Proposition 5.1.

Neither $\widehat{P}$ nor the increment argument supplies a coercive position bound. The constant-history kernel remains, and the descent kernel also contains ramp and cycle directions. Strict storage coercivity cannot be imported to equality by continuity.

Corollary 5.1 (Scalar endpoint boundedness)**.** *For* (1) *on* $\mathbb{R} \times \mathbb{R}$*, assume a KKT pair and* (85)*. Every initialized orbit is bounded, with the explicit constants of Theorem 6 obtained from any uniform increment bounds in Proposition 5.1.*

Proof. Proposition 5.1 supplies the increment hypothesis, and Theorem 6 supplies the scalar bounds. ∎

5.4 Finite products in matched diagonal coordinates

The scalar conclusion also applies to a finite orthogonal product, provided both objectives separate in the same coordinates that diagonalize the coupling.

Corollary 5.2 (Separable diagonal systems)**.** *Let X and Y be real d-dimensional Hilbert spaces with $1 \le d < \infty$. Suppose orthonormal bases $(e_i)_{i=1}^d$ and $(g_i)_{i=1}^d$, real numbers ℓ_i, and proper lower-semicontinuous convex scalar functions f_i, h_i satisfy*

$$Le_i = \ell_i g_i, \qquad f(x) = \sum_{i=1}^{d} f_i(\langle x, e_i\rangle), \qquad h(y) = \sum_{i=1}^{d} h_i(\langle y, g_i\rangle). \tag{90}$$

Assume a KKT pair exists, meaning $-L^*y_* \in \partial f(x_*)$ and $Lx_* \in \partial h(y_*)$, and run (1) *from any initial pair.*

For any $\theta \in \mathbb{R}$ and $\tau, \sigma > 0$, bounded successive increments imply bounded positions. In particular, every such orbit is bounded under the global endpoint condition (85)*. Zero diagonal entries are allowed. If $L = 0$ altogether, every orbit is bounded for all these parameters without an increment hypothesis.*

Proof. Write $x_{n,i} = \langle x_n, e_i\rangle$ and $y_{n,i} = \langle y_n, g_i\rangle$. Orthonormality and finite separability split the proximal minimizations into independent coordinates. Also $L^*g_i = \ell_i e_i$, so the actual coordinate updates are

$$\begin{aligned} x_{n,i} &= \operatorname{prox}_{\tau f_i}(x_{n-1,i} - \tau\ell_i y_{n-1,i}),\\ y_{n,i} &= \operatorname{prox}_{\sigma h_i}\big(y_{n-1,i} + \sigma\ell_i(x_{n,i} + \theta(x_{n,i} - x_{n-1,i}))\big). \end{aligned}$$

Varying just coordinate i in each KKT subgradient inequality yields

$$-\ell_i y_{*,i} \in \partial f_i(x_{*,i}), \qquad \ell_i x_{*,i} \in \partial h_i(y_{*,i}).$$

When $\ell_i \neq 0$, global increment bounds give coordinate increment bounds, and Theorem 6 bounds that coordinate pair.

When $\ell_i = 0$, the two coordinates follow uncoupled proximal iterations about minimizers. For example,

$$\frac{x_{n-1,i} - x_{n,i}}{\tau} \in \partial f_i(x_{n,i}), \qquad 0 \in \partial f_i(x_{*,i}).$$

Adding the two subgradient inequalities gives

$$(x_{n-1,i} - x_{n,i})(x_{n,i} - x_{*,i}) \ge 0.$$

Thus

$$\begin{aligned} |x_{n,i} - x_{*,i}|^2 &\le (x_{n-1,i} - x_{*,i})(x_{n,i} - x_{*,i})\\ &\le |x_{n-1,i} - x_{*,i}|\,|x_{n,i} - x_{*,i}|, \end{aligned}$$

which implies $|x_{n,i} - x_{*,i}| \le |x_{n-1,i} - x_{*,i}|$, including the zero-distance case. The same argument applies to y. Hence every zero-coupling coordinate is bounded without assuming bounded increments.

There are finitely many coordinates. Summing their squared coordinate bounds gives finite bounds on both full norms. This proves the conditional statement and the assertion for $L = 0$. At the endpoint, Proposition 5.1 supplies bounded increments for the full Hilbert-space iteration, and the conditional statement applies. It is not necessary for each nonzero ℓ_i to satisfy the endpoint equality separately. ∎

The simultaneous separability hypothesis in (90) is essential to this reduction. Diagonalizing L alone does not separate the proximal maps of arbitrary objectives. In particular, the corollary makes no assertion for a general rank-one operator with nonseparable objectives.

5.5 Exact continuation of the endpoint storage

For the scalar map (8), retain the numerator normalization of the endpoint certificate:

$$P_0 = r_* P(0) \in \mathbb{R}^{8\times 8}, \qquad C_0 = r_* C_8(0) \in \mathbb{R}^{8\times 8}, \qquad D_0 = r_* D(0) \in \mathbb{R}^{10\times 10}. \tag{91}$$

Here P_0 is precisely the constant numerator coefficient in (79). The formal lower-bound matrix C in (82) is 10×10 with its final pair of rows and columns zero; C_0 denotes its numerator-scaled 8×8 block, not that full matrix. The parameter $r_0 = r_* - 10^{-5}$ of (78) continues to denote the distinct strict certificate.

All state vectors below use interleaved slots. Thus for $z_j = (x_j, y_j)$ set

$$h_n = (x_n, y_n, \dots, x_{n+3}, y_{n+3})^{\mathsf{T}}, \qquad W_n = (x_n, y_n, \dots, x_{n+4}, y_{n+4})^{\mathsf{T}}.$$

The species order of the multiplier arrays is primal first, then dual, as in Section 5; it is distinct from the interleaved state-slot order. Let $\widehat{\mu}_e = r_* \mu_e(0)$ and $\widehat{\lambda}_e = r_* \lambda_e(0)$. The endpoint identities have no storage value term and read

$$\begin{aligned} E_n^0 &= h_n^{\mathsf{T}} P_0 h_n = h_n^{\mathsf{T}} C_0 h_n + \sum_e \widehat{\mu}_e S_{e,n}(r_*), \\ E_n^0 - E_{n+1}^0 &= W_n^{\mathsf{T}} D_0 W_n + \sum_e \widehat{\lambda}_e S_{e,n}(r_*). \end{aligned} \tag{92}$$

Every active edge has both endpoints among the four local outputs $1, 2, 3, 4$ (among $1, 2, 3$ for the lower identity). The identities therefore require no value or subgradient at a raw input. We use them at global indices $n \ge 1$.

For a symmetric slot matrix M, let M_x retain its primal–primal entries and set its other entries to zero. In the interleaved convention these are exactly the entries whose two zero-based indices are even. Put

$$\begin{gathered} \alpha = \frac{r_*}{r}, \qquad \varepsilon = 1 - \alpha, \\ P_r = P_0 - \varepsilon (P_0)_x, \qquad C_r^{\mathrm{mat}} = C_0 - \varepsilon (C_0)_x, \qquad D_r = D_0 - \varepsilon (D_0)_x. \end{gathered} \tag{93}$$

Lemma 5 (Primal-block continuation)**.** *With the fixed numerator multipliers above and the actual slacks evaluated at r, one has exactly*

$$\begin{aligned} E_n^r &:= h_n^{\mathsf{T}} P_r h_n = h_n^{\mathsf{T}} C_r^{\mathrm{mat}} h_n + \sum_e \widehat{\mu}_e S_{e,n}(r), \\ E_n^r - E_{n+1}^r &= W_n^{\mathsf{T}} D_r W_n + \sum_e \widehat{\lambda}_e S_{e,n}(r). \end{aligned} \tag{94}$$

There are finite constants $K_C \geq 0$ and $K_D > 0$, depending only on the endpoint matrices, such that

$$0 \preceq (C_0)_x \preceq K_C C_0, \qquad 0 \leq W^{\mathsf{T}}(D_0)_x W \leq K_D J(W), \quad J(W) = \sum_{i=1}^{4} \|z_i - z_{i-1}\|^2 . \tag{95}$$

Consequently, for $r \geq r_*$ with $\varepsilon K_C \leq 1/2$, actual windows satisfy

$$C_r^{\mathrm{mat}} \succeq \tfrac{1}{2} C_0 \succeq 0, \qquad E_n^r \geq 0, \qquad E_n^r - E_{n+1}^r \geq W_n^{\mathsf{T}} D_0 W_n - \varepsilon K_D J(W_n). \tag{96}$$

Proof. At a local output i the actual primal residual is

$$p_i(r) = \alpha \frac{x_{i-1} - x_i}{r_*} - y_{i-1},$$

whereas $q_i = y_{i-1} - y_i + (9x_i - x_{i-1})/8$ is independent of α. For a primal edge the only α-dependent part of $-p_i(r)(x_j - x_i)$ is its primal–primal block. The dual slacks have no primal–primal block and are independent of r. Thus scaling that block in every term of (92), while keeping all multipliers fixed, gives (94) coefficient by coefficient. The function-value coefficients are independent of α and still cancel exactly. Equivalently, every coefficient is affine in α; reconstruction of its constant and linear coefficients establishes the identity for every $r > 0$. This is an algebraic identity, not a positivity extrapolation from two parameter samples.

The matrix C_0 is positive semidefinite with kernel exactly the constant four-pair histories, by (84). Its extended principal block $(C_0)_x$ is positive semidefinite and annihilates that same constant-history space. To see the latter directly, separate the two constant columns of Z_8: on the primal constant column, $(C_0)_x$ selects the primal rows of the zero vector $C_0 Z_8$; on the dual constant column it is zero. On the orthogonal complement of constants, let $\lambda_C > 0$ be the least eigenvalue of C_0. Then $K_C = \|(C_0)_x\|_{\mathrm{op}} / \lambda_C$ proves the first inequality in (95); subtraction of the constant projection extends it to every history. It also covers $(C_0)_x = 0$.

The same reasoning shows that $(D_0)_x \succeq 0$ and that it annihilates constant five-pair histories. Subtracting the constant history z_0 from an arbitrary W, Cauchy–Schwarz gives

$$\sum_{i=1}^{4} \|z_i - z_0\|^2 \leq \sum_{i=1}^{4} i \sum_{j=1}^{i} \|z_j - z_{j-1}\|^2 \leq 10 J(W) \leq 16 J(W).$$

For example $K_D = 1+16\,\|(D_0)_x\|_{\mathrm{op}}$ is valid. This estimate is unrestricted: it needs neither feasibility nor boundedness. Nonnegativity of the actual convexity slacks and their fixed multipliers now proves (96). Also P_0 and $(P_0)_x$ annihilate constants, so P_r does too, and P_r is uniformly bounded on any compact positive parameter interval. ∎

No positivity of D_r to the right of r_* is asserted. Its negative perturbation is controlled by $J(W)$, not by $W^{\mathsf{T}} D_0 W$ alone. The surviving ramp kernel prevents the latter from coercively controlling all first increments. Appendix B gives the full finite matrix and scalar-order conventions needed for the next argument.

6 Uniform scalar and summable coordinate bounds

Choose one of the two fixed values of r and assume $k^2 \le r$. Set

$$X_j = \sqrt{r}\,u_j, \qquad Y_j = v_j, \qquad \kappa = \frac{k}{\sqrt{r}}, \qquad A_r(X) = A(X/\sqrt{r}). \tag{97}$$

The actual subgradients in (75) become

$$\frac{X_{j-1} - X_j}{r} - \kappa Y_{j-1} \in \partial A_r(X_j), \quad Y_{j-1} - Y_j + \frac{\kappa}{8}(9X_j - X_{j-1}) \in \partial B(Y_j). \tag{98}$$

Embed the two scalar species into $\mathbb{R}^2$ by

$$J(X) = (\kappa X, \sqrt{1-\kappa^2}X), \qquad I(Y) = (Y, 0).$$

These maps are isometries and $\langle J(X), I(Y)\rangle = \kappa XY$. Substitute $J(X_j), I(Y_j)$ into the formal slots and keep the actual values $A(u_j), B(v_j)$. Every same-species and cross-species pairing then reproduces exactly an actual convexity slack from (98). Anchor slacks are legitimate because both anchored functions have zero as a minimizer. This argument does not assert that a formal vector gradient is a subgradient of an invented function on $\mathbb{R}^2$.

For $n \ge 4$, let w_n be the eight embedded slots at actual indices $n-3, \dots, n$, and let ϕ_n contain their actual function values. Define

$$E_n = \langle w_n, P w_n\rangle + c^{\mathsf{T}} \phi_n. \tag{99}$$

For vector slots, the first term means $\sum_{a,b} P_{ab}\,\langle w_{n,a}, w_{n,b}\rangle$. Every index is positive, so all values used here are finite. The lower-bound identity and the descent identity, applied also to the five-pair history ending at $n+1$, give

$$E_n \ge \langle w_n, C_8 w_n\rangle \ge 0, \qquad E_{n+1} \le E_n. \tag{100}$$

Indeed the difference between either side is the corresponding nonnegative matrix form plus nonnegative multiples of actual convexity slacks. Scalar positive-semidefinite matrices remain positive-semidefinite on vector-valued slots by applying their quadratic forms to each orthogonal component.

Let

$$S = \sum_{j=1}^{4} 64^j = 17\,043\,520. \tag{101}$$

From (74), $r < 3$, and the isometries,

$$\|w_4\|^2 = \sum_{j=1}^{4} (r u_j^2 + v_j^2) \le 3Sd^2.$$

The entrywise bound on P in (84) implies $|\langle w, Pw\rangle| \le 3\|w\|^2$, also for vector slots. For example, use the triangle inequality followed by Cauchy–Schwarz and $\|P\|_{\mathrm{op}} \le \sum_{a,b} |P_{ab}|$. By (77), every value in ϕ_4 is at most $4Sd^2$, so $|c^{\mathsf{T}}\phi_4| \le 4Sd^2 \sum_a |c_a|$. Therefore

$$E_4 \le 9Sd^2 \quad (t = 0), \qquad E_4 \le 13Sd^2 \quad (t = \delta). \tag{102}$$

The second estimate explicitly controls the signed value terms in the strict storage, without assuming a finite initial objective gap.

6.1 Endpoint increments and scalar barriers

Use $r = r_*$ and $t = 0$ for every $0 \le k \le \sqrt{r_*}$. Since C_8 annihilates constant histories, subtract the final embedded pair from each pair of w_n without changing its quadratic form. The last pair is then zero, and the leading six slots are acted on by C_6. From (84),

$$\begin{aligned} \langle w_n, C_8 w_n\rangle &\ge \beta_{\mathrm{c}} \sum_{j=n-3}^{n-1} \big(r_* |u_j - u_n|^2 + |v_j - v_n|^2\big) \\ &\ge \beta_{\mathrm{c}} \big(|\Delta u_n|^2 + |\Delta v_n|^2\big). \end{aligned} \tag{103}$$

The first inequality applies the scalar matrix bound to each orthogonal component of the embedded vectors; the second uses $r_* > 1$. Equations (100) and (102) show that

$$|\Delta u_n|^2 + |\Delta v_n|^2 \le 9S \cdot 10^{10} d^2 \qquad (n \ge 4).$$

For $n = 1, 2, 3$, the Euclidean increment norm is at most $(8^n + 8^{n-1})d \le 576d$. Thus the fixed integer

$$M = 2 \cdot 10^9 \tag{104}$$

satisfies $M \ge 576$ and $M^2 \ge 9S \cdot 10^{10}$, and gives

$$|\Delta u_n| \le Md, \qquad |\Delta v_n| \le Md \qquad (n \ge 1). \tag{105}$$

These bounds are obtained for each scalar orbit from its own initial distance. A global bound on the norms of Hilbert-space increments would not provide the summable coordinate right-hand side needed later.

6.1.1 Apply scalar trapping for coupling at least one

Assume $1 \le k \le \sqrt{r_*} < 2$. Apply Theorem 5 to A, B in (73), with extrapolation $1/8$ and both increment bounds $A_0 = B_0 = Md$. Its constants become

$$\begin{aligned} C_0 &= B_0/k + A_0/8, \quad X = A_0 + \max\{|u_0|, C_0\}, \\ z_0 &= v_0 - ku_0/8, \qquad T = A_0/k + B_0 + kX/8, \\ Z &= \max\{|z_0|, T\}, \quad Y = Z + kX/8. \end{aligned} \tag{106}$$

Consequently $|u_n| \le X$, $|v_n - ku_n/8| \le Z$ and $|v_n| \le Y$ for every index. This invokes the full first-output, threshold-tie and shifted-dual arguments already proved, rather than a new order assumption.

6.1.2 Uniform numerical constants

The restrictions $k \ge 1$ and $k < 2$ in (106) give

$$X \le C_X d, \quad C_X = 1 + \frac{17}{8}M < 5{\cdot}10^9, \qquad Y \le C_Y d, \quad C_Y = \frac{5}{4} + 2M + \frac{C_X}{2} < 7{\cdot}10^9. \tag{107}$$

For the second estimate, use $|z_0| \le |v_0| + |u_0|/4 \le 5d/4$ and $\max\{s,t\} \le s + t$ for nonnegative s, t. Therefore

$$u_n^2 + v_n^2 \le (C_X^2 + C_Y^2)d^2 < 10^{20}d^2 \qquad (1 \le k \le \sqrt{r_*}). \tag{108}$$

The constants are deliberately conservative and are not claimed to be sharp.

6.2 A fixed strict certificate for small couplings

For every $0 \le k \le 1$, use the single parameter $t = \delta$, so $r = r_0 > 1 \ge k^2$. The transfer in Section 6 applies. Its strict coercivity gives

$$E_n \ge \beta_{\mathrm{c}} \|w_n\|^2 \ge \beta_{\mathrm{c}}(u_n^2 + v_n^2) \qquad (n \ge 4).$$

By (100) and (102),

$$u_n^2 + v_n^2 \le 13S \cdot 10^{10}d^2 \qquad (n \ge 4).$$

The first three indices are bounded by 64^3d^2 through (74), and the initial index by d^2. Since

$$\max\{64^3, 13S \cdot 10^{10}\} < 10^{20},$$

the same scalar envelope as (108) holds for $0 \le k \le 1$. No constant has been divided by a coupling approaching zero, and no coordinate-dependent choice of a near-critical storage parameter is involved. Combined with (108), we have proved

$$\sup_{n \ge 0}(u_n^2 + v_n^2) \le 10^{20}(u_0^2 + v_0^2) \qquad (0 \le k \le \sqrt{r_*}). \tag{109}$$

Only the two fixed exact certificates, actual convexity slacks, and the scalar barrier argument enter this estimate.

6.3 The summable envelope on the Hilbert space

Proof of Theorem 2. Apply (109) to each exact coordinate recursion obtained in Section 4. The sequence

$$d_i^2 = \frac{|x_{0,i} - x_{*,i}|^2}{\tau} + \frac{|y_{0,i} - y_{*,i}|^2}{\sigma}$$

is nonnegative and summable. For every coordinate, the scalar bound already proves

$$\sup_{n\geq 0}\left(\frac{|x_{n,i} - x_{*,i}|^2}{\tau} + \frac{|y_{n,i} - y_{*,i}|^2}{\sigma}\right) \leq 10^{20} d_i^2.$$

This is (5) and its coordinatewise supremum version. Sum these pointwise bounds to obtain (6). Each actual iterate belongs to $\mathcal{H} \times \mathcal{H}$ by the genuine proximal construction, and its squared weighted distance is at most the sum of this envelope. There is no interchange of a sum with a supremum.

If $L = 0$, the two proximal iterations decouple, and (2) says that every anchor coordinate minimizes the corresponding scalar function. Nonexpansiveness relative to that fixed point gives $|x_{n,i} - x_{*,i}| \leq |x_{0,i} - x_{*,i}|$ and its dual counterpart. Thus $C = 1$ suffices. No part of the argument requires a nonzero lower bound on $|\ell_i|$, attainment of $\|L\|$, or finitely many coordinates. ∎

7 Compactness stability and convergence

Use the equivalent weighted product norm

$$\|(u, v)\|_w^2 = \frac{\|u\|^2}{\tau} + \frac{\|v\|^2}{\sigma}.$$

The summable envelope has a consequence stronger than boundedness in an infinite-dimensional space.

Corollary 7.1 (Relative norm compactness)**.** *Under the hypotheses of Theorem 2, the paired orbit* $\{(x_n, y_n) : n \geq 0\}$ *has compact closure in the product norm.*

Proof. Set $z_n = (x_n - x_*, y_n - y_*)$ and

$$e_i = C\left(\frac{|x_{0,i} - x_{*,i}|^2}{\tau} + \frac{|y_{0,i} - y_{*,i}|^2}{\sigma}\right).$$

Then $\sum_i e_i < \infty$. If P_m retains the first m coordinates in both components, the coordinate envelope gives

$$\sup_{n\geq 0} \|(I - P_m)z_n\|_w^2 \leq \sum_{i>m} e_i \longrightarrow 0. \tag{110}$$

The projected orbit $\{P_m z_n : n \geq 0\}$ is bounded in a finite-dimensional space and hence totally bounded. For any $\varepsilon > 0$, choose m so that the square root of the right-hand side of

(110) is less than $\varepsilon/2$. A finite $\varepsilon/2$-net for the projected orbit, with centers extended by zero tails, is a finite ε-net for the full centered orbit. Thus the latter is totally bounded. Completeness of the weighted product space gives compact closure. Translation by (x_*, y_*) and equivalence of the weighted and usual product norms prove the claim. ∎

Uniform tails are essential in this proof: boundedness in ℓ^2 alone would not imply (110) or relative norm compactness.

Corollary 7.2 (KKT stability and a conditional convergence criterion)**.** *Fix the objectives and parameters in Theorem 2. Every KKT pair p is a Lyapunov stable fixed point of the actual paired map. More precisely, for every initial pair z_0 and every $n \geq 0$,*

$$\|z_n - p\|_w \leq 10^{10} \|z_0 - p\|_w . \tag{111}$$

If an orbit has a subsequence z_{n_j}, with $n_j \to \infty$, converging in norm to a KKT pair p, then the entire orbit converges in norm to p.

Proof. The objective class (4) is independent of a chosen anchor. At any KKT pair, the component supporting slopes belong to ℓ^2 and the component base values have finite sums, so the argument of Section 4 re-anchors the functions at that pair with no change in C. The alternative form (7) is also an instance of (4). For example, take

$$a = -L^* y_*, \quad \alpha = f(x_*) + \langle L^* y_*, x_* \rangle , \quad \phi_i(t) = F_i(t - x_{*,i}),$$

with the analogous dual definition. Thus the same re-anchoring argument applies to that presentation as well.

At a KKT pair $p = (x^p, y^p)$, proximal optimality gives

$$x^p = \operatorname{prox}_{\tau f}(x^p - \tau L^* y^p), \qquad y^p = \operatorname{prox}_{\sigma h}(y^p + \sigma L x^p).$$

The extrapolation at a stationary primal point equals that point, so p is fixed by (1). Applying Theorem 2 with anchor p and taking square roots proves (111). The usual stability definition follows by choosing initial radius $\varepsilon/10^{10}$ for a prescribed radius ε.

The map (1) is autonomous on the current pair: its dual step uses only the newly computed primal point and the old primal point already in that pair. Consequently any tail beginning at index m is an actual orbit with initialization z_m. Reapplying (111) gives

$$\sup_{n \geq m} \|z_n - p\|_w \leq 10^{10} \|z_m - p\|_w . \tag{112}$$

For a norm-convergent subsequence $z_{n_j} \to p$, choose j so that the right-hand side with $m = n_j$ is less than ε. The whole remaining tail is then within ε of p. This proves convergence in the weighted norm and hence in the usual product norm. ∎

The conditional criterion by itself does not identify a cluster point as KKT. Relative compactness supplies cluster points, but their KKT status requires an additional argument. Neither corollary alone is an unconditional convergence claim.

7.1 Convergence from the endpoint kernel

The endpoint storage contains more information than bounded increments. We now identify its limiting kernel dynamics and use scalar monotonicity to eliminate their non-constant periodic part. This section supplies the extra argument that the compactness and stability consequences alone do not provide.

7.1.1 The linked cycle matrix

Throughout this section put $r = r_*$. In the normalized scalar map (73) at $k = \sqrt{r}$, set $X_n = \sqrt{r}\,u_n$, $Y_n = v_n$, and $\mathcal{F}(X) = A(X/\sqrt{r})$. The actual paired map is

$$X_n = \operatorname{prox}_{r\mathcal{F}}(X_{n-1} - rY_{n-1}), \qquad Y_n = \operatorname{prox}_B\left(Y_{n-1} + \frac{9X_n - X_{n-1}}{8}\right). \tag{113}$$

It is continuous, being a composition of Lipschitz proximal maps and linear maps. At every positive index its actual subgradients are

$$p_n = \frac{X_{n-1} - X_n}{r} - Y_{n-1} \in \partial\mathcal{F}(X_n), \qquad q_n = Y_{n-1} - Y_n + \frac{9X_n - X_{n-1}}{8} \in \partial B(Y_n). \tag{114}$$

Define

$$c = \frac{9r}{16} - \frac{1}{2}, \qquad d = \frac{c^2 + c + 1}{r} = \frac{11}{16} + \frac{3}{2r}, \qquad \mathsf{A} = \begin{pmatrix} c & -r \\ d & -1 - c \end{pmatrix}. \tag{115}$$

Using $81r^2 - 176r - 192 = 0$, direct exact calculation gives

$$\mathsf{A}^2 + \mathsf{A} + I_2 = 0, \qquad \mathsf{A}^3 = I_2, \qquad \det(I_2 - \mathsf{A}) = 3. \tag{116}$$

The endpoint descent matrix of Section 5 has the equivalent kernel description

$$\ker D = \left\{(a + jb + \mathsf{A}^j v)_{j=0}^4 : a, b, v \in \mathbb{R}^2\right\}. \tag{117}$$

This is an exact finite identity. The public scalar convergence checker reconstructs the endpoint matrices and verifies (116)–(117) with the same coefficient file. It uses the numerator-scaled matrices rP, rD, rC_8; the positive scale changes none of their kernels. It checks zero action on the displayed six kernel columns, an invertible four-column complement, and exact positive LDL^{T} pivots on that complement.

For clarity, the recurrence content of (117) is also verified directly. For a sequence of pairs z_j, put

$$\rho_j = z_{j+3} - (\mathsf{A} + 2I_2)z_{j+2} + (2\mathsf{A} + I_2)z_{j+1} - \mathsf{A}z_j. \tag{118}$$

The 4×10 matrix mapping a five-pair history to (ρ_0, ρ_1) has rank four and annihilates the six columns in (117); its kernel is therefore exactly $\ker D$.

Lemma 6 (Bounded kernel dynamics)**.** *If a bounded forward sequence $(z_j)_{j\geq 0}$ has every five-pair history in $\ker D$, then $z_j = a + \mathsf{A}^j v$ for fixed $a, v \in \mathbb{R}^2$. In particular it is three-periodic.*

Proof. Every ρ_j in (118) vanishes. With $w_j = z_{j+1} - \mathsf{A}z_j$, this says $\Delta^2 w_j = 0$, so $w_j = w_0 + jh$ for some fixed h. Since $I_2 - \mathsf{A}$ is invertible, choose

$$b = (I_2 - \mathsf{A})^{-1}h, \qquad a = (I_2 - \mathsf{A})^{-1}(w_0 - b).$$

Then $z_j - a - jb$ obeys the recurrence $v_{j+1} = \mathsf{A}v_j$. Thus $z_j = a + jb + \mathsf{A}^j v$. Since $\mathsf{A}^3 = I_2$, the subsequence $z_{3j} = a + 3jb + v$ is bounded only if $b = 0$. ∎

7.1.2 The period three theorem removes the critical cycle

The endpoint-scaled map (113) is exactly (29) with $f = \mathcal{F}$, $h = B$, $\tau = r = r_*$, $\sigma = L = 1$, and $\theta = 1/8$. Thus the product in Theorem 7 is precisely r_*, not $\sqrt{r_*}$. The functions remain fixed proper lower-semicontinuous convex functions under the nonzero scalar change of variable. Since

$$155^2 - 16^2 \cdot 91 = 729 > 0, \qquad 16\sqrt{91} < 155, \qquad r_* < 3,$$

every actual three-periodic orbit of this scaled map is constant by Theorem 7. No solution anchor is needed for this exclusion. The kernel lemma alone does not assert that its abstract sequence is an actual proximal orbit; the next proof establishes that fact by taking a continuous forward limiting orbit before invoking the exclusion. Appendix A records the earlier specialized exact identities, which are subsumed here by the all-$\theta \geq 0$ theorem.

7.1.3 Cluster points at equality are fixed

Lemma 7 (The scalar equality case)**.** *For the scalar normalized map* (73) *with* $k = \sqrt{r_*}$ *and a KKT anchor, the full orbit converges to a KKT pair.*

Proof. The uniform scalar envelope proves boundedness. The endpoint storage has zero value coefficients, so it is a continuous quadratic function of a four-pair history. Its nonnegative descent identity implies convergence of E_n to a finite limit and

$$\langle W_n, DW_n \rangle \longrightarrow 0, \tag{119}$$

where W_n is the five-pair history at actual indices $n-3, \ldots, n+1$. Choose any subsequence $z_{n_j} \to \bar{z}$ with $n_j \to \infty$. If $\mathcal{T}$ is the actual continuous map (113), then for every fixed $m \geq 0$,

$$z_{n_j+m} = \mathcal{T}^m(z_{n_j}) \longrightarrow \mathcal{T}^m(\bar{z}) = \widetilde{z}_m.$$

The limiting forward orbit lies in the compact closure of the bounded original orbit. Polynomial continuity and (119) place every five-pair window of this orbit in $\ker D$. Lemma 6 makes it three-periodic, and the verified application of Theorem 7 makes it constant. Therefore $\mathcal{T}(\bar{z}) = \bar{z}$, so every cluster point is a KKT pair.

Choose one such point and undo the fixed scalar normalization. The homogeneous bound at every KKT anchor, together with the restart argument of Corollary 7.2, forces the entire orbit to converge to that point. The limiting argument uses no convergence of function values and no limiting passage in a convexity slack. The limiting forward orbit supplies its own actual subgradients through (114). ∎

7.1.4 Every smaller coupling from the same kernel

The same endpoint kernel gives a direct argument for every $0 \le k < \sqrt{r}$, including zero. No strictly subcritical convergence theorem is needed. Use $X_n = \sqrt{r}\,u_n$, $Y_n = v_n$, $\kappa = k/\sqrt{r}$, and

$$\lambda = \sqrt{1-\kappa^2} > 0, \qquad U_n = (\kappa X_n, \lambda X_n), \qquad V_n = (Y_n, 0).$$

The exact transfer of Section 6 makes every formal slack an actual scalar convexity slack. In particular, pairing the formal primal gradient with $U_j - U_n$ gives

$$\left(\frac{X_{n-1} - X_n}{r} - \kappa Y_{n-1}\right)(X_j - X_n) = [u_{n-1} - u_n - k v_{n-1}](u_j - u_n),$$

and the dual pairing gives

$$\left[v_{n-1} - v_n + \frac{k}{8}(9u_n - u_{n-1})\right](v_j - v_n).$$

These are the actual scalar subgradient pairings. Thus the nonnegative decreasing endpoint storage and (119) remain valid on the vector-valued slots. The two spatial components are not assumed to be separate proximal orbits.

The envelope again makes the original scalar orbit bounded. For any cluster point, continuity gives its bounded forward limiting orbit as before. Each five-pair lifted history has zero D quadratic form. This form is the sum of the two nonnegative scalar forms corresponding to the two spatial components, so each component history lies in $\ker D$. Lemma 6 applies to these component sequences purely as a kernel statement.

For the second component, write

$$(\lambda \widetilde{X}_n, 0) = a' + \mathsf{A}^n v'.$$

Averaging its three phases and using $I_2 + \mathsf{A} + \mathsf{A}^2 = 0$ gives $a'_Y = 0$. Its dual coordinate at phase zero gives $v'_Y = 0$, and that at phase one gives

$$0 = (\mathsf{A}v')_Y = dv'_X - (1+c)v'_Y = dv'_X.$$

Since $d = 11/16 + 3/(2r) > 0$, also $v'_X = 0$. Hence this component is constant, and $\lambda > 0$ implies that $\widetilde{X}_n$ is constant. For the first component write

$$(\kappa \widetilde{X}_n, \widetilde{Y}_n) = a'' + \mathsf{A}^n v''.$$

Its primal coordinate is constant, including when $\kappa = 0$. Averaging the phases identifies that coordinate with a''_X; phases zero and one then give $v''_X = 0$ and

$$0 = (\mathsf{A}v'')_X = cv''_X - rv''_Y = -rv''_Y.$$

Therefore $v''_Y = 0$, and $\widetilde{Y}_n$ is constant as well. The limiting actual orbit is constant, so its initial cluster point is a KKT fixed point. The same homogeneous stability and restart argument as in Lemma 7 proves full scalar convergence.

We have now proved convergence for every scalar coupling $0 \le k \le \sqrt{r_*}$ under the KKT assumption. The only additional matrix signs used in the strict case are $r > 0$ and $d > 0$, immediate from their exact formulas. In particular this argument covers a zero diagonal entry directly and introduces no lower bound on positive couplings.

7.1.5 Strong convergence of the countable diagonal system

Proof of Theorem 1. The genuine proximal separation in Section 4 makes every paired coordinate an actual scalar orbit. Its normalized coupling k_i lies in $[0, \sqrt{r_*}]$ and its KKT anchor is supplied by the global pair. The scalar equality argument and Section 7.1.4 therefore give a limit $(x_{\infty,i}, y_{\infty,i})$ in every coordinate.

With e_i as in the proof of Corollary 7.1, passing to the coordinate limit in the envelope yields

$$\frac{|x_{\infty,i} - x_{*,i}|^2}{\tau} + \frac{|y_{\infty,i} - y_{*,i}|^2}{\sigma} \le e_i, \qquad \sum_i e_i < \infty.$$

The limit sequences thus belong to $\mathcal{H}$, because their differences from the original $\mathcal{H}$ anchor have summable squares. Moreover,

$$\frac{|x_{n,i} - x_{\infty,i}|^2}{\tau} + \frac{|y_{n,i} - y_{\infty,i}|^2}{\sigma} \le 4e_i. \tag{120}$$

Each left-hand side tends to zero at fixed i. Dominated convergence for the nonnegative series, equivalently a finite-coordinate and uniform-tail argument, gives

$$\frac{\|x_n - x_\infty\|^2}{\tau} + \frac{\|y_n - y_\infty\|^2}{\sigma} \longrightarrow 0.$$

No uniform convergence rate or coordinate-independent convergence time has been used.

Finally, the full Hilbert-space CP map $\mathcal{T}$ is continuous, since L is bounded and both full proximal maps are nonexpansive. The relation $z_{n+1} = \mathcal{T}z_n$ and strong convergence of both sequences to z_∞ imply $\mathcal{T}z_\infty = z_\infty$. Proximal optimality at that fixed point yields

$$-L^*y_\infty \in \partial f(x_\infty), \qquad Lx_\infty \in \partial h(y_\infty).$$

This establishes the full Hilbert KKT inclusions without inferring global subdifferential membership merely by summing coordinate inclusions. Finite objective values at the limit follow from actual proximal optimality. ∎

7.2 The same interval in matched scalar coordinates

The universal interval and anchor constant in Theorem 3 transfer to the original scalar-separable model. Two different normalizations are needed. The first is harmless even at zero coupling; the second is used only where the coupling is bounded away from zero. This distinction preserves the summable physical-coordinate envelope.

Corollary 7.3 (Matched-separable interval extension)**.** *Assume exactly the objective class* (4) *on real ℓ^2, a bounded real diagonal L, and a full Hilbert KKT pair. Let η, B_{cont} be as in Theorem 3. For fixed positive steps satisfying*

$$\theta = \frac{1}{8}, \qquad \tau\sigma \left\|L\right\|^2 \le r_* + \eta < 3, \tag{121}$$

every actual orbit (1) *from an arbitrary Hilbert-space initialization converges strongly to a full Hilbert KKT pair.*

Put

$$C_{\mathrm{env}} = \max\{10^{20}, 3B_{\mathrm{cont}}^2\}, \qquad \left\|(x,y)\right\|_w^2 = \frac{\left\|x\right\|^2}{\tau} + \frac{\left\|y\right\|^2}{\sigma}. \tag{122}$$

For every KKT pair $z_* = (x_*, y_*)$, every restart index $m \ge 0$, and every coordinate i,

$$\begin{aligned} &\sup_{n\ge m}\left(\frac{|x_{n,i}-x_{*,i}|^2}{\tau} + \frac{|y_{n,i}-y_{*,i}|^2}{\sigma}\right) \\ &\qquad \le C_{\mathrm{env}}\left(\frac{|x_{m,i}-x_{*,i}|^2}{\tau} + \frac{|y_{m,i}-y_{*,i}|^2}{\sigma}\right). \end{aligned} \tag{123}$$

Consequently the sum of these coordinatewise suprema is at most $C_{\mathrm{env}}\left\|z_m - z_*\right\|_w^2$, and

$$\sup_{n\ge m}\left\|z_n - z_*\right\|_w \le \sqrt{C_{\mathrm{env}}}\left\|z_m - z_*\right\|_w. \tag{124}$$

The constants are independent of the objectives, entries, steps, anchor, initialization, restart, and number of coordinates. The original squared constant 10^{20} and stability factor 10^{10} remain available when $\tau\sigma\left\|L\right\|^2 \le r_*$; if $L = 0$, the squared constant is 1. The same conclusions hold in finite or countable matched orthonormal scalar coordinates, provided both objectives separate in those coordinates.

Proof. *Parameter-independent domains and genuine proximal maps.* The properness, anchor representation, and infinite-product proximal identity proved in Section 4 do not use any upper bound on $\tau\sigma\left\|L\right\|^2$. They therefore apply under (121). To make the domain point explicit, the nonnegative sums in (4) are increasing suprema of lower-semicontinuous convex finite sums, and each has a finite point. Their continuous affine terms preserve properness and lower semicontinuity. Thus the genuine Hilbert proximal maps are defined everywhere before any coordinate argument is made.

At any global KKT pair, both component sums have finite values. One-coordinate variation in the full subgradient inequalities gives the supporting slopes

$$\alpha_i^* = -\ell_i y_{*,i} - a_i \in \partial\phi_i(x_{*,i}), \qquad \gamma_i^* = \ell_i x_{*,i} - b_i \in \partial\psi_i(y_{*,i}).$$

Both slope sequences belong to ℓ^2. Subtracting their affine supports defines the nonnegative functions F_i, H_i of (71), vanishing at zero. The finite-prefix identities pass to (7), because the anchor component sums are finite and the slope pairings are absolutely convergent by Cauchy–Schwarz. This holds as an extended-real identity at every point; it subtracts no two infinite quantities.

For completeness, the product-proximal argument uses actual minimizers. If $Q_i \geq 0$, $Q_i(0) = 0$, and $p_i = \operatorname{prox}_{\lambda Q_i}(w_i)$ with $w \in \ell^2$, then $|p_i| \leq |w_i|$, so $p \in \ell^2$. Scalar comparison with zero gives

$$\sum_i \left(Q_i(p_i) + \frac{|p_i - w_i|^2}{2\lambda} \right) \leq \frac{\|w\|^2}{2\lambda} < \infty.$$

Summing the scalar minimization inequalities against any Hilbert competitor is legitimate termwise with nonnegative objectives. It proves that p is the full proximal minimizer, unique by strict convexity of the quadratic. Completing the continuous affine squares in (7) therefore gives, for $\xi_{n,i} = x_{n,i} - x_{*,i}$ and $\zeta_{n,i} = y_{n,i} - y_{*,i}$,

$$\begin{aligned} \xi_{n,i} &= \operatorname{prox}_{\tau F_i}(\xi_{n-1,i} - \tau \ell_i \zeta_{n-1,i}), \\ \zeta_{n,i} &= \operatorname{prox}_{\sigma H_i}\left(\zeta_{n-1,i} + \frac{\sigma \ell_i}{8}(9\xi_{n,i} - \xi_{n-1,i}) \right). \end{aligned} \tag{125}$$

The primal affine completion adds $\tau L^* y_*$ to the original input; the dual completion subtracts $\sigma L x_*$. The anchor component of $(9x_n - x_{n-1})/8$ is exactly x_*. These observations verify both cancellations and the update order.

Signed physical normalization. Set $s_i = \operatorname{sign}(\ell_i)$ when $\ell_i \neq 0$, and $s_i = 1$ otherwise. Define

$$u_{n,i} = \frac{\xi_{n,i}}{\sqrt{\tau}}, \qquad v_{n,i} = \frac{s_i \zeta_{n,i}}{\sqrt{\sigma}}, \qquad k_i = \sqrt{\tau\sigma}|\ell_i|, \qquad r_i = k_i^2. \tag{126}$$

Let $A_i(u) = F_i(\sqrt{\tau}u)$ and $D_i(v) = H_i(s_i\sqrt{\sigma}v)$. Each is proper lower-semicontinuous convex, nonnegative, and zero at zero. Suppressing i, the actual primal quadratic becomes

$$\frac{|\sqrt{\tau}u - [\sqrt{\tau}u_{n-1} - \tau\ell s\sqrt{\sigma}v_{n-1}]|^2}{2\tau} = \frac{1}{2}|u - u_{n-1} + kv_{n-1}|^2,$$

and the dual quadratic becomes

$$\frac{|s\sqrt{\sigma}v - [s\sqrt{\sigma}v_{n-1} + \sigma\ell\sqrt{\tau}(9u_n - u_{n-1})/8]|^2}{2\sigma} = \frac{1}{2}|v - v_{n-1} - k(9u_n - u_{n-1})/8|^2.$$

Thus every paired coordinate follows the genuine unit-step map

$$u_n = \operatorname{prox}_A(u_{n-1} - kv_{n-1}), \qquad v_n = \operatorname{prox}_D\left(v_{n-1} + \frac{k}{8}(9u_n - u_{n-1}) \right). \tag{127}$$

Its selected residuals are

$$a_n = u_{n-1} - u_n - kv_{n-1} \in \partial A(u_n), \qquad d_n = v_{n-1} - v_n + k(9u_n - u_{n-1})/8 \in \partial D(v_n).$$

Indeed a_n is $\sqrt{\tau}$ times the centered physical primal residual and d_n is $s\sqrt{\sigma}$ times the centered physical dual residual. The invertible linear subgradient rules

$$\partial A(u) = \sqrt{\tau}\,\partial F(\sqrt{\tau}u), \qquad \partial D(v) = s\sqrt{\sigma}\,\partial H(s\sqrt{\sigma}v)$$

show that $-kv \in \partial A(u)$, $ku \in \partial D(v)$ are equivalent to the centered physical KKT conditions. The rule for $s = -1$ is a set equality, not an inequality multiplied without reversing its sign. At $\ell = 0$, the same formulas give $k = 0$ and two independent proximal iterations. No division by a coupling has occurred.

High modes and the second normalization. Only for $r_i \geq r_*$ define, coordinate by coordinate,

$$X_n = ku_n, \qquad Y_n = v_n, \qquad g(X) = A(X/k), \qquad h_c(Y) = D(Y), \qquad r = k^2. \tag{128}$$

Here $k > 0$ and $r \in [r_*, r_* + \eta]$. The primal minimization becomes exactly

$$A(U/k) + \tfrac{1}{2}|U/k - u_{n-1} + kv_{n-1}|^2 = g(U) + \frac{|U - X_{n-1} + rY_{n-1}|^2}{2r}.$$

The dual input is $Y_{n-1} + (9X_n - X_{n-1})/8$. Consequently

$$X_n = \operatorname{prox}_{rg}(X_{n-1} - rY_{n-1}), \qquad Y_n = \operatorname{prox}_{h_c}\left(Y_{n-1} + \frac{9X_n - X_{n-1}}{8}\right),$$

which is precisely the canonical map of Theorem 3. Its primal step is $r = k^2$, and its primal objective is $g = A(\cdot/k)$. The actual residual transformation is

$$\frac{X_{n-1} - X_n}{r} - Y_{n-1} = \frac{a_n}{k}, \qquad Y_{n-1} - Y_n + \frac{9X_n - X_{n-1}}{8} = d_n.$$

Since $\partial g(X) = k^{-1}\partial A(X/k)$, its KKT conditions $-Y \in \partial g(X)$, $X \in \partial h_c(Y)$ are exactly those of (127). The functions and this invertible scalar transformation are fixed throughout the coordinate orbit; zero remains a KKT pair.

On this branch $1 < r_* \leq k^2 < 3$, so

$$u^2 + v^2 \leq X^2 + Y^2 = k^2u^2 + v^2 \leq 3(u^2 + v^2). \tag{129}$$

The two differences are $(r-1)u^2$ and $(3-r)u^2 + 2v^2$, respectively, and are nonnegative. The scalar theorem thus proves coordinate convergence and

$$\sup_n(u_n^2 + v_n^2) \leq B_{\text{cont}}^2(X_0^2 + Y_0^2) \leq 3B_{\text{cont}}^2(u_0^2 + v_0^2).$$

No inverse of the small-mode multiplier is used, and no new canonical Hilbert proximal problem is assembled.

Low modes and a summable envelope. For $r_i < r_*$, use directly the physical scalar envelope (109) and scalar convergence established in Section 7.1.4. These are scalar

statements for (127), and do not require already knowing a high-product Hilbert theorem. They give convergence and the squared constant 10^{20} uniformly as $k_i \downarrow 0$. At $k_i = 0$, nonexpansiveness about the minimizing anchor gives the sharper constant 1. At $r_i = r_*$ either branch is valid.

Combining the two branches with (122), put

$$d_i^2 = \frac{|x_{0,i} - x_{*,i}|^2}{\tau} + \frac{|y_{0,i} - y_{*,i}|^2}{\sigma}, \qquad e_i = C_{\mathrm{env}} d_i^2.$$

Then every centered coordinate square is at most e_i at every time, and $\sum_i e_i = C_{\mathrm{env}} \|z_0 - z_*\|_w^2 < \infty$. Summing the separately established coordinatewise supremum bounds proves the asserted envelope for $m = 0$; no exchange of an infinite sum and a supremum is made. Every $r_i \le \tau\sigma \|L\|^2$, even when the operator norm is not attained. Signed, arbitrarily small, and zero entries have all been covered. If the global product is at most r_*, all coordinates may instead use the old scalar bound, preserving 10^{20} throughout that original range; if $L = 0$, all may use 1.

Strong convergence and the full KKT inclusions. Let $(x_{\infty,i}, y_{\infty,i})$ be the coordinate limits. Passing to each scalar limit preserves its bound by e_i, so both limit sequences belong to ℓ^2 relative to the given ℓ^2 anchor. Moreover

$$\frac{|x_{n,i} - x_{\infty,i}|^2}{\tau} + \frac{|y_{n,i} - y_{\infty,i}|^2}{\sigma} \le 4e_i. \tag{130}$$

Given $\epsilon > 0$, first choose a finite prefix so that the sum of $4e_i$ outside it is smaller than $\epsilon/2$. Coordinate convergence makes the sum of the finitely many remaining left sides smaller than $\epsilon/2$ for all sufficiently large n. Therefore $z_n \to z_\infty$ in the weighted Hilbert norm, and hence in the ordinary product norm because the positive steps are fixed.

Let T be the genuine Hilbert map (1). It is continuous: its primal output difference is bounded by $\|\Delta x\| + \tau \|L\| \|\Delta y\|$, and its dual output difference by

$$\|\Delta y\| + \frac{\sigma \|L\|}{8}(9 \|\Delta x_{\mathrm{out}}\| + \|\Delta x\|).$$

It follows that $Tz_\infty = \lim_n Tz_n = \lim_n z_{n+1} = z_\infty$. Proximal optimality at this full fixed point, where the extrapolated primal point equals x_∞, gives

$$-L^* y_\infty \in \partial f(x_\infty), \qquad Lx_\infty \in \partial h(y_\infty).$$

These are full Hilbert inclusions and supply finite objective values at the limit. They have not been inferred merely by summing coordinate inclusions.

The same envelope gives relative norm compactness: the centered tail squared norms are uniformly bounded by $\sum_{i>M} e_i \to 0$, while each finite coordinate projection is bounded and admits finite nets. Extending such nets with zero tails proves total boundedness in the complete Hilbert product.

Arbitrary anchors, restarts, and matched bases. The anchor was arbitrary, and its representation was proved without the old product restriction. Starting the autonomous map at z_m gives exactly its actual tail with unchanged functions and parameters. Repeating the same argument therefore proves (123) for every m, its summed form, and (124). The radius $\epsilon/\sqrt{C_{\mathrm{env}}}$ proves Lyapunov stability of every KKT pair. On the original range the same argument retains the numerical factor 10^{10}.

Finally let $U : \ell^2(J) \to X$ and $V : \ell^2(J) \to Y$ be fixed onto orthogonal maps, with $J = \mathbb{N}$ or $\{1, \dots, d\}$, such that $V^*LU = \operatorname{diag}(\ell_i)$. Assume $f \circ U, h \circ V$ have the stated scalar-separable class. Substitution in the defining minimization proves

$$\operatorname{prox}_{\tau f}(Uw) = U \operatorname{prox}_{\tau(f\circ U)}(w), \qquad \operatorname{prox}_{\sigma h}(Vw) = V \operatorname{prox}_{\sigma(h\circ V)}(w).$$

Both actual updates and the full KKT conditions are thereby conjugate to the diagonal problem, with all norms and bounds preserved. For a finite sum of arbitrary proper lower-semicontinuous convex scalar components, a KKT pair supplies finite component values and supporting slopes; subtracting those finitely many affine supports places the system in the same nonnegative anchored class. Thus the finite matched presentation is included without a new restriction on the unshifted components. ∎

This corollary requires scalar separability of both objectives in the matched coordinates. Diagonalizing L alone does not supply that hypothesis, and the argument gives no coupled-block or arbitrary nonseparable Hilbert theorem. The new constant is existential through B_{cont}. The scalar square-sum constant H_{cont} has not been combined with an unproved uniform low-mode counterpart: no full Hilbert increment-square sum or convergence rate is asserted. Strong convergence does, of course, imply vanishing Hilbert increments. All steps and products remain fixed in time.

7.3 Anchored geometry and uniform compensation

We now prove the scalar estimate that compensates the possible negative part of the continued dissipation. Fix a KKT pair (x_*, y_*) for (8), and replace f, h by

$$f_*(u) = f(x_* + u) - f(x_*) + y_* u, \qquad h_*(v) = h(y_* + v) - h(y_*) - x_* v.$$

Both functions are proper lower-semicontinuous convex, are nonnegative, and vanish at zero. Direct proximal optimality shows that the centered coordinates follow exactly the same map, with $0 \in \partial f_*(0)$ and $0 \in \partial h_*(0)$. We henceforth write f, h for these fixed centered functions.

For an interleaved five-state window $W = (z_0, \dots, z_4)$, $z_i = (x_i, y_i)$, define only the four output residuals

$$p_i = \frac{x_{i-1} - x_i}{r} - y_{i-1}, \qquad q_i = y_{i-1} - y_i + \frac{9x_i - x_{i-1}}{8}, \qquad 1 \le i \le 4. \tag{131}$$

Let $\mathcal{C}_r \subset \mathbb{R}^{10}$ be the set specified by

$$\begin{aligned} (x_i - x_j)(p_i - p_j) \geq 0, \qquad (y_i - y_j)(q_i - q_j) \geq 0 \quad (1 \leq i, j \leq 4), \\ x_i p_i \geq 0, \qquad y_i q_i \geq 0 \quad (1 \leq i \leq 4). \end{aligned} \tag{132}$$

Every actual centered window belongs to $\mathcal{C}_r$, by monotonicity of the two fixed subdifferentials including their common anchored graph points $(0, 0)$. No subgradient or anchor inequality at input index zero is assumed. The set is homogeneous under every real scaling, including simultaneous sign reversal. Its joint dependence on (r, W) is closed for $r > 0$, because its finitely many inequalities have continuous coefficients. It must not be confused with C_r^{mat} in (93).

These finite constraints also have a literal fixed-function interpretation. For either species, append $(0, 0)$ to the prescribed graph points and group their equal abscissae as $t_1 < \cdots < t_m$, with smallest and largest prescribed slopes l_j, u_j. Monotonicity implies $u_j \leq l_{j+1}$. A nondecreasing piecewise-constant derivative equal to u_j on (t_j, t_{j+1}), to l_1 to the left of t_1, and to u_m to the right of t_m has an integral whose subdifferential contains every prescribed point. It is a finite convex function. The two residual inclusions then reconstruct all four outputs uniquely. Thus ties and vertical graph segments require neither differentiability nor a changing objective.

The exact kernel description (117), with numerator-scaled D_0, and the four-output order lemma proved in Appendix B give

$$W \in \mathcal{C}_{r_*} \cap \ker D_0 \quad \Longrightarrow \quad z_i = a + ib, \qquad b_x b_y = 0. \tag{133}$$

The anchor inequalities further restrict these surviving ramps, but do not eliminate their nonzero speeds.

Lemma 8 (Complete anchored axis-ramp geometry)**.** *Set*

$$Q_r(x, y) = \frac{x^2}{r} - xy + y^2. \tag{134}$$

For $0 < r < 4$ this is positive definite. Suppose $z_i = a + ib$ belongs to $\mathcal{C}_r$ and $b_x b_y = 0$. If $b = 0$, then $a_x a_y = 0$. If $b \neq 0$, simultaneously reverse signs if needed so that its nonzero component is $s > 0$.

For a primal ramp $b = (s, 0)$, write $v = y_i/s$ and $w_i = x_i/s = w_4 + i - 4$. The exact admissibility conditions are

$$-1/r \leq v \leq 0, \qquad \begin{cases} w_4 \leq 0, & v = 0, \\ w_4 \leq -1/8, & v < 0. \end{cases} \tag{135}$$

For a dual ramp $b = (0, s)$, write $v = x_i/s$ and $w_i = y_i/s = w_4 + i - 4$. The exact conditions are

$$0 \leq v \leq 1, \qquad \begin{cases} w_4 \leq 0, & v < 1, \\ w_4 \leq 1, & v = 1. \end{cases} \tag{136}$$

In both cases $Q_r(z_i) \le Q_r(z_{i-1})$ *for* $1 \le i \le 4$, *and*

$$Q_r(z_1) - Q_r(z_4) \ge \min\left\{\frac{27}{4r}, 6\right\} \|b\|^2 . \tag{137}$$

Proof. The determinant of the matrix of Q_r is $1/r - 1/4 > 0$, and its first diagonal entry is positive. On a constant history $p_i = -a_y$ and $q_i = a_x$, so the two anchor inequalities require both signs of $a_x a_y$, giving $a_x a_y = 0$.

For a primal ramp, $p_i/s = -1/r - v$ and $q_i/s = w_i + 1/8$. The anchor constraints are exactly

$$w_i(-1/r - v) \ge 0, \qquad v(w_i + 1/8) \ge 0.$$

If $v > 0$, they put all four w_i in $[-1/8, 0]$, impossible since $w_4 - w_1 = 3$. If $v < -1/r$, they require simultaneously $w_i \ge 0$ and $w_i \le -1/8$. For $-1/r \le v \le 0$ they reduce precisely to (135); in particular the boundary $v = -1/r$ still imposes $w_4 \le -1/8$. Each stated condition is also sufficient, and pairwise same-species monotonicity holds automatically on any axis ramp.

For a dual ramp, $p_i/s = 1 - w_i$ and $q_i/s = v - 1$. Thus anchoring is equivalent to

$$v(w_i - 1) \le 0, \qquad w_i(v - 1) \ge 0.$$

If $v < 0$, these force $w_i \ge 1$ and $w_i \le 0$. If $v > 1$, they place every w_i in $[0, 1]$, again contradicting their separation. The remaining conditions are exactly (136), including both endpoints, and they are sufficient.

The one-transition differences are respectively

$$s^2\left(\frac{2w_i - 1}{r} - v\right), \qquad s^2(2w_i - 1 - v).$$

The classifications make these nonpositive for all four transitions. For the three transitions between outputs, direct summation gives respectively

$$Q_r(z_4) - Q_r(z_1) = s^2\left(\frac{6w_4 - 9}{r} - 3v\right) \le -\frac{27}{4r}s^2,$$

$$Q_r(z_4) - Q_r(z_1) = s^2(6w_4 - 9 - 3v) \le -6s^2.$$

For the first bound, $v = 0$ gives the stronger $-9s^2/r$; otherwise use $w_4 \le -1/8$ and $-3v \le 3/r$. The second follows directly from the two alternatives in (136). This proves (137). The same calculation over all four transitions gives the valid, but unused, bound $Q_r(z_0) - Q_r(z_4) \ge \min\{13/r, 12\} \|b\|^2$ on these exact ramps. ∎

Example 1 (Why Euclidean monotonicity is too strong)**.** One stronger route considered in this argument was monotonicity of the ordinary Euclidean distance to the anchor. It fails even on a finite actual axis ramp. Let $s > 0$, $f = \iota_{[0,s]}$, $h = 0$, and $z_0 = (s, -3s)$. The origin is KKT. For every $r > 0$, projection onto $[0, s]$ and the literal dual update give

$$z_i = (s, (-3 + i)s), \qquad 0 \le i \le 4.$$

The last transition $(s,0) \mapsto (s,s)$ raises the Euclidean squared distance from s^2 to $2s^2$. The tilted distance is constant on this transition and decreases on the preceding ones. This is a counterexample to the stronger estimate attempted here, not to the convergence theorem. Nonzero finite ramps genuinely survive the order constraints.

Lemma 9 (Quantitative neighborhoods of constant axes)**.** *Let $a > 0$ and $r \in [2,3]$. Suppose $W \in \mathcal{C}_r$ and every coordinate of every z_j, $0 \le j \le 4$, differs by less than $a/100$ from the corresponding coordinate of one of*

$$(a,0), \quad (-a,0), \quad (0,a), \quad (0,-a).$$

Then the three output-to-output transitions satisfy

$$Q_r(z_{i-1}) - Q_r(z_i) \ge \|z_i - z_{i-1}\|^2, \qquad i = 2,3,4. \tag{138}$$

The relative number $1/100$ is a spatial neighborhood radius, not a step-size continuation length.

Proof. First take the positive primal axis. At outputs $x_i > 99a/100$ and

$$|q_i - a| < \frac{13a}{400},$$

so $q_i > 0$. Anchoring gives $y_i \ge 0$ and $p_i \ge 0$. For $i = 2,3,4$, set

$$x = x_i > 0, \quad u = y_{i-1} \ge 0, \quad v = y_i \ge 0, \quad t = x_{i-1} - x_i \ge ru \ge 0.$$

The preceding dual value is an anchored output precisely because $i \ge 2$. Exact expansion gives

$$\begin{aligned} Q_r(z_i) - Q_r(z_{i-1}) &= x(-2t/r - v + u) - t^2/r + ut + v^2 - u^2 \\ &\le -xt/r - v(x - v). \end{aligned}$$

Indeed the difference between the last expression and the preceding equality is $(x+t)(t/r - u) + u^2 \ge 0$. If $v \le x/2$, it follows that

$$Q_r(z_{i-1}) - Q_r(z_i) \ge x(t/r + v/2).$$

On the other hand,

$$\|z_i - z_{i-1}\|^2 = t^2 + (v-u)^2 \le (1 + 2/r^2)t^2 + 2v^2.$$

When $t, v \le \epsilon x$, the right side is at most $\epsilon x[(1 + 2/r^2)t + 2v]$. Since $r + 2/r \le 4$ on $[2,3]$, taking $\epsilon \le 1/4$ makes it at most $x(t/r + v/2)$. In the stated neighborhood, $t/x < 2/99$ and $v/x < 1/99$, so this applies with $\epsilon = 1/4$.

For the positive dual axis, $y_i > 99a/100$ and

$$|p_i + a| < \frac{2a}{100r} + \frac{a}{100} \le \frac{2a}{100},$$

so $p_i < 0$. Anchoring now gives $x_i \le 0$ and $q_i \ge 0$. For $i = 2, 3, 4$ put

$$X = -x_i \ge 0, \quad U = -x_{i-1} \ge 0, \quad Y = y_i > 0, \quad s = q_i \ge 0, \quad V = X + U,$$

$$d = y_i - y_{i-1} = -\frac{9}{8}X + \frac{1}{8}U - s.$$

The exact expansion, retaining the cross term with its sign, is

$$Q_r(z_i) - Q_r(z_{i-1}) = \frac{X^2 - U^2}{r} + Y[(X - U) + 2d] + Ud - d^2. \tag{139}$$

Here $+Ud$ comes from subtracting $U(Y - d)$ and cannot be omitted. Since

$$(X - U) + 2d = -\tfrac{5}{4}X - \tfrac{3}{4}U - 2s, \qquad |d| \le \tfrac{9}{8}V + s,$$

one obtains

$$\begin{aligned} Q_r(z_i) - Q_r(z_{i-1}) &\le -Y(\tfrac{3}{4}V + 2s) + (1/r + 9/8)V^2 + Vs \\ &\le -Y(\tfrac{3}{4}V + 2s) + \tfrac{13}{8}V^2 + Vs. \end{aligned}$$

If $V \le \epsilon Y$ and $\epsilon \le \min\{1, 3/13\}$, this implies

$$Q_r(z_{i-1}) - Q_r(z_i) \ge Y(\tfrac{3}{8}V + s) \ge \tfrac{3}{8}Y(V + s).$$

Also

$$\|z_i - z_{i-1}\|^2 \le [1 + (9/8)^2](V + s)^2 = \frac{145}{64}(V + s)^2.$$

Thus $V + s \le \epsilon Y$ with $\epsilon \le 24/145$ gives the desired descent. In the stated neighborhood,

$$V < \frac{2a}{100}, \qquad 0 \le s < \frac{13a}{400}, \qquad \frac{V + s}{Y} < \frac{7}{132} < \frac{1}{10}.$$

The choice $\epsilon = 1/10$ meets all three restrictions. Finally, simultaneous reversal of all state and residual signs preserves Q_r and every product in (132), proving the negative-axis cases. No anchor at z_0 was used, and no estimate for its first transition is asserted. ∎

Lemma 10 (Uniform compensation near the endpoint)**.** *There exist $\delta_{\mathrm{comp}} > 0$ and $C_{\mathrm{comp}} > 0$, with $[r_* - \delta_{\mathrm{comp}}, r_* + \delta_{\mathrm{comp}}] \subset [2, 3]$, such that for every r in that interval and every $W \in \mathcal{C}_r$,*

$$Q_r(z_1) - Q_r(z_4) \ge \frac{1}{2}\sum_{i=2}^{4} \|z_i - z_{i-1}\|^2 - C_{\mathrm{comp}} W^{\mathsf{T}} D_0 W. \tag{140}$$

The dissipation matrix remains fixed at r_*; the feasible constraints and Q_r use the actual parameter r.

Proof. Write

$$\mathcal{B}(W) = \sum_{i=2}^{4} \|z_i - z_{i-1}\|^2, \quad d_0(W) = W^{\mathsf{T}} D_0 W, \quad G_r(W) = Q_r(z_1) - Q_r(z_4) - \tfrac{1}{2}\mathcal{B}(W).$$

On $\mathcal{C}_{r_*} \cap \ker D_0$, (133) and Lemma 8 leave only anchored axis ramps. A nonzero speed gives

$$G_{r_*}(W) \geq \left(\min\{27/(4r_*), 6\} - \tfrac{3}{2}\right) \|b\|^2 > 0,$$

because $2 < r_* < 3$. At zero speed the only possibilities are constant-axis histories.

Let $S = \{W : \|W\| = 1\}$. Its four constant-axis histories repeat the states $(\pm 1/\sqrt{5}, 0)$ or $(0, \pm 1/\sqrt{5})$. Let $\mathcal{U}$ be the union of their open relative coordinate neighborhoods from Lemma 9. For every $r \in [2, 3]$ and $W \in \mathcal{C}_r \cap \mathcal{U}$,

$$G_r(W) \geq \tfrac{1}{2}\mathcal{B}(W) \geq 0.$$

Choose first $\beta > 0$ so that $I = [r_* - \beta, r_* + \beta] \subset [2, 3]$. The set

$$\mathcal{K} = \{(r, W) \in I \times S : W \in \mathcal{C}_r,\ W \notin \mathcal{U},\ G_r(W) \leq 0\}$$

is compact by the joint closedness already established. Its fiber $\mathcal{K}_*$ at $r = r_*$ contains no point with $d_0(W) = 0$. If that fiber is nonempty, set $m_* = \min_{\mathcal{K}_*} d_0 > 0$ and $\tau_{\text{comp}} = m_*/2$; if it is empty, set $\tau_{\text{comp}} = 1$. In either case there is $0 < \delta_{\text{comp}} \leq \beta$ such that

$$(r, W) \in \mathcal{K}, \quad |r - r_*| \leq \delta_{\text{comp}} \quad \Longrightarrow \quad d_0(W) \geq \tau_{\text{comp}}.$$

Otherwise a sequence with $r \to r_*$ and $d_0 < \tau_{\text{comp}}$ would have a compact limit in the fiber, contradicting either emptiness or its lower bound m_*.

The constants δ_{comp} and τ_{comp} have now been fixed, before choosing a compensation coefficient. Uniformly on S and $r \in [2, 3]$, one has $0 \leq Q_r(z) \leq 3\|z\|^2/2$ and $\mathcal{B}(W) \leq 4$, hence $G_r(W) \geq -7/2$. Choose

$$C_{\text{comp}} = \frac{7}{2\tau_{\text{comp}}}.$$

If $G_r(W) \geq 0$, then $G_r(W) + C_{\text{comp}} d_0(W) \geq 0$ immediately. If $G_r(W) < 0$, the local lemma excludes $\mathcal{U}$, so the preceding compact-complement bound gives the same conclusion. This proves (140) on S. Homogeneity gives it for every nonzero window, and the zero window is immediate.

The quantitative local neighborhoods are essential: nonnegativity on the exact kernel, by itself, would not justify a uniform quadratic compensation bound near its constant directions. ∎

Example 2 (Why compensation starts at an output)**.** Another stronger route considered here would replace z_1 in (140) by the raw input z_0. Even the weaker inequality

$Q_{r_*}(z_0) - Q_{r_*}(z_4) \geq -Cd_0(W)$ fails for every fixed C. Take $f = 0$, $h = \iota_{\{0\}}$, anchor $(0, 0)$, and $z_0 = (a, -e)$ with $a, e > 0$. The first output is $(a + re, 0)$ and is fixed thereafter. Thus

$$Q_r(z_4) - Q_r(z_0) = ae + (r - 1)e^2.$$

At $r = r_*$, subtract the repeated constant $(a + r_*e, 0)$ from W. What remains is eV, where the first pair of V is $(-r_*, -1)$ and all later pairs vanish. Since D_0 annihilates constants, $d_0(W) = e^2V^\mathsf{T}D_0V$. At fixed a, its quadratic scale cannot absorb the positive linear term ae as $e \downarrow 0$. This actual fixed-function example explains both the output-only compensation and the finite startup treatment below.

7.4 Summation, finite startup, and scalar convergence

Proof of Theorem 3. Choose the constants of Lemmas 5 and 10 first. A valid strictly positive choice is

$$\eta = \frac{1}{2}\min\left\{\delta_{\text{comp}},\ 3 - r_*,\ \frac{r_*}{2(1 + K_C)},\ \frac{r_*}{16C_{\text{comp}}K_D}\right\}. \tag{141}$$

This formula belongs to the original compactness proof and supplies an existential choice of η. It is not a numerical computation of the separate choice η_{eff} in Theorem 9; no comparison between these two particular choices is needed. Fix $r \in [r_*, r_* + \eta]$ for the entire orbit. Since $\varepsilon = (r - r_*)/r \leq \eta/r_*$, the compensation lemma applies, $\varepsilon K_C \leq 1/2$, and

$$t_r := C_{\text{comp}}\varepsilon K_D \leq \frac{1}{16}. \tag{142}$$

Center the orbit at any KKT pair as in Section 7.3, so that the anchor is zero.

Put $a_j = \|z_j - z_{j-1}\|^2$, $j \geq 1$. Apply (140) to the actual window $W_n = (z_n, \ldots, z_{n+4})$, for $n \geq 1$, and use (96). The result is

$$\begin{aligned}\tfrac{1}{2}(a_{n+2} + a_{n+3} + a_{n+4}) \leq\ & Q_r(z_{n+1}) - Q_r(z_{n+4}) \\ & + C_{\text{comp}}(E_n^r - E_{n+1}^r) + t_r \sum_{j=n+1}^{n+4} a_j.\end{aligned} \tag{143}$$

In particular the output-to-output compensation begins with a_{n+2}, while the storage perturbation also contains a_{n+1}.

Sum over $n = 1, \ldots, N$, where $N \geq 3$. The quadratic terms cancel exactly to

$$Q_r(z_2) + Q_r(z_3) + Q_r(z_4) - Q_r(z_{N+2}) - Q_r(z_{N+3}) - Q_r(z_{N+4}),$$

and the storage terms cancel to $C_{\text{comp}}(E_1^r - E_{N+1}^r)$. Every increment a_j with $3 \leq j \leq N+4$ occurs at least once in the left sum. Every increment in the perturbation sum occurs at most four times. The only perturbation increment outside $3, \ldots, N + 4$ is a_2, occurring

exactly once. Therefore (142) leaves a coefficient at least $1/2-4t_r \geq 1/4$ on every retained increment and yields

$$\begin{aligned} &Q_r(z_{N+2}) + Q_r(z_{N+3}) + Q_r(z_{N+4}) + C_{\mathrm{comp}}E^r_{N+1} + \frac{1}{4}\sum_{j=3}^{N+4} a_j \\ &\quad \leq Q_r(z_2) + Q_r(z_3) + Q_r(z_4) + C_{\mathrm{comp}}E^r_1 + t_r a_2. \end{aligned} \tag{144}$$

All three terminal quadratic terms, the nonnegative terminal storage, and all final perturbation increments have been retained. This is a summed inequality; it does not presume a one-step Lyapunov function with an unverified terminal correction.

For $r \in [2,3]$, Young's inequality $|xy| \leq (2/7)x^2 + (7/8)y^2$ gives

$$\frac{1}{21}\left\|(x,y)\right\|^2 \leq \frac{x^2}{21} + \frac{y^2}{8} \leq Q_r(x,y) \leq \frac{3}{2}\left\|(x,y)\right\|^2. \tag{145}$$

Also $E^r_{N+1} \geq 0$ by (96). Thus (144), whose right side uses only a finite prefix of actual outputs, already proves boundedness of the tail and summability of its squared increments. Boundedness has not been an input to the argument.

We make its uniform startup control explicit. Both centered proximal maps fix zero and are nonexpansive. If the old pair has maximum norm at most R, the new primal coordinate has absolute value at most $4R$, and the new dual coordinate at most

$$R + \frac{9(4R)+R}{8} = \frac{45}{8}R < 6R.$$

Writing $R_0 = \|z_0\|$, it follows that

$$\|z_j\|^2 \leq 2\,6^{2j}R_0^2 \quad (j \geq 0), \qquad a_j \leq 4(6^{2j} + 6^{2j-2})R_0^2 \quad (j \geq 1). \tag{146}$$

These estimates require no initial subgradients or function values. In particular E^r_1 is evaluated on the genuine positive-index outputs $z_1, \ldots, z_4$; every active slack in its identity has finite values and an actual subgradient selection.

Set

$$L_P = \sup_{r \in [r_*, r_*+\eta]} \|P_r\|_{\mathrm{op}} < \infty$$

and define the finite constant

$$A_{\mathrm{init}} = 3\sum_{j=2}^{4} 6^{2j} + 2C_{\mathrm{comp}}L_P \sum_{j=1}^{4} 6^{2j} + \frac{1}{4}(6^4 + 6^2). \tag{147}$$

The upper bound on Q_r, the estimate $|E^r_1| \leq L_P \sum_{j=1}^4 \|z_j\|^2$, and $t_r \leq 1/16$ show that the right side of (144) is at most $A_{\mathrm{init}}R_0^2$. Valid choices are consequently

$$\begin{aligned} B^2_{\mathrm{cont}} &= \max\{2\,6^8,\ 21A_{\mathrm{init}}\}, \\ H_{\mathrm{cont}} &= 4A_{\mathrm{init}} + 4\big[(6^2+1) + (6^4+6^2)\big]. \end{aligned} \tag{148}$$

Indeed every state of index at least five occurs in one of the three terminal terms of (144), while (146) covers indices zero through four. Letting $N \to \infty$ bounds $\sum_{j=3}^{\infty} a_j$ by $4A_{\text{init}}R_0^2$; the two explicit additions in H_{cont} cover a_1 and a_2. Thus

$$\sup_{n\geq 0} \|z_n\| \leq B_{\text{cont}} \|z_0\|, \qquad \sum_{n=1}^{\infty} a_n \leq H_{\text{cont}} \|z_0\|^2.$$

Neither constant depends on the functions, the chosen KKT pair, the initialization, or the particular fixed r in the interval.

It remains to establish convergence of the entire trajectory. Square summability gives $z_{n+1}-z_n \to 0$. The actual paired map $T_r : \mathbb{R}^2 \to \mathbb{R}^2$ in (8) is continuous as a composition of proximal and linear maps, and its fixed points are precisely the KKT pairs. Boundedness in $\mathbb{R}^2$ supplies a subsequence $z_{n_k} \to \bar{z}$. Hence

$$T_r\bar{z} - \bar{z} = \lim_{k\to\infty} (z_{n_k+1} - z_{n_k}) = 0,$$

so $\bar{z}$ is a KKT pair. The preceding uniform bound may be centered at this pair and restarted at n_k. The recursion is autonomous, and its input is the current pair alone, so the restarted orbit is exactly the original tail. Therefore

$$\|z_n - \bar{z}\| \leq B_{\text{cont}} \|z_{n_k} - \bar{z}\| \qquad (n \geq n_k).$$

For any tolerance, choose k large enough to make the right side smaller than that tolerance. The entire subsequent tail then lies in that ball. This proves full convergence without uniqueness of the KKT set or a rate.

Finally, reanchoring at any KKT pair and restarting at any $m \geq 0$ preserves the same functions up to their supporting affine translation and leaves the fixed algorithmic parameters unchanged. Applying both uniform bounds to that genuine tail gives (9) and (10) with the same constants. This completes the theorem. ∎

8 An effective scalar continuation interval

The compactness proof above is retained as an independent argument. We now make the continuation length effective, before constructing a genuine state Lyapunov function. Throughout this section all coordinates are interleaved, and all matrices have the numerator normalization of Section 5. Write $R = r_*$ and

$$\begin{gathered} M_F = 1000, \quad L_F = 50001, \quad T_D = 100000, \quad T_C = 1000, \\ K_C = 1000, \quad K_D = 17, \quad \delta = \frac{10^{-50}}{100002}, \quad \tau_D = \frac{\delta^2}{100000}, \quad C_H = \frac{400000}{\delta^2}. \end{gathered} \tag{149}$$

The subscript H distinguishes this explicit compensation constant from a compact subcritical constant introduced later. The coefficient details below are part of the proof; the finite checkers reconstruct their exact premises from the public raw tables.

Theorem 9 (Effective continuation)**.** *The conclusion of Theorem 3 holds with*

$$\eta_{\text{eff}} = 10^{-118}. \tag{150}$$

Its constants can be chosen independently of the fixed scalar objectives, KKT anchor, initialization, fixed $r \in [R, R + \eta_{\text{eff}}]$*, and restart. In particular the fixed-step matched-separable transfer in Corollary 7.3 holds with this continuation length. This is an effective existence bound, not a practically significant enlargement, and it does not settle the whole interval* $(R, 3)$*.*

8.1 Exact complements and approximate order

Center at any KKT pair. For $W = (z_0, \dots, z_4)$ and $1 \le i \le 4$, put

$$p_i = \frac{x_{i-1} - x_i}{r} - y_{i-1}, \qquad q_i = y_{i-1} - y_i + \frac{9x_i - x_{i-1}}{8}. \tag{151}$$

Let $\mathcal{F}_r$ be the homogeneous closed set given by all same-species pairwise monotonicity inequalities and the eight anchor inequalities

$$(x_i - x_j)(p_i - p_j) \ge 0, \quad (y_i - y_j)(q_i - q_j) \ge 0, \qquad x_i p_i \ge 0, \quad y_i q_i \ge 0.$$

Only outputs $1, \dots, 4$ occur. No objective value, graph inclusion, or anchor inequality is imposed at the raw input z_0.

The endpoint matrices P_0, C_0, D_0 and their nonnegative slack multipliers give

$$\begin{aligned} E_n^0 &= h_n^T P_0 h_n = h_n^T C_0 h_n + \sum_e \mu_e S_e, \\ E_n^0 - E_{n+1}^0 &= W_n^T D_0 W_n + \sum_e \lambda_e S_e. \end{aligned} \tag{152}$$

Here $h_n = (z_n, \dots, z_{n+3})$. The lower identity uses only output indices $1, 2, 3$, and the drop identity only $1, 2, 3, 4$, relative to the initial state of each window. The literal supports are verified from the raw multipliers. Thus these identities hold from an arbitrary initial state, including one outside either objective domain.

The exact facts used here are $C_0 \succeq 0$, with kernel the constant four-pair histories, and $D_0 \succeq 0$, of rank four, with kernel

$$z_i = a + (i-1)b + A^{i-1}v, \qquad A = \begin{pmatrix} c & -R \\ d & -1-c \end{pmatrix}, \quad c = \frac{9R}{16} - \frac{1}{2}, \quad d = \frac{11}{16} + \frac{3}{2R}. \tag{153}$$

One has $A^2 + A + I = 0$, and the maximum absolute row sums of I, A, A^2 are less than six. The power $i - 1$ places actual outputs $1, \dots, 4$ at certificate indices $0, \dots, 3$; negative powers mean $A^{-1} = A^2$.

Let D_4 be the leading four-by-four block of D_0, and C_6 the leading six-by-six block of C_0. Both are positive definite. If E_k denotes the first k columns of the corresponding identity matrix, exact complementarity gives a unique decomposition

$$W = E_4 u + K, \qquad K \in \ker D_0.$$

Put $e = \|u\|$ and $d(W) = W^T D_0 W$. Exact inversion in $\mathbb{Q}(\sqrt{91})$ gives

$$\operatorname{tr}(D_4^{-1}) \le T_D, \quad \operatorname{tr}(C_6^{-1}) \le T_C, \quad \operatorname{tr}((C_6)_x) \le 1, \quad \operatorname{tr}((D_0)_x) \le 1. \tag{154}$$

Here a subscript x retains only primal/primal entries and extends them by zero. Consequently

$$e \le \sqrt{T_D d(W)}, \qquad \|W - K\|_\infty \le e. \tag{155}$$

This is a coordinate complement, not an orthogonal projection: $d(W) = u^T D_4 u$ directly proves the estimate. Subtracting a repeated last pair from a four-state history, and using the common constant kernel, gives

$$0 \preceq (C_0)_x \preceq K_C C_0, \qquad 0 \le W^T (D_0)_x W \le K_D J(W), \quad J(W) = \sum_{i=1}^{4} \|z_i - z_{i-1}\|^2 . \tag{156}$$

Indeed the first estimate follows from $T_C \operatorname{tr}((C_6)_x) \le 1000$. For the second, subtract z_0 and bound squared partial sums; even $\sum_i \|z_i - z_0\|^2 \le 10 J(W)$ suffices for the conservative choice $K_D = 17$.

Normalize $W \in \mathcal{F}_r$ by $\|W\| = 1$, and initially take $r \in [2, 3]$. Sort the four actual outputs of each species by coordinate, sorting within each coordinate tie by residual. Monotonicity then orders residuals across distinct-coordinate groups too. All twelve adjacent coordinate and residual order forms are nonnegative, including ties. They select one of the $24^2 = 576$ complete permutation cones. On K at parameter R, their errors are bounded by $2e$ for coordinate differences, $4e + |r - R|$ for primal residual differences, and $13e/2$ for dual residual differences. The parameter contribution uses

$$|x_{i-1} - x_i|\,|1/r - 1/R| \le |r - R|/2.$$

Thus each kernel form is at least $-(7e + |r - R|)$.

For every cone, the exact Farkas table expresses each of $v_x, -v_x, v_y, -v_y$ as a nonnegative combination of these twelve forms, with weight sum at most M_F. All 2304 identities, their signs and their complete permutation coverage are checked. It follows that

$$|v_x|, |v_y| \le M_F(7e + |r - R|).$$

Removing the cycle in (153) leaves a ramp $R_i = a + (i - 1)b$ with

$$\left\|W - (R_i)_{i=0}^4\right\|_\infty \le e + 6M_F(7e + |r - R|) \le L_F(e + |r - R|) =: \xi. \tag{157}$$

No unstated inverse bound for the ramp coordinates is used.

8.2 Robust anchored axis geometry

Assume $\xi \le 1$. Each actual coordinate has magnitude at most one, and each ramp coordinate at most two. At output endpoints $1, 4$, product perturbations are at most 24ξ for primal order and 39ξ for dual order. For example, coordinate differences change by at most 2ξ, primal residual differences by 4ξ, and dual residual differences by $13\xi/2$; expansion of the products gives these constants. The ramp endpoint products are $-9b_xb_y$ and $9b_xb_y$. Hence

$$|b_xb_y| \le 100\xi/9.$$

At least one slope has magnitude at most $4\sqrt{\xi}$. Remove that slope about the center $i = 2$, retaining R_2, to obtain an axis ramp T. This gives

$$\|W - T\|_\infty \le \xi + 8\sqrt{\xi} \le 9\sqrt{\xi} =: \lambda. \tag{158}$$

If a slope vanishes there is nothing to remove; if both vanish either axis designation is allowed. For $\lambda \le 1$, coordinate magnitudes of T are at most two, and direct expansion of the literal anchor products gives

$$T_{i,x}p_i(T; r) \ge -20\lambda, \qquad T_{i,y}q_i(T; r) \ge -20\lambda. \tag{159}$$

The respective sharper errors are 6λ and $39\lambda/4$.

Put

$$\begin{gathered} Q_r(x, y) = x^2/r - xy + y^2, \qquad B(W) = \sum_{i=2}^{4} \|z_i - z_{i-1}\|^2, \\ G_r(W) = Q_r(z_1) - Q_r(z_4) - B(W)/2. \end{gathered} \tag{160}$$

Quadratic expansion also gives

$$|G_r(W) - G_r(T)| \le 100\lambda. \tag{161}$$

The two Q terms cost at most 15λ; the six squared coordinate increments, divided by two, cost at most 36λ.

We need a margin that survives approximate anchoring. Let $s > 0$ be the axis speed, reverse all signs if necessary, and suppose every anchor product is at least $-ts^2$, where $t = 10^{-4}$. For a primal ramp write $T_{i,y} = sv$, $T_{i,x} = sw_i$, so $w_4 - w_1 = 3$. Its inequalities are

$$w_i(-1/r - v) \ge -t, \qquad v(w_i + 1/8) \ge -t.$$

If $v > 0$, they give $w_4 \le 3t$, $w_1 \ge -1/8 - t/v$, so $v \le t$. If $v < -1/r$, write $k = -1/r - v > 0$: then $w_1 \ge -t/k$, $w_4 \le -1/8 + 3t$, giving $k \le t$. Thus $-1/r - t \le v \le t$. If $v \le -1/100$, then $w_4 \le -1/8 + 1/100$ and

$$\frac{Q_r(T_4) - Q_r(T_1)}{s^2} = \frac{6w_4 - 9}{r} - 3v \le -\frac{9}{4} + \frac{3}{100} + 3t < -2.$$

If $v > -1/100$, then $-1/r - v < -1/6$, so $w_4 \le 6t$ and the same difference is at most $-3 + 18t + 3/100 < -2$.

For a dual ramp write $T_{i,x} = sv$, $T_{i,y} = sw_i$. The inequalities are

$$v(w_i - 1) \le t, \qquad w_i(v - 1) \ge -t.$$

If $v < 0$, put $k = -v$; then $w_1 \ge 1 - t/k$, $w_4 \le t$, whence $k \le t$. If $v > 1$, then $w_4 \le 1 + t$, $w_1 \ge -t/(v-1)$, whence $v - 1 \le t$. Thus $-t \le v \le 1 + t$. For $v \le 1 - 1/100$, $w_4 \le 1/100$ and

$$\frac{Q_r(T_4) - Q_r(T_1)}{s^2} = 6w_4 - 9 - 3v \le -9 + 6/100 + 3t < -2.$$

For $v > 1 - 1/100$, $w_4 \le 1 + 2t$ gives an upper bound $-6 + 3/100 + 12t < -2$. All divisions were restricted to cases with the displayed strict sign. Since $B(T) = 3s^2$, in every case

$$G_r(T) \ge s^2/2. \tag{162}$$

8.3 Small slopes and constant axes

Set $\beta = 10^{-8}$, $\lambda_0 = 10^{-24}$, and suppose $\|W - T\|_\infty \le \lambda_0$ and $s < \beta$. With $a = T_0$ and $h = \lambda_0 + 4\beta$,

$$\|W - (a, a, a, a, a)\|_\infty \le h.$$

Since $\|W\| = 1$ and $10(1/4 + h)^2 < 1$, some coordinate of a has magnitude greater than $1/4$; both are at most two. At output one, the residuals differ from $-a_y, a_x$ by at most $2h, 13h/4$. The corresponding anchor products differ from $-a_x a_y, a_x a_y$ by at most $4h, 21h/4$. Thus $|a_x a_y| \le 7h$ and the smaller coordinate has magnitude at most $28h$. Deleting that coordinate leaves a signed nonzero constant axis a_*, with

$$\|W - (a_*, \ldots, a_*)\|_\infty \le 29h < 1/400 < |a_*|/100.$$

Here a_* denotes the retained two-dimensional axis state, and $|a_*|$ its nonzero coordinate magnitude.

For completeness the relative $1/100$ local descent is explicit. Near a positive primal axis, $x_i, q_i > 0$, hence $p_i, y_i \ge 0$. At transitions $i = 2, 3, 4$, put $x = x_i$, $u = y_{i-1}$, $v = y_i$, and $t_0 = x_{i-1} - x_i \ge ru \ge 0$. Expansion yields

$$Q_r(z_{i-1}) - Q_r(z_i) \ge x(t_0/r + v/2), \qquad \|z_i - z_{i-1}\|^2 \le (1 + 2/r^2)t_0^2 + 2v^2.$$

Now $t_0/x < 2/99 < 1/4$, $v/x < 1/99 < 1/4$, and $r + 2/r \le 4$, which prove descent by at least the increment square. Near a positive dual axis put $X = -x_i$, $U = -x_{i-1}$, $Y = y_i$, $V = X + U$, $s_0 = q_i$, and

$$d_0 = y_i - y_{i-1} = -(9/8)X + (1/8)U - s_0.$$

Anchoring gives $X, U, s_0 \geq 0$. Retaining the term Ud_0 in the exact expansion gives

$$Q_r(z_i) - Q_r(z_{i-1}) \leq -Y[(3/4)V + 2s_0] + (13/8)V^2 + Vs_0.$$

The neighborhood implies $(V + s_0)/Y < 7/132 < 1/10$. Using $1/10 < 3/13$ and $1/10 < 24/145$ yields

$$Q_r(z_{i-1}) - Q_r(z_i) \geq (3/8)Y(V + s_0), \qquad \|z_i - z_{i-1}\|^2 \leq (145/64)(V + s_0)^2.$$

These prove the same descent. Negative axes follow by simultaneous reversal. Both endpoints are anchored outputs because $i \geq 2$. Summation gives

$$G_r(W) \geq B(W)/2 \geq 0. \tag{163}$$

8.4 The explicit compensation and absorption

If $|r - R| \leq \delta$ and $d(W) \leq \tau_D$ on the unit sphere, (155) gives $e \leq \delta$. Hence $\xi \leq 10^{-50}$ and $\lambda \leq 9 \cdot 10^{-25} < \lambda_0$. If $s < \beta$, (163) applies. Otherwise the normalized anchor defect in (159) is at most

$$20\lambda_0/\beta^2 = 2 \cdot 10^{-7} < 10^{-4}.$$

Equations (162) and (161) then give $G_r(W) \geq \beta^2/2 - 100\lambda_0 > 0$. Thus $G_r \geq 0$ whenever $d \leq \tau_D$. On the full unit sphere, $0 \leq Q_r \leq 3\|z\|^2/2$, $B \leq 4$, so $G_r \geq -7/2$. For $d > \tau_D$, the choice $C_H = 4/\tau_D$ gives $G_r + C_H d > 0$. Homogeneity, including $W = 0$, proves

$$Q_r(z_1) - Q_r(z_4) \geq \frac{1}{2}B(W) - C_H W^T D_0 W \quad (W \in \mathcal{F}_r,\ |r - R| \leq \delta). \tag{164}$$

This is a two-sided parameter neighborhood, an important fact for the later overlap construction.

For every $r > 0$, put $\varepsilon_r = 1 - R/r$ and

$$P_r = P_0 - \varepsilon_r(P_0)_x, \quad C_r = C_0 - \varepsilon_r(C_0)_x, \quad D_r = D_0 - \varepsilon_r(D_0)_x. \tag{165}$$

For every $r > 0$, keeping all numerator slack multipliers fixed scales exactly the primal/primal matrix blocks; both affine coefficients in the literal residual identities are reconstructed exactly. Therefore, for $r \geq R$ on actual frozen windows,

$$E_n^r \geq 0 \quad (\varepsilon_r K_C \leq 1/2), \qquad E_n^r - E_{n+1}^r \geq d(W_n) - \varepsilon_r K_D J(W_n). \tag{166}$$

A sufficient interval length is any positive number at most

$$\min\{\delta,\ 1/K_C,\ 1/(8C_H K_D),\ 1/1000\}. \tag{167}$$

Exact root signs give $2 + 1/1000 < R < 3 - 1/1000$ and $[R - \delta, R + \delta] \subset [2, 3]$. Exact rational comparison gives that the smallest positive integer N for which 10^{-N} satisfies every bound in (167) is $N = 118$. For this choice $\varepsilon_r \leq \eta_{\text{eff}}/2$ and

$$\varepsilon_r K_C \leq 1/2, \qquad t_r = C_H \varepsilon_r K_D \leq 1/16.$$

Proof of Theorem 9. The preceding compensation and continued storage satisfy precisely the summation hypotheses of the retained proof of Theorem 3. To exhibit complete constants, write $a_j = \|z_j - z_{j-1}\|^2$. Summing from $n = 1$ to $N \geq 3$, without dropping final increments, yields

$$\begin{aligned} \sum_{j=N+2}^{N+4} Q_r(z_j) + C_H E_{N+1}^r + \frac{1}{4}\sum_{j=3}^{N+4} a_j \\ \leq \sum_{j=2}^{4} Q_r(z_j) + C_H E_1^r + t_r a_2. \end{aligned} \tag{168}$$

The left side is nonnegative and $Q_r(z) \geq \|z\|^2/21$. Centered proximals fix zero; the literal map sends a state of maximum-coordinate norm R_0 to a state with coordinates bounded by $4R_0, 45R_0/8 < 6R_0$. Thus $\|z_j\|^2 \leq 2 \cdot 6^{2j} \|z_0\|^2$. Put $L_P = \|P_0\|_F + \|(P_0)_x\|_F$ and

$$A_0 = 3\sum_{j=2}^{4} 6^{2j} + 2C_H L_P \sum_{j=1}^{4} 6^{2j} + (6^4 + 6^2)/4.$$

The right side of (168) is at most $A_0 \|z_0\|^2$. Valid constants, including the initial prefix, are

$$B_0^2 = \max\{2 \cdot 6^8, 21A_0\}, \qquad H_0 = 4A_0 + 4[(6^2 + 1) + (6^4 + 6^2)]. \tag{169}$$

Consequently the orbit is bounded and its first differences are square summable. Continuity of the fixed proximal map makes every cluster point fixed, hence KKT. Restart the bound with constant B_0 at a subsequence converging to such a point; its right side tends to zero, proving convergence of the whole orbit. This uses no uniqueness or implication from cycle exclusion. The same constants apply at every KKT anchor and restart. The matched-separable consequence follows from the already retained proof with $B_{\text{cont}} = B_0$. ∎

9 A strict finite-shadow state Lyapunov function

Fix the two scalar objectives and a KKT anchor $a = (x_*, y_*)$. The centered functions

$$f_a(u) = f(x_* + u) - f(x_*) + y_* u, \qquad h_a(v) = h(y_* + v) - h(y_*) - x_* v$$

have zero in their subdifferentials at zero. Direct proximal optimality shows that the centered literal map has exactly the formula

$$T_r(x, y) = (u, v), \qquad u = \operatorname{prox}_{rf_a}(x - ry), \quad v = \operatorname{prox}_{h_a}\big(y + (9u - x)/8\big). \tag{170}$$

The anchor is fixed for every $r > 0$. All norms below are Euclidean, and constants do not depend on the objectives or anchor.

Theorem 10 (Finite-shadow energy on the high interval)**.** *For $r \in I_H = [R, R + \eta_{\text{eff}}]$, define genuine frozen shadow states $z^{(j)} = T_r^j z$ and $H_r(z) = (z^{(0)}, \dots, z^{(3)})$. Then*

$$V_r^H(z) = \sum_{j=1}^{3} Q_r(z^{(j)}) + C_H E_r(H_r(z)) + \frac{1}{8} \|T_r z - z\|^2, \qquad E_r(h) = h^T P_r h, \tag{171}$$

satisfies, for all states,

$$\frac{1}{29} \|z\|^2 \le V_r^H(z) \le U_H \|z\|^2, \qquad V_r^H(z) - V_r^H(T_r z) \ge \frac{1}{16} \|T_r z - z\|^2. \tag{172}$$

There are explicit finite constants U_H, L_H given below such that

$$|V_r^H(z) - V_s^H(z)| \le L_H |r - s| \, \|z\|^2, \tag{173}$$

$$|V_r^H(z) - V_r^H(w)| \le U_H (\|z\| + \|w\|) \, \|z - w\|. \tag{174}$$

The energy is generally nonlinear in the state z; it is a quadratic expression in its genuine finite shadow history. Its zero-decrease set is exactly the fixed-point set.

Proof. Every term in (171) is nonnegative by (166). In particular, with $u = T_r z$,

$$V_r^H(z) \ge \frac{\|u\|^2}{21} + \frac{\|u - z\|^2}{8} \ge \frac{\|z\|^2}{29},$$

where the difference in the last inequality equals $\frac{29}{168} \|u - (21/29)z\|^2$. Let $a_i = \left\|z^{(i)} - z^{(i-1)}\right\|^2$. The three Q terms telescope under one frozen step. Combining (164), (166), and the correction $(a_1 - a_2)/8$ gives

$$\begin{aligned} V_r^H(z) - V_r^H(T_r z) &\ge (1/8 - t_r) a_1 + (3/8 - t_r) a_2 + (1/2 - t_r)(a_3 + a_4) \\ &\ge a_1/16 + (5/16) a_2 + (7/16)(a_3 + a_4). \end{aligned} \tag{175}$$

This proves strict descent at every nonfixed state. At a fixed state all shadows are identical, so the decrease is zero. No terminal negative energy is left for the subsequent product argument.

For continuity, both centered proximal maps fix zero and are nonexpansive. For $r \in [2, 3]$, the primal output has state Lipschitz constant at most four and the dual output at most six. Thus

$$\|T_r z - T_r w\| \le 10 \|z - w\|, \qquad \|T_r z\| \le 10 \|z\|. \tag{176}$$

If $u_s = \text{prox}_{s f_a}(x - sy)$ and $p_s = (x - u_s)/s - y \in \partial f_a(u_s)$, then $u_s = \text{prox}_{r f_a}(u_s + r p_s)$. Therefore

$$|u_r - u_s| \le |1 - r/s| \, |x - u_s| \le (5/2) |r - s| \, \|z\|.$$

The dual difference is at most $9/8$ times this quantity. In particular

$$\|T_r z - T_s z\| \le (85/16) |r - s| \, \|z\| \le 6 |r - s| \, \|z\|. \tag{177}$$

These literal one-step estimates do not infer any two-orbit long-time Lipschitz bound from anchored stability.

Put $R_H = 1 + 10 + 100 + 1000 = 1111$, $S'_H = 6 + 120 + 1800 = 1926$, and $M_P = 16$. Writing $(P_0)_{ij} = a_{ij} + b_{ij}\sqrt{91}$, the exact raw coefficient sum is

$$1 + 2\sum_{i,j}(|a_{ij}| + 10|b_{ij}|) = \frac{26247397032610153}{1680534540000000} < 16. \tag{178}$$

Since $\sqrt{91} < 10$ and $|\varepsilon_r - \varepsilon_s| \le 3|r-s|/4$, this proves

$$\|P_r\|_{\mathrm{op}} \le M_P, \qquad \|P_r - P_s\|_{\mathrm{op}} \le M_P|r-s|.$$

Induction in (176)–(177) gives, for $1 \le j \le 3$,

$$\left\|T_r^j z\right\| \le 10^j \|z\|, \qquad \left\|T_r^j z - T_s^j z\right\| \le c_j|r-s| \, \|z\|, \qquad c_j = 6j10^{j-1}.$$

Consequently the history norm is at most $R_H \|z\|$, its parameter difference at most $S'_H|r-s| \, \|z\|$, and its state Lipschitz constant at most R_H. Define

$$\begin{aligned} U_H &= \frac{3}{2}\sum_{j=1}^{3} 10^{2j} + C_H M_P R_H^2 + \frac{121}{8}, \\ L_H &= \sum_{j=1}^{3}\left(3 \cdot 10^j c_j + \frac{10^{2j}}{4}\right) + C_H M_P(2R_H S'_H + R_H^2) + \frac{33}{2}. \end{aligned} \tag{179}$$

The matrix of Q_r has norm at most $3/2$ and parameter difference norm at most $|r-s|/4$. Its jth shadow term therefore costs the corresponding summand of L_H. Adding and subtracting $H_s(z)^T P_r H_s(z)$ gives storage cost at most $C_H M_P(2R_H S'_H + R_H^2)|r-s| \, \|z\|^2$. The increment vectors have norm at most $11 \|z\|$ and difference at most $6|r-s| \, \|z\|$, giving $33/2$ for the last term. This proves (173) and the upper bound. The same expansion at fixed r, using history state Lipschitz constant R_H and increment Lipschitz constant 11, proves (174) with the displayed U_H. ∎

The finite-shadow construction applies even when the input state is outside the objective domains. Only its exact proximal outputs need graph inclusions. A mixed-parameter actual history is never substituted into the frozen identities. From coercivity and (174) one also obtains

$$\left|\sqrt{V_r^H(z)} - \sqrt{V_r^H(w)}\right| \le \sqrt{29}\, U_H \, \|z - w\|, \tag{180}$$

by dividing by $\sqrt{V_r^H(z)} + \sqrt{V_r^H(w)}$; the case $z = w = 0$ is immediate.

10 Subcritical compact scalar bands

The high energy has explicit constants. To cover any fixed positive compact lower range we also need a subcritical construction; its constants may depend on the positive lower endpoint. Fix $0 < m < R$ and $I_L = [m, R]$, retaining (151), $\mathcal{F}_r$, and (165). Set $D_x = (D_0)_x$, $d_r(W) = W^T D_r W$, and retain B, G_r from (160).

Lemma 11 (Compact subcritical compensation)**.** *There is a finite $C_L > 0$, depending only on m and the fixed coefficient tables, such that*

$$Q_r(z_1) - Q_r(z_4) \geq B(W)/2 - C_L d_r(W) \quad (r \in [m, R],\ W \in \mathcal{F}_r). \tag{181}$$

No uniformity as m tends to zero is asserted.

Proof. For $r < R$, writing $t = R/r - 1 > 0$ gives

$$D_r = D_0 + tD_x \succeq 0, \qquad \ker D_r = \ker D_0 \cap \ker D_x. \tag{182}$$

Both summands are positive semidefinite. Exact reconstruction gives

$$\operatorname{rank} D_x = 3, \qquad \operatorname{rank} \begin{pmatrix} D_0 \\ D_x \end{pmatrix} = 6.$$

The ten-by-four matrix with paired rows $[I_2, iI_2]$, $0 \leq i \leq 4$, has rank four and is annihilated by both matrices. Thus the intersection is exactly the affine ramps $z_i = a + ib$. Equivalently, on the two-column cycle matrix with paired rows A^i, the product with D_x has rank two, so the cycle part is eliminated exactly.

For a ramp the primal and dual order products are $-(i-j)^2 b_x b_y$ and $(i-j)^2 b_x b_y$. Feasibility forces $b_x b_y = 0$. At $r = R$ the retained four-output scalar rigidity gives the same axis-ramp conclusion despite the endpoint rank jump. Thus, throughout I_L,

$$W \in \mathcal{F}_r,\ d_r(W) = 0 \quad \Longrightarrow \quad W \text{ is an axis ramp.} \tag{183}$$

For an anchored primal ramp with positive speed s, write $y_i = sv$, $x_i = sw_i$, $w_4 - w_1 = 3$. The exact anchors give $-1/r \leq v \leq 0$, with $w_4 \leq 0$ at $v = 0$ and $w_4 \leq -1/8$ at $v < 0$. For a dual ramp they give $0 \leq v \leq 1$, with $w_4 \leq 0$ at $v < 1$ and $w_4 \leq 1$ at $v = 1$. The excluded ranges either force contradictory signs or confine four points spanning length three to an interval of length at most one. Direct summation, including these boundary cases, gives

$$Q_r(z_1) - Q_r(z_4) \geq \min\{27/(4r), 6\} \|b\|^2 .$$

Consequently every nonconstant feasible zero-kernel window satisfies

$$G_r(W) \geq \left(\frac{27}{4R} - \frac{3}{2} \right) \|b\|^2 > 0. \tag{184}$$

A constant feasible ramp has $a_x a_y = 0$, so the only remaining zero-margin histories are constant axes.

We give neighborhoods uniform in $r \in [m, R]$, rather than assume a spectral gap near the endpoint. Define

$$\begin{gathered} A_m = 1/m + 9/8, \qquad e_p = \min\{1/4, 1/(3 + 2/m)\}, \\ e_d = \min\{1, 3/(8A_m), 24/145\}, \\ \rho = \min\{1/100, e_p/4, 1/[4(1 + 2/m)], e_d/8\} > 0. \end{gathered} \tag{185}$$

Suppose every coordinate of W is within ρa of a repeated signed axis of magnitude $a > 0$. Near the positive primal axis, $x_i > (1-\rho)a$ and $|q_i - a| < (13/4)\rho a$, so $q_i > 0$. Anchors imply $y_i, p_i \geq 0$. For $i = 2, 3, 4$, put $x = x_i$, $u = y_{i-1}$, $v = y_i$, $t = x_{i-1} - x_i \geq ru \geq 0$. The exact expansion gives

$$Q_r(z_i) - Q_r(z_{i-1}) \leq -xt/r - v(x - v).$$

The upper bound minus the exact expression is $(x + t)(t/r - u) + u^2 \geq 0$. Further,

$$t/x < 2\rho/(1-\rho) \leq e_p, \qquad v/x < \rho/(1-\rho) \leq e_p \leq 1/4.$$

Since $e_p(r + 2/r) \leq 1$,

$$\|z_i - z_{i-1}\|^2 \leq (1 + 2/r^2)t^2 + 2v^2 \leq x(t/r + v/2) \leq Q_r(z_{i-1}) - Q_r(z_i).$$

Near the positive dual axis, $y_i > (1-\rho)a$ and $|p_i + a| < \rho a(1 + 2/m) \leq a/4$. Thus $x_i \leq 0, q_i \geq 0$. Put $X = -x_i$, $U = -x_{i-1}$, $Y = y_i$, $V = X + U$, $s = q_i$, and $d = y_i - y_{i-1} = -(9/8)X + (1/8)U - s$. The exact expansion is

$$\begin{aligned} Q_r(z_i) - Q_r(z_{i-1}) &= (X^2 - U^2)/r + Y[(X - U) + 2d] + Ud - d^2 \\ &\leq -Y[(3/4)V + 2s] + A_m V^2 + Vs. \end{aligned}$$

Now $V < 2\rho a$, $s < (13/4)\rho a$, and $(V + s)/Y < 7\rho \leq e_d$. Hence $A_m V^2 \leq (3/8)YV$ and $Vs \leq Ys$, while

$$\|z_i - z_{i-1}\|^2 \leq (145/64)(V + s)^2 \leq (3/8)Y(V + s).$$

This again gives descent by the increment square. Negative axes follow by simultaneous sign reversal. Only output-to-output transitions were used. In all these neighborhoods

$$G_r(W) \geq B(W)/2 \geq 0. \tag{186}$$

On the unit sphere in $\mathbb{R}^{10}$, let $\mathcal{U}$ be the union of the four open coordinate neighborhoods of the repeated normalized axis states $(\pm 1/\sqrt{5}, 0)$ and $(0, \pm 1/\sqrt{5})$, each of radius $\rho/\sqrt{5}$. This set is independent of r. The joint set

$$\mathcal{K}_0 = \{(r, W) : r \in [m, R],\ \|W\| = 1,\ W \in \mathcal{F}_r,\ W \notin \mathcal{U},\ G_r(W) \leq 0\}$$

is compact. It contains no point with $d_r(W) = 0$, by (183)–(184); the constant axes are inside $\mathcal{U}$. Put $\tau = \min_{\mathcal{K}_0} d_r > 0$ if the set is nonempty and $\tau = 1$ otherwise. With

$$q_+ = 1/m + 3/2, \qquad C_L = \max\{1, (q_+ + 2)/\tau\},$$

we have $0 \leq Q_r(z) \leq q_+ \|z\|^2$ and $G_r \geq -(q_+ + 2)$ on the sphere. If $G_r < 0$, the local lemma excludes $\mathcal{U}$, so compensation follows from $d_r \geq \tau$. If $G_r \geq 0$, it follows from $D_r \succeq 0$. Homogeneity proves (181), including the zero history. ∎

Theorem 11 (Subcritical finite-shadow energy)**.** *On* $[m, R]$ *the energy*

$$V_r^L(z) = \sum_{j=1}^{3} Q_r(T_r^j z) + C_L E_r(H_r(z)) + \frac{1}{8} \|T_r z - z\|^2 \tag{187}$$

has constants $U_L, S_L, L_L < \infty$*, depending only on* m *and the coefficient tables, such that*

$$\|z\|^2 /42 \le V_r^L(z) \le U_L \|z\|^2 , \tag{188}$$

$$V_r^L(z) - V_r^L(T_r z) \ge \|T_r z - z\|^2 /8, \tag{189}$$

$$|V_r^L(z) - V_r^L(w)| \le S_L(\|z\| + \|w\|) \|z - w\| , \tag{190}$$

$$|V_r^L(z) - V_s^L(z)| \le L_L |r - s| \|z\|^2 . \tag{191}$$

These statements hold after centering at any KKT anchor and for arbitrary raw inputs.

Proof. The exact identities, with output-only supports, give

$$E_r(H_r(z)) \ge H_r(z)^T C_r H_r(z) \ge 0, \qquad E_r(h_0) - E_r(h_1) \ge d_r(W),$$

because $C_r = C_0 + (R/r - 1)(C_0)_x \succeq 0$. Subtract the two energies and use (181). The full drop is

$$V_r^L(z) - V_r^L(T_r z) \ge a_1/8 + 3a_2/8 + a_3/2 + a_4/2.$$

Since $r \le R < 3$, Young's inequality gives $Q_r(u) \ge \|u\|^2 /21$. Retaining this term at $u = T_r z$ and the residual correction gives the stated weaker coercivity $1/42$ (indeed $1/29$ also follows from completion of squares).

The literal state Lipschitz constant can be taken as $L_T = 7$, since the primal and dual constants are 4 and $45/8$, and $\sqrt{4^2 + (45/8)^2} < 7$. The resolvent identity used above, now with $r, s \ge m$, gives

$$\|T_r z - T_s w\| \le L_T \|z - w\| + H_T |r - s| \|w\| , \qquad H_T = 8/m.$$

Induction yields

$$\left\|T_r^j z - T_s^j w\right\| \le L_T^j \|z - w\| + j H_T L_T^{j-1} |r - s| \|w\| . \tag{192}$$

The Q matrices have norm at most q_+ and parameter Lipschitz constant $1/m^2$. The P matrices have bounds

$$P_{\max} = \|P_0\|_{\mathrm{op}} + (R/m - 1) \|(P_0)_x\|_{\mathrm{op}} , \qquad R \|(P_0)_x\|_{\mathrm{op}} /m^2$$

for their norm and parameter Lipschitz constant respectively. A valid upper constant is

$$U_L = q_+ \sum_{j=1}^{3} L_T^{2j} + C_L P_{\max} \sum_{j=0}^{3} L_T^{2j} + (L_T + 1)^2/8.$$

Apply, term by term, the elementary quadratic estimate

$$|u^T A u - v^T B v| \le \|A\| (\|u\| + \|v\|) \|u - v\| + \|A - B\| \|v\|^2$$

and (192). This provides finite S_L, L_L for all states and parameters on the displayed compact interval. It requires no differentiability of either objective and no finite objective values at inputs. ∎

11 One energy across the entire compact band

Separate lower and upper bounds do not by themselves permit a parameter path to cross their boundary repeatedly. We now establish an actual overlap and a Lipschitz partition. Fix

$$0 < m \le R + \eta_{\mathrm{eff}}, \qquad J = [m, R + \eta_{\mathrm{eff}}].$$

The positive upper extension is unchanged as m varies.

Lemma 12 (Left extension of the high energy)**.** *The same V_r^H, U_H, and L_H from Theorem 10 satisfy its conclusions on*

$$J_H = [R - \delta, R + \eta_{\mathrm{eff}}]. \tag{193}$$

Proof. The compensation (164) is valid for $|r - R| \le \delta$. For $r < R$, $\varepsilon_r < 0$, and

$$C_r = C_0 + |\varepsilon_r|(C_0)_x \succeq 0, \qquad D_r = D_0 + |\varepsilon_r|(D_0)_x \succeq D_0.$$

Thus the shadow storage is nonnegative and its drop is at least $W^T D_0 W$. For $r \ge R$ use (166). These two cases give (175) with

$$t_r = \max\{0, C_H \varepsilon_r K_D\} \le 1/16.$$

In particular we do not multiply the upper bound on $(D_0)_x$ by a negative coefficient and reverse its role. Coercivity follows as before. Since $J_H \subset [2, 3]$, $|\varepsilon_r| < 1$, and $|\varepsilon_r - \varepsilon_s| \le 3|r - s|/4$, all literal-map and matrix estimates in the high proof, and hence its constants, remain valid. ∎

Put

$$\rho_0 = R - 3\delta/4, \qquad \rho_1 = R - \delta/4, \qquad \chi(r) = \begin{cases} 0, & r \le \rho_0, \\ (r - \rho_0)/(\rho_1 - \rho_0), & \rho_0 \le r \le \rho_1, \\ 1, & r \ge \rho_1. \end{cases} \tag{194}$$

Both joins lie strictly in $(R - \delta, R)$, and χ has Lipschitz constant $2/\delta$. If $m \ge \rho_1$, use $V_r = V_r^H$ throughout J. If $m < \rho_1$, invoke the lower construction on $[m, R]$ and set

$$V_r = \begin{cases} V_r^L, & m \le r \le \rho_0, \\ (1 - \chi(r))V_r^L + \chi(r)V_r^H, & \max\{m, \rho_0\} \le r \le \rho_1, \\ V_r^H, & \rho_1 \le r \le R + \eta_{\mathrm{eff}}. \end{cases} \tag{195}$$

Empty branches are omitted. The two energies are evaluated together only on their common domain, and their weights at the joins give continuity without requiring that the two formulas agree.

Theorem 12 (Uniform positive energy on a compact band)**.** *At every KKT anchor the energy* (195)*, or its high-only branch, has constants $U, S, L < \infty$ depending only on m and the fixed tables such that, for every $r, s \in J$ and every centered z, w,*

$$a \|z\|^2 \le V_r(z) \le U \|z\|^2, \qquad a = 1/42, \tag{196}$$

$$V_r(z) - V_r(T_r z) \ge d \|T_r z - z\|^2, \qquad d = 1/16, \tag{197}$$

$$|V_r(z) - V_r(w)| \le S(\|z\| + \|w\|) \|z - w\|, \tag{198}$$

$$|V_r(z) - V_s(z)| \le L|r - s| \|z\|^2. \tag{199}$$

Proof. In the mixed case take

$$U = \max\{U_L, U_H\}, \quad S = \max\{S_L, U_H\}, \quad L = \max\{L_L, L_H\} + (2/\delta)(U_L + U_H). \tag{200}$$

Coercivity, upper bounds, state continuity and drop follow by nonnegative convex combination at the same fixed r and the same map T_r. Inside the overlap, add and subtract the energies at s with weights evaluated at r. The energy variation costs $\max\{L_L, L_H\}|r - s| \|z\|^2$, and the weight variation costs $(2/\delta)(U_L + U_H)|r - s| \|z\|^2$. On an unmixed branch use its own estimate. For parameters in different branches insert the intervening joins; the subinterval lengths sum to $|r - s|$. This proves global parameter continuity without evaluating any family beyond its valid domain. In the high-only case take $U = S = U_H$, $L = L_H$ and the same weaker a, d. ∎

The compact construction is an existence construction through C_L, although η_{eff} and the high constants are explicit. It makes no uniform assertion as $m \downarrow 0$.

12 Finite parameter variation and summable state errors

We prove the scalar robustness theorem stated at the start of the article. The argument first establishes boundedness using an everywhere nonnegative energy, then proves dissipation and identifies a KKT cluster point. This order avoids using bounded coefficients before boundedness has been proved.

12.1 A positive-energy transmission argument

Consider a finite-dimensional family of maps on a compact parameter interval. Suppose each map fixes an anchor, the maps are uniformly state Lipschitz with constant M_T and jointly continuous, and an anchor-centered energy satisfies (196)–(199) with $a, d > 0$. Suppose also that every fixed point of any map is a common fixed point and that the same estimates hold at each such anchor. For the actual recursion

$$z_n = T_{r_n} z_{n-1} + b_n, \qquad D = \sum_{n \ge 1} |r_{n+1} - r_n| < \infty, \quad E = \sum_{n \ge 1} \|b_n\| < \infty, \tag{201}$$

we give the full convergence argument.

Coercivity and state comparison imply

$$|\sqrt{V_r(z)} - \sqrt{V_r(w)}| \le K \|z - w\|, \qquad K = S/\sqrt{a}. \tag{202}$$

Divide the quadratic comparison by the sum of square roots, which is at least $\sqrt{a}(\|z\| + \|w\|)$; if both states vanish there is no division. Put

$$\begin{gathered} q_n = |r_{n+1} - r_n|, \quad g_n = 1 + (L/a)q_n, \quad \Pi_0 = 1, \quad \Pi_n = \prod_{j=1}^{n} g_j, \\ \Pi_\infty \le \exp((L/a)D), \qquad F_n = V_{r_{n+1}}(z_n), \quad F_0 = V_{r_1}(z_0), \quad v_n = T_{r_n} z_{n-1}. \end{gathered} \tag{203}$$

The definition of F_n is for $n \ge 1$, with the separately stated initial value. Parameter comparison at the same actual state gives $V_{r_{n+1}}(z_n) \le g_n V_{r_n}(z_n)$. Apply (202) from v_n to z_n, and then fixed-parameter descent, to obtain

$$\sqrt{F_n} \le \sqrt{g_n}\big[\sqrt{F_{n-1}} + K \|b_n\|\big]. \tag{204}$$

Dividing by $\sqrt{\Pi_n}$ and telescoping proves

$$\begin{aligned} \sqrt{F_n} &\le \sqrt{\Pi_n}\left[\sqrt{F_0} + K \sum_{j=1}^{n} \frac{\|b_j\|}{\sqrt{\Pi_{j-1}}}\right] \\ &\le \sqrt{\Pi_\infty}\,[\sqrt{U}\,\|z_0\| + KE], \\ \sup_n \|z_n\| &\le B := a^{-1/2}\sqrt{\Pi_\infty}\,[\sqrt{U}\,\|z_0\| + KE]. \end{aligned} \tag{205}$$

All actual states are therefore bounded before we use an error term with a bounded state coefficient. The exact outputs have norm at most $M_T B$. State continuity now bounds the potential error from v_n to z_n by $S(1 + M_T)B\|b_n\|$. Descent and parameter comparison give

$$F_n \le g_n \left[F_{n-1} - d\|v_n - z_{n-1}\|^2 + S(1 + M_T)B\|b_n\|\right]. \tag{206}$$

The bracket bounds the nonnegative quantity $V_{r_n}(z_n)$, so there is no unproved sign assumption. Divide by Π_n and telescope. Since $F_N \ge 0$ and $\Pi_{n-1} \le \Pi_\infty$,

$$\sum_{n\ge1} \|v_n - z_{n-1}\|^2 \le \frac{\Pi_\infty}{d}[U\|z_0\|^2 + S(1 + M_T)BE]. \tag{207}$$

Absolute summability implies $\sum \|b_n\|^2 \le E^2$. Hence the actual increments satisfy

$$\sum_{n\ge1} \|z_n - z_{n-1}\|^2 \le \frac{2\Pi_\infty}{d}[U\|z_0\|^2 + S(1 + M_T)BE] + 2E^2 < \infty. \tag{208}$$

For $E = 0$ the factor two is unnecessary. These are anchored estimates, not arbitrary two-trajectory Lipschitz estimates.

Finite total variation gives $r_n \to r_\infty$ in the closed interval. Choose a bounded subsequence $z_{n_j} \to z_*$. Equation (208) gives $z_{n_j-1} \to z_*$ as well, while $b_{n_j} \to 0$. Joint continuity in the actual recursion yields $z_* = T_{r_\infty} z_*$, hence a common fixed point. Center the energy at this particular point and restart (205) at any k:

$$\sup_{n \geq k} \|z_n - z_*\| \leq a^{-1/2} e^{LD/(2a)} \left[\sqrt{U}\, \|z_k - z_*\| + K \sum_{j>k} \|b_j\| \right]. \tag{209}$$

It includes $n = k$ because $a \leq U$. Along the convergent subsequence both terms on the right tend to zero. Thus the entire orbit converges to z_*. No uniqueness, Euclidean Fejer monotonicity, or summability of $|r_n - r_\infty|$ has been assumed.

12.2 Application to the scalar proximal map

Proof of Theorem 4. For every $r > 0$, the fixed points of the literal map are precisely the KKT pairs, since

$$x = \operatorname{prox}_{rf}(x - ry) \iff -y \in \partial f(x), \qquad y = \operatorname{prox}_h(y + x) \iff x \in \partial h(y).$$

Center at any such pair using the fixed supporting-affine translations at the start of Section 9. On J, $r < 3$, so the direct state estimate has constant $M_T = 10$. For $r, s \geq m$, the same resolvent identity as before gives

$$|u_r - u_s| \leq |r - s|\, |x - u_s|/s \leq (5/m)|r - s|\, \|z\|\,.$$

The dual difference is at most $9/8$ times the primal difference. Therefore

$$\|T_r z - T_s z\| \leq \frac{5\sqrt{145}}{8m} |r - s|\, \|z\| \leq \frac{8}{m} |r - s|\, \|z\|\,. \tag{210}$$

Together with the state estimate this proves joint continuity. Theorem 12 supplies all positive-energy hypotheses. The transmission argument proves whole-orbit KKT convergence and actual-increment square summability.

An error may put the actual x_n outside $\operatorname{dom} f$ or y_n outside $\operatorname{dom} h$, as can happen for indicator objectives. Nothing requires their graph feasibility. At such a raw state the potential is defined by genuine exact frozen shadows; all objective values and graph inclusions used by the identities concern those exact outputs. Even in the error-free variable-parameter case, only the first frozen output equals the next actual state. Later mixed-parameter outputs are never used as a single frozen history. ∎

Proof of Corollary 1.1. Let $(u_n, v_n) = T_{r_n}(z_{n-1})$ be the exact full output at the actual previous state. The first displayed update in the corollary gives $x_n - u_n = e_n^x$. Nonexpansiveness of the second proximal map gives

$$\left| \operatorname{prox}_h \left(y_{n-1} + \frac{9x_n - x_{n-1}}{8} \right) - v_n \right| \leq \frac{9}{8} |e_n^x|.$$

Writing $b_n = z_n - (u_n, v_n)$ and using the Euclidean triangle inequality therefore gives

$$\|b_n\| \le \frac{\sqrt{145}}{8}|e_n^x| + |e_n^y|. \tag{211}$$

The full-state errors are absolutely summable. Apply Theorem 4. This reduction uses the actual perturbed x_n in the dual extrapolation, rather than silently replacing it by the exact primal output. ∎

13 Scope and exact verification

The all-real-θ trapping theorem is conditional on bounded increments. Its no-KKT endpoint increment input is valid on arbitrary Hilbert spaces, but the position barriers use scalar order. The Hilbert strong-convergence theorems require both objectives to separate into matched scalar coordinates. Diagonalizing a coupling alone does not separate arbitrary proximal maps. Thus no nonseparable finite-dimensional, general Hilbert, unrelated compact-coupling, or coupled-block convergence theorem follows.

The original endpoint kernel proof itself supplies convergence and vanishing increments, without a rate or square-sum conclusion. Theorem 3 adds uniform first-increment square summability for the normalized scalar map on its right neighborhood, including the endpoint; Theorem 9 supplies the concrete length 10^{-118}. Theorem 4 further proves actual-increment square summability on every compact positive scalar band under its parameter-variation and error assumptions. Corollary 7.3 extends only the same matched scalar-separable Hilbert model; it makes no new full-Hilbert increment-square-sum or rate assertion. The constants 10^{20} and 10^{10} remain valid on the original range, while the extension uses $C_{\mathrm{env}} = \max\{10^{20}, 3B_{\mathrm{cont}}^2\}$ and $\sqrt{C_{\mathrm{env}}}$. Diagonalizing a general coupling cannot make nonseparable objectives separable.

At $\theta = 1/8$, convergence throughout $(r_*, 3)$ remains unresolved. Theorem 7 concerns period three for all $\theta \ge 0$ and has no negative-θ conclusion. Theorem 8 rules out every nonconstant orbit whose period divides six on $[297/100, 3)$; the four- and five-period bridges imply least period at least seven there. These statements do not exclude all longer cycles, bounded aperiodic dynamics or escape. The witness at $r = 3$ shows a sharp open endpoint for period-dividing-six exclusion, through its least period three; it does not identify a least-period-six onset.

The four original standard-library checkers reconstruct endpoint support and positivity, both fixed storages and their explicit constants, the endpoint recurrence and Appendix A, and the full period-three classification and witnesses. The added standard-library checkers reconstruct the continued storage, four-output Farkas identities, all six-period interval certificates and both proximal normalizations. The narrow five-period bridge and independent cross-checks in the new robustness suite use the declared SymPy

dependency. The new suite also reconstructs the quantitative coefficient chain, start-up supports, finite-shadow coefficients, subcritical common kernel and compact partition algebra, with negative controls. These are exact, solver-free replays: no saved result, numerical eigenvalue or discovery trajectory is a proof input. The manuscript supplies the analytic domain, local geometry, compactness, exhaustive-order and infinite-orbit arguments.

The public `RESULT_MAP.md` and `SEMANTIC_LABEL_MAP.json` locate the retained and added results. Historical `SOURCE_PINS.json` entries remain historical identifiers; `PUBLIC_INPUTS.json` and `CURRENT_SOURCE_DEPENDENCIES.json` describe the current release-local mathematical closure. No earlier PDF or companion source tree is needed. Run `bash verify.sh` for the exact suite and `bash build.sh` for the manuscript; the source manifest covers every other public member. The A4 article style, blank identity fields and date, Keywords, MSC, and absence of page numbers are preserved.

A Specialized critical cycle identities

For independent finite verification, we retain the specialized identities underlying the endpoint exclusion. Their conclusion is subsumed by Theorem 7, and the main convergence proof does not depend on this second exclusion argument.

Consider an actual three-periodic orbit of (113) with the form in Lemma 6. Its constant center cancels from differences of subgradients. Every centered cycle can be parameterized by real a, b as

$$\begin{aligned} &(c_0, c_1, c_2) = (1, -1/2, -1/2), \qquad (s_0, s_1, s_2) = (0, 1/2, -1/2), \\ &X_i = ac_i + bs_i, \\ &Y_i = a\left(\frac{9}{16}c_i + \frac{3s_i}{2r}\right) + b\left(\frac{9}{16}s_i - \frac{c_i}{2r}\right). \end{aligned} \tag{212}$$

The map $(a, b) \mapsto (X_0, Y_0)$ is invertible, and (212) satisfies $z_{i+1} = \mathsf{A}z_i$, including the wrap from phase two to phase zero.

Put $\omega = 10 - \sqrt{91} > 0$. Direct substitution into (114), with cyclic indices, gives

$$\begin{aligned} (p_0 - p_1)(X_0 - X_1) &= \frac{\omega}{32}(9a^2 - b^2), \\ (p_1 - p_2)(X_1 - X_2) &= \frac{\omega}{16}b(b - 3a), \\ (p_2 - p_0)(X_2 - X_0) &= \frac{\omega}{16}b(b + 3a). \end{aligned} \tag{213}$$

All three quantities must be nonnegative by monotonicity of the scalar subdifferential. The first gives $|b| \le 3|a|$. The last two give $b^2 \ge 3|ab|$, so either $b = 0$ or $|b| \ge 3|a|$. Therefore

$$b = 0, \qquad b = 3a, \qquad \text{or} \qquad b = -3a. \tag{214}$$

A one-step time shift sends

$$(a, b) \longmapsto \left(\frac{-a+b}{2}, \frac{-3a-b}{2} \right).$$

If $b = -3a$, one shift gives $(-2a, 0)$; if $b = 3a$, two shifts give $(-2a, 0)$. Thus a nonzero cycle may be shifted to have $b = 0$ and $a \neq 0$. But at that phase the dual monotonicity pairing is

$$(q_1 - q_2)(Y_1 - Y_2) = \frac{9(9 - \sqrt{91})}{32r} a^2 < 0, \tag{215}$$

contradicting $q_i \in \partial B(Y_i)$. Hence the cycle amplitude is zero.

The scalar convergence checker verifies each coefficient in (213), the phase-shift identities, and (215). The sign reduction from the three inequalities is the analytic argument above. All graph points in the cyclic calculation are actual positive-index outputs: the graph point at phase zero can be taken from index three of the already periodic orbit. No subgradient at an arbitrary initialization is required.

B Endpoint matrices and four-output order

This appendix specifies the finite proof data used by the scalar continuation, including the output-index change in the order certificates. The matrices use interleaved state slots; the multiplier arrays use primal-species edges followed by dual-species edges. Keeping these two conventions distinct is necessary for direct reconstruction.

B.1 Reconstructing the fixed numerator matrices

Work in the ordered field $\mathbb{Q}(\sqrt{91})$, with the positive square root. A serialized pair $[a, b]$ in `exact_certificate.json` means $a + b\sqrt{91}$ with rational entries. The field element $r_* = (88 + 16\sqrt{91})/81$ satisfies

$$81 r_*^2 - 176 r_* - 192 = 0, \qquad 2 < r_* < 3.$$

The 36 entries of the constant array `P_numerator[0]` fill the upper triangle of P_0 in lexicographic order $0 \le i \le j < 8$, and symmetry fills the lower triangle. This is exactly $r_* P(0)$, not $P(0)$ itself. The constant arrays of `lambda` and `mu` are multiplied by r_* once to give $\widehat{\lambda}$ and $\widehat{\mu}$.

Let $e_{i,s} \in \mathbb{R}^{10}$ select coordinate $2i + s$ in zero-based indexing, $0 \le i \le 4$, $s \in \{0, 1\}$, so that $s = 0$ means primal and $s = 1$ means dual. Set $e_{\varnothing,s} = 0$. The residual coefficient vectors at $1 \le i \le 4$ are

$$\begin{aligned} g_{i,0}(\alpha) &= \frac{\alpha}{r_*}(e_{i-1,0} - e_{i,0}) - e_{i-1,1}, \\ g_{i,1}(\alpha) &= e_{i-1,1} - e_{i,1} + \frac{9}{8} e_{i,0} - \frac{1}{8} e_{i-1,0}, \qquad g_{\varnothing,s} = 0. \end{aligned} \tag{216}$$

At $\alpha = r_*/r$, these are the actual residuals of (131). For an edge $e = (i, j, s)$ put $\delta_e = e_{j,s} - e_{i,s}$ and

$$Q_e(\alpha) = -\frac{1}{2}\left(g_{i,s}(\alpha)\delta_e^{\mathsf{T}} + \delta_e g_{i,s}(\alpha)^{\mathsf{T}}\right). \tag{217}$$

Then $S_e = W^{\mathsf{T}} Q_e(\alpha) W + v_e^{\mathsf{T}}\phi$, where v_e adds the value at (j, s) and subtracts that at (i, s), with the anchor value zero. No $g_{0,s}$ is defined.

For each species, list the 25 descent edges as

$$(\varnothing, j)\ (j = 0, \dots, 4), \quad (i, \varnothing)\ (i = 1, \dots, 4), \quad (i, j)\ (i = 1, \dots, 4;\ j = 0, \dots, 4;\ j \neq i),$$

with the last group lexicographic. Concatenate the entire primal list and then the entire dual list. This gives the 50 entries of λ. Remove every edge touching index four, preserving the remaining order, to obtain the 32 entries of μ. Thus the x-then-y ordering applies to species edges, not to the raw matrix slots.

Let $O, N : \mathbb{R}^{10} \to \mathbb{R}^8$ select, respectively, pairs $0, 1, 2, 3$ and $1, 2, 3, 4$. With P_r as in (93), reconstruction is

$$\begin{aligned} \widetilde{C}(\alpha) &= O^{\mathsf{T}} P_r O - \sum_e \widehat{\mu}_e Q_e(\alpha), \\ D(\alpha) &= O^{\mathsf{T}} P_r O - N^{\mathsf{T}} P_r N - \sum_e \widehat{\lambda}_e Q_e(\alpha). \end{aligned} \tag{218}$$

The final pair of rows and columns of $\widetilde{C}(\alpha)$ vanish. Its leading 8×8 block is C_r^{mat}, and $D(\alpha) = D_r$. At $\alpha = 1$, these give C_0 and D_0. The endpoint storage value coefficients vanish, and both weighted value vectors sum to zero. Every nonzero multiplier is positive and has two ordinary output endpoints, with indices at most four (at most three for μ). Thus the formal identities become actual nonnegative-slack identities without a value at the initial input or an anchor slack.

For clarity, the positive-semidefinite checks concern C_0, D_0 and their extended principal primal blocks, not P_0. Exact symmetric Schur elimination gives

$$\operatorname{rank} C_0 = 6, \quad \operatorname{rank} D_0 = 4, \quad \operatorname{rank}(C_0)_x = \operatorname{rank}(D_0)_x = 3,$$

with positive pivots and an exactly zero final residual in each case. In this elimination, permute a nonzero diagonal entry to the leading position, verify that it is positive in the ordered field, and replace the remaining matrix by its exact Schur complement. Repeating and verifying a zero residual gives a congruence to a positive diagonal block plus zero, hence proves positive semidefiniteness. No floating-point eigenvalue test is involved. The two constant-history columns are annihilated by C_0 and P_0; rank six then identifies the full kernel of C_0.

The cycle matrix is the same A as in (115), evaluated at r_*:

$$\mathsf{A} = \begin{pmatrix} 9r_*/16 - 1/2 & -r_* \\ 11/16 + 3/(2r_*) & -1/2 - 9r_*/16 \end{pmatrix}, \qquad \mathsf{A}^2 + \mathsf{A} + I_2 = 0, \qquad \det \mathsf{A} = 1. \tag{219}$$

Construct a 10×6 matrix K whose two rows at pair i are the block row $[I_2,\ iI_2,\ \mathsf{A}^i]$, for $0 \le i \le 4$. Exact multiplication gives $D_0 K = 0$, and exact elimination gives $\operatorname{rank} K = 6$. Together with $\operatorname{rank} D_0 = 4$, this proves precisely

$$\ker D_0 = \{(a + ib + \mathsf{A}^i v)_{i=0}^4 : a, b, v \in \mathbb{R}^2\}. \tag{220}$$

In (218), each coefficient is affine in α. Reconstructing the constant coefficient and the coefficient of α verifies exactly $C_r^{\mathrm{mat}} = C_0 - (1-\alpha)(C_0)_x$ and $D_r = D_0 - (1-\alpha)(D_0)_x$. All value coefficients remain zero. Evaluation at $\alpha = 0$ can be used to read the constant coefficient; it is an algebraic operation and does not posit an algorithm with an infinite step.

These operations completely specify the matrix proof from the common raw coefficient file. The public checker `check_anchored_storage.py` reconstructs the slacks, value cancellations, exact ranks and kernels, and affine continuation, and writes fresh exact results. Sign tests for $a + b\sqrt{91}$ use rational signs when equal and comparison of a^2 with $91b^2$ when they oppose. The continuous compactness and orbit arguments are supplied in the main text rather than inferred from these finite checks.

B.2 Four-output order and complete cone coverage

Lemma 13 (Four-output order rigidity)**.** *Let $z_i = a + ib + \mathsf{A}^i v$, $0 \le i \le 4$, with $a, b, v \in \mathbb{R}^2$. Form the literal residuals* (131) *at $r = r_*$ and at outputs $i = 1, 2, 3, 4$. If*

$$(x_i - x_j)(p_i - p_j) \ge 0, \qquad (y_i - y_j)(q_i - q_j) \ge 0 \qquad (1 \le i, j \le 4),$$

then $v = 0$ and $b_x b_y = 0$. Conversely every affine axis ramp, with $v = 0$ and $b_x b_y = 0$, satisfies these pairwise inequalities. No anchor hypothesis is required for this lemma.

Proof. For a scalar graph list (X_i, P_i), pairwise monotonicity is equivalent to the existence of a permutation ordering both lists nondecreasingly. For necessity, first sort by X, and within a tied-X block sort by P. Between distinct X blocks, strict order of the abscissae and pairwise monotonicity force the same weak order of ordinates. For sufficiency, two lists nondecreasing along one permutation have same-sign or zero differences for every pair. Adjacent comparisons imply all comparisons by summation. This argument includes arbitrary ties, vertical and flat graph pieces, and constant species, and uses no division by a coordinate difference.

We first describe the saved finite certificate in its own indexing. Its four output labels $0, 1, 2, 3$ mean states at times $0, 1, 2, 3$, preceded by input time -1. Use

$$z_i = a + ib + \mathsf{A}^i v \qquad (-1 \le i \le 3), \qquad \omega = (b_x, b_y, v_x, v_y)^{\mathsf{T}}.$$

The inverse at time -1 is $\mathsf{A}^{-1} = \mathsf{A}^2$, by (219). The center a contributes the common constants $a_x, a_y, -a_y, a_x$ to x, y, p, q, respectively, and disappears from all differences.

For a pair (π, ρ) of permutations of $\{0,1,2,3\}$, define the 12×4 matrix $M^{\mathrm{old}}_{\pi,\rho}$ to have, in order, the coefficient rows of

$$\begin{aligned} x_{\pi(j+1)} - x_{\pi(j)}, \quad j = 0,1,2; \qquad & p_{\pi(j+1)} - p_{\pi(j)}, \quad j = 0,1,2; \\ y_{\rho(j+1)} - y_{\rho(j)}, \quad j = 0,1,2; \qquad & q_{\rho(j+1)} - q_{\rho(j)}, \quad j = 0,1,2. \end{aligned} \tag{221}$$

Each difference is later minus earlier in the relevant permutation. The joint monotonicity set is exactly the union of the $24^2 = 576$ closed cones

$$\{\omega : M^{\mathrm{old}}_{\pi,\rho}\omega \geq 0\}.$$

The permutations for the two species are independent. The cones may overlap on their faces; uniqueness is unnecessary.

For every one of these permutation pairs, the finite data supply four nonnegative multiplier vectors λ, one for each $j \in \{3,4\}$ and $\epsilon \in \{-1,1\}$, such that

$$(M^{\mathrm{old}}_{\pi,\rho})^{\mathsf{T}}\lambda = \epsilon e_j, \qquad \lambda \geq 0. \tag{222}$$

Here e_j uses one-based mathematical coordinates of ω. On the cone, multiplying its inequalities by λ proves $\epsilon\omega_j \geq 0$. The two signs therefore force both components of the old v to be zero. The exact schema and all finite verification obligations are given below.

There is a necessary index change before using these identities on outputs $1,2,3,4$. If old time k denotes new time $k+1$, then

$$z_{k+1} = (a+b) + kb + \mathsf{A}^k(\mathsf{A}v), \qquad a_{\mathrm{old}} = a_{\mathrm{new}} + b_{\mathrm{new}}, \quad b_{\mathrm{old}} = b_{\mathrm{new}}, \quad v_{\mathrm{old}} = \mathsf{A}v_{\mathrm{new}}. \tag{223}$$

Let $T = \mathrm{diag}(I_2, \mathsf{A})$. Center shifts still cancel, and direct residual construction gives

$$M^{\mathrm{new}}_{\pi,\rho} = M^{\mathrm{old}}_{\pi,\rho}T, \qquad \lambda^{\mathsf{T}}M^{\mathrm{new}}_{\pi,\rho} = \epsilon e_j^{\mathsf{T}}T. \tag{224}$$

Thus the same multipliers force both signs of each component of $\mathsf{A}v_{\mathrm{new}}$, not immediately of v_{new}. Since $\det \mathsf{A} = 1$, this implies $v_{\mathrm{new}} = 0$. In particular the saved output labelled zero has not been mistaken for the arbitrary initial input of the present theorem.

After $v = 0$, the actual residuals at the present outputs are

$$p_i = -b_x/r_* - a_y - (i-1)b_y, \qquad q_i = -b_y + a_x + ib_x + b_x/8.$$

For every distinct pair of outputs,

$$(x_i - x_j)(p_i - p_j) = -(i-j)^2 b_x b_y, \qquad (y_i - y_j)(q_i - q_j) = (i-j)^2 b_x b_y.$$

The two inequalities are therefore equivalent to $b_x b_y = 0$, also proving the converse. Anchor restrictions on these ramps are the separate content of Lemma 8. ∎

B.3 Finite certificate schema and replay

The file `kernel_farkas_certificates.json` is a list of 576 records. Each record has exactly the keys `x_order`, `y_order`, and `certificates`. The first two are permutations of the integer labels $0, 1, 2, 3$. Every ordered pair of permutations occurs exactly once. Each `certificates` list has four records with keys `coordinate`, `sign`, `support`, and `weights`:
(1) `coordinate` is 2 or 3 in zero-based indexing of (b_x, b_y, v_x, v_y), and `sign` is -1 or $+1$. Every target/sign pair occurs exactly once per cone.
(2) `support` is a list of distinct integers between 0 and 11, selecting the rows in (221). The `weights` list has the same length, at most four, and is paired entrywise with this support. Unlisted weights are zero.
(3) Each weight is an exact expression built from integers, rational arithmetic, and `sqrt(91)`. It is parsed in $\mathbb{Q}(\sqrt{91})$; division requires a nonzero exact denominator. Its sign is established in the positive real embedding by rational square comparison, with no floating-point approximation.

The standalone `check_scalar_order.py` reconstructs all graph rows directly from the states and actual residuals, checks the complete permutation-pair set, validates distinct supports and complete target coverage, and checks all 2304 identities (222). It then independently reconstructs the present output rows and verifies both identities in (224), including all 2304 transformed target identities. These are the same certificate witnesses in two coordinate conventions, not two unrelated rigidity assumptions. There are 7480 listed weights, all strictly positive; support sizes are two for 394 identities, three for 948, and four for 962. All target and row coefficients are checked exactly.

A finite identity $M^{\mathsf{T}}\lambda = \epsilon e_j$ with $\lambda \geq 0$ is its own proof on the associated closed cone: it is an explicit nonnegative combination of that cone's defining inequalities. Complete coverage of all 576 cones therefore closes the argument for every real graph list, including all faces and tied configurations. No numerical feasibility failure, optimizer, or sample orbit is used. The checker also tests the common-order equivalence on all 75^2 pairs of weak-order types of four labels; the sorting argument above supplies its general proof independently of that supplemental enumeration.

This finite rigidity statement removes the oscillatory component and leaves affine axis ramps. It supplies neither first-increment coercivity of D_0 nor a convergence theorem by itself. The quantitative axis neighborhoods, compact-complement compensation, exact continuation, and global summation in Sections 5.5–7.4 provide those additional analytic steps in their stated scalar scope.

C Exact periodic-order certificates

This appendix gives the complete finite specification used in Theorem 8 and the exact short-period identities used in Section 3.6. The data are finite rational weights, rather

than feasibility reports. The analytic implications, including weak-order coverage and the equality cases, were proved in the body of the manuscript.

C.1 Rows, labels and rational encoding

Use the coordinate vectors e_i^x, e_i^w of $\mathbb{R}^{12}$, with all subscripts modulo six, and define the row coefficient vectors

$$\begin{aligned} X_i &= e_i^x, & W_i &= e_i^w, \\ P_i^0 &= e_{i-1}^x - e_i^x - e_{i-1}^w, & Q_i^0 &= 8(e_{i-1}^w - e_i^w), \quad Q_i^1 = 9e_i^x - e_{i-1}^x. \end{aligned} \tag{225}$$

Thus $(P_i^0)^{\mathsf{T}}\mathbf{z} = P_i$ and $(Q_i^0 + rQ_i^1)^{\mathsf{T}}\mathbf{z} = Q_i$ in (55). For $0 \le k < 5$, the rows of M_0 in the four successive blocks are

$$(X_{\pi_{k+1}} - X_{\pi_k})^{\mathsf{T}}, \quad (P^0_{\pi_{k+1}} - P^0_{\pi_k})^{\mathsf{T}}, \quad (W_{\rho_{k+1}} - W_{\rho_k})^{\mathsf{T}}, \quad (Q^0_{\rho_{k+1}} - Q^0_{\rho_k})^{\mathsf{T}}.$$

The first fifteen rows of M_1 vanish; its last five rows are $(Q^1_{\rho_{k+1}} - Q^1_{\rho_k})^{\mathsf{T}}$. These integer formulas specify the entire affine matrix $M(r) = M_0 + rM_1$, including both translation directions.

List the 64 normalized primal representatives $\pi \le R(\pi)$ in lexicographic order, with $\pi_0 = 0$, using (57). List all 720 permutations ρ lexicographically. The gzip-compressed UTF-8 file `six_interval_certificates.jsonl.gz` contains exactly one JSON object for every pair of zero-based indices (p, q) in these lists. Each object has exactly four fields:

1. `p` and `q`: the primal and dual order indices;
2. `s`: distinct integer support indices in $\{0, \dots, 39\}$;
3. `w`: equally many nonnegative rational strings.

An index $s < 20$ selects entry s of u; an index $s \ge 20$ selects entry $s - 20$ of v. Omitted entries are exactly zero. Each rational string denotes an integer or a quotient of integers with a strictly positive denominator. The verifier rejects duplicate JSON keys, duplicate support indices, invalid labels and negative weights. Every required (p, q) occurs once and only once. No discovery file, optimizer or saved result is read to reconstruct these objects.

C.2 The exact coefficient tests

For one record let $D > 0$ be the least common multiple of its rational denominators, and set $U = Du$, $V = Dv$. All entries of U, V are nonnegative integers. Equation (58) is equivalent to the following three integer vector equations:

$$\begin{aligned} M_0^{\mathsf{T}}(300U - 297V) &= -300D\, d_\pi, \\ M_0^{\mathsf{T}}\big(100(V-U)\big) + M_1^{\mathsf{T}}(300U - 297V) &= 100D\, d_\pi, \\ M_1^{\mathsf{T}}\big(100(V-U)\big) &= 0. \end{aligned} \tag{226}$$

The supplied `check_six_periods.py` rebuilds the rows from (225) and checks every coordinate of these equations. There are $46\,080 \cdot 3 \cdot 12 = 1\,658\,880$ such scalar coefficient identities. The records contain 690 088 explicitly stored nonnegative weights, with support sizes from 3 to 25; the largest common denominator is 176133351776184.

For comparison, the separately supplied author checker `verify_six_interval.py` sets $x_0 = w_0 = 0$ and checks the equivalent ten-coordinate endpoint form. Let $a = 297/100$, $b = 3$, $t = (r-a)/(b-a)$, and let $\widehat{M}(r)$ be $M(r)$ with columns x_0, w_0 deleted. With $A = \widehat{M}(a)$, $B = \widehat{M}(b)$ and e selecting x_{π_5} in these ten coordinates, it verifies

$$A^{\mathsf{T}}u = -e, \qquad A^{\mathsf{T}}v + B^{\mathsf{T}}u = -e, \qquad B^{\mathsf{T}}v = 0. \tag{227}$$

Indeed, $\widehat{M}(r) = (1-t)A + tB$, and multiplying by $(1-t)u + tv$ gives $-(1-t)e$. Since $300 - 100r = 3(1-t)$ and $100r - 297 = 3t$, multiplication by three recovers (58) in the reduced coordinates. For integer checking this version forms $100A$ and tests

$$(100A)^{\mathsf{T}}U = -100De, \quad (100A)^{\mathsf{T}}V + 100B^{\mathsf{T}}U = -100De, \quad B^{\mathsf{T}}V = 0.$$

It checks $46\,080 \cdot 3 \cdot 10 = 1\,382\,400$ identities. The twelve-coordinate and ten-coordinate counts are two representations of the same certificate, and are not added as independent constraints.

The independent checker also verifies the complete raw-order coverage in Lemma 1, 576 literal residual scaling equalities, 1296 rotation-covariance coefficient equalities, translation invariance of the rows, and every coefficient of (59). It retains all 4096 binary tie/flat-graph regression cases; the real-valued ordering proof is the one in Lemma 1, not this finite test. Four deliberate negative controls are rejected: an altered weight, a negative weight, a duplicate support index, and an invalid primal label. Both checkers recompute six literal steps of the boundary witness and the exact comparisons $297/100 < r_* < 3$.

Both six-period checkers use only the Python standard library and exact integer or rational arithmetic in proof tests. The independent checker checks its local certificate pin before and after the run. Its result is written to `six_period_results.json`; the author checker writes `verification_summary.json`. Neither saved result is a proof input. The source archive's current input pins and manifest identify the data and scripts used in these replays.

C.3 Exact five-period algebra and signs

For completeness, the shift expansions in (67) can be checked using only real 2×2 matrices. Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad R = \cos\phi\, I + \sin\phi\, J, \qquad \phi = 2\pi k/5.$$

Writing $\operatorname{sym} A = (A + A^{\mathsf{T}})/2$, direct multiplication gives, for $j = 1, 2, 3, 4$,

$$\begin{aligned} \operatorname{sym}\big((R^{-1} - I)^{\mathsf{T}}(R^j - I)\big) &= D_{kj} I, \\ (R^{-1})^{\mathsf{T}}(R^j - I) &= \operatorname{Re} Z_{kj}\, I + \operatorname{Im} Z_{kj}\, J, \\ \big((1+\theta)I - \theta R^{-1}\big)^{\mathsf{T}}(R^j - I) &= \operatorname{Re} W_{kj}\, I + \operatorname{Im} W_{kj}\, J. \end{aligned} \tag{228}$$

If $c = \langle u, v\rangle$ and $d = \langle Ju, v\rangle$, then $\langle u, Jv\rangle = -d$. The last two matrix identities therefore give respectively $-\operatorname{Re} Z_{kj} c - \operatorname{Im} Z_{kj} d$ and $\operatorname{Re} W_{kj} c - \operatorname{Im} W_{kj} d$ as the primal and dual cross terms, exactly as used in the proof.

The positive signs in (70) are fully algebraic. For the three minus-sign numerators, the required rational square comparisons are

$$\begin{aligned} 6750^2 - 5 \cdot 2720^2 &= 8570500 > 0, \\ 575^2 - 5 \cdot 218^2 &= 93005 > 0, \\ 3980285^2 - 5 \cdot 1556275^2 &= 3732709303100 > 0. \end{aligned} \tag{229}$$

All the rational terms being compared are positive, so squaring preserves the relevant order. These identities prove positivity without numerical approximations to $\sqrt{5}$.

The directory `short_periods/` contains the two scripts `verify_independent.py` and `verify_corollaries_independent.py`, which retain the exact real-matrix and sign checks. They use SymPy, tested here at version 1.14.0. With $a = \cos(2\pi/5)$ and $b = \sin(2\pi/5)$, the first checker reduces polynomial identities modulo $4a^2 + 2a - 1 = 0$ and $a^2 + b^2 - 1 = 0$. It verifies roots-of-unity, orthogonality and all cyclic normal-test identities in both modes, with symbolic extrapolation. The second imports that helper and computes the weighted coefficients in (69) exactly. Its signs lie in $\mathbb{Q}(\sqrt{5})$ and are decided by rational square comparisons. The scripts' broader auxiliary algebra tests are retained, but Lemma 3 uses only the $\theta = 1/8$, $r_{\mathrm{c}} = 31/10$ exclusion weights and their positive-definite mode forms. No auxiliary existence result or comparison with a different admissible class is a premise of the scalar corollary. Fresh console output is retained in `five_period_results.txt`.

Thus the periodic proofs have a closed source dependency: the six-period rational data, the two standard-library six-period checkers, and the two exact short-period scripts with their declared SymPy dependency. The four-period proof is the analytic shift argument in Lemma 2. No earlier report, discovery log, numerical search or external source tree is needed to reproduce the assertions used in Corollary 3.1.

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